

Replicating Anomalies

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Most anomalies fail to hold up to currently acceptable standards for empirical finance. With microcaps mitigated via NYSE breakpoints and value-weighted returns, 65% of the 452 anomalies in our extensive data library, including 96% of the trading frictions category, cannot clear the *single* test hurdle of the absolute *t*-value of 1.96. Imposing the higher multiple test hurdle of 2.78 at the 5% significance level raises the failure rate to 82%. Even for replicated anomalies, their economic magnitudes are much smaller than originally reported. In all, capital markets are more efficient than previously recognized. (*JEL* C58, G12, G14, G17, M41)

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This paper replicates the bulk of the published anomalies literature in finance and accounting by compiling an extensive data library with 452 anomaly variables. We adopt a common set of replication procedures. To ensure

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the reliability of the replicated anomalies, we control for microcaps (stocks smaller than the 20th percentile of the market equity for NYSE stocks) via portfolio sorts with NYSE breakpoints and value-weighted returns. We treat an anomaly as a replication success if the average return of its high-minus-low decile is significant at the 5% threshold (the absolute t -value, $|t|, \geq 1.96$).

Our key finding is that most anomalies fail to replicate, falling short of currently acceptable standards for empirical finance. First, of the 452 anomalies, 65% cannot clear the single test hurdle of $|t| \geq 1.96$. The key word is “microcaps.” Microcaps represent only 3.2% of the aggregate market capitalization but 60.7% of the number of stocks. Microcaps have the highest equal-weighted returns and the largest cross-sectional dispersions in returns and in anomaly variables. Many original studies overweight microcaps via equal-weighted returns and often with NYSE-Amex-NASDAQ breakpoints in portfolio sorts. Hundreds of studies perform cross-sectional regressions of returns on anomaly variables, mostly with ordinary least squares, which are highly sensitive to microcap outliers.

Second, regardless of microcaps, most anomalies fail to replicate if we adjust for multiple testing. With NYSE-Amex-NASDAQ breakpoints and equal-weighted returns, which assign maximum weights to microcaps in portfolio sorts, the failure rate across the 452 anomalies is 41.4% with the single test hurdle of $|t| \geq 1.96$, but 52% with the multiple test hurdle of 2.78, both at the 5% significance level. For cross-sectional regressions with ordinary least squares, which assign maximum weights to microcaps in regressions, the failure rate is 41.8% for single tests but 51.5% for multiple tests.

Third, the large-scale replication failure is not due to our extended samples through December 2016. Repeating our tests on the shorter samples in the original studies, we find that 65.3% of the anomalies cannot clear the single test hurdle of $|t| \geq 1.96$ with NYSE breakpoints and value-weighted returns. The failure rate drops to 43.1% if we allow microcaps to run amok with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns. However, the failure rate rises to 56.2% with the multiple test hurdle of $|t| \geq 2.78$. These results are quantitatively similar to those in the extended samples.

The biggest casualty of our replication is the trading frictions literature. In the category that contains mostly liquidity, market microstructure, and other trading frictions variables, 102 of 106 anomalies (96%) fail to replicate in *single* tests. Prominent anomalies that fail to replicate include the Jegadeesh (1990) short-term reversal; the Datar, Naik, and Radcliffe (1998) share turnover; the Chordia, Subrahmanyam, and Anshuman (2001) coefficient of variation for dollar trading volume; the Amihud (2002) absolute return-to-volume; the Easley, Hvidkjaer, and O'Hara (2002) probability of informed trading; the Pastor and Stambaugh (2003) liquidity beta; the Acharya and Pedersen (2005) liquidity betas; the Ang et al. (2006) idiosyncratic volatility, total volatility, and systematic volatility; the Liu (2006) number of zero daily trading

volume; the Bali, Cakici, and Whitelaw (2011) maximum daily return; the Corwin and Schultz (2012) high-low bid-ask spread; the Adrian, Etula, and Muir (2014) financial intermediary leverage beta; and the Kelly and Jiang (2014) tail risk.

Maximally weighting microcaps via NYSE-Amex-NASDAQ breakpoints and equal-weighted returns in sorts or cross-sectional regressions with ordinary least squares does not cure the replication failure of the trading frictions literature. In total, 60.4% of the anomalies in sorts and 62.3% in cross-sectional regressions cannot clear the single test hurdle. The equal-weighted sorts revive short-term reversal, share turnover, dollar trading volume, absolute return-to-volume, and the number of zero trading days, but not the probability of informed trading, the Pastor and Stambaugh (2003) liquidity beta, the Acharya and Pedersen (2005) liquidity betas, idiosyncratic volatility, the high-low bid-ask spread, or the intermediary leverage beta.

Other influential anomalies that fail to replicate include the Bhandari (1988) debt-to-market; the Lakonishok, Shleifer, and Vishny (1994) 5-year sales growth; the La Porta (1996) long-term analysts' forecasts; several of the Abarbanell and Bushee (1998) fundamental signals; the O-score and the Z-score studied in Dichev (1998); the Piotroski (2000) fundamental score; the Diether, Malloy, and Scherbina (2002) dispersion in analysts' forecasts; the Gompers, Ishii, and Metrick (2003) corporate governance index; the Francis et al. (2004) earnings attributes, including persistence, smoothness, value relevance, and conservatism; the Francis et al. (2005) accrual quality; the Richardson et al. (2005) total accruals; the Campbell, Hilscher, and Szilagyi (2008) failure probability; and the Fama and French (2015) operating profitability.

Even for replicated anomalies, their economic magnitudes are much smaller than originally reported. Famous examples include the Jegadeesh and Titman (1993) price momentum; the Lakonishok, Shleifer, and Vishny (1994) cash flow-to-price; the Sloan (1996) operating accruals; the Chan, Jegadeesh, and Lakonishok (1996) earnings momentum formed on standardized unexpected earnings, abnormal stock returns around earnings announcements, and revisions in analysts' earnings forecasts; the Cohen and Frazzini (2008) customer momentum; and the Cooper, Gulen, and Schill (2008) asset growth.

We follow the replication literature in economics in defining replication as "any study whose primary purpose is to establish the correctness of a previous study" (The Replication Network¹). Hamermesh (2007) distinguishes three categories of replication. Pure replication (reproduction) is redoing a prior study in exactly the same way. Statistical replication is the same empirical model but different sample from the same underlying population. Scientific replication is different sample, different population, and similar, but not identical, statistical

¹ See <https://replicationnetwork.com>.

model. Hamermesh (2007, p. 716) argues that scientific replication “appears much more suited in type to our methods of research and, indeed, comprises most of what economists view as replication.” The crux is that unlike natural sciences, economics, finance, and accounting are mostly observational in nature. As such, it is critical to evaluate the reliability of published results against “similar, but not identical,” specifications.

In our large-scale replication, we utilize the same population, different and same samples, as well as similar, but not identical, methods. We closely follow the variable definitions from the original studies. While reporting results from a variety of different procedures, we emphasize sorts with NYSE breakpoints and value-weighted returns as well as cross-sectional regressions with weighted least squares, because of their economic importance and statistical reliability. Most important, the eight replication articles in the May 2017 issue of *American Economic Review* all adopt the same definition of replication as we do.²

Finally, we explore the commonality in the 158 replicated anomalies with $|t| \geq 1.96$ in sorts with NYSE breakpoints and value-weighted returns. We show that our ex ante economic categorization of anomalies is largely consistent with ex post statistical clustering and principle component analysis.

Our major contribution is to provide a direct, large-scale replication in finance and accounting. Using a multiple testing framework, Harvey, Liu, and Zhu (2016, p. 5) cast doubt on the credibility of the anomalies literature, concluding that “most claimed research findings in financial economics are likely false.” However, they do not attempt replication. Failing to replicate most of the published anomalies, our extensive evidence lends support to their conclusion.

1. Motivating Replication

In a pioneering meta-study in finance, Harvey, Liu, and Zhu (2016) present a multiple testing framework to derive threshold levels to account for data mining. The threshold cutoff increases over time as more anomalies are data mined. Reevaluating 296 significant anomalies in past published studies, they report that 80–158 (27%–53%) are likely false discoveries, depending on the specific adjustment for multiple testing.³ Two publication biases are likely responsible

² For example, Berry et al. (2017, p. 27) define replication as “any project that reports results that speak directly to the veracity of the original paper’s main hypothesis.” Hamermesh (2017, p. 38) writes: “Applied microeconomics is not a laboratory science—at its best it consists of the generation of new ideas describing economic behavior, independent of time or space. The empirical validity of these ideas, after their relevance is first demonstrated for a particular time and place, can only be usefully replicated at other times and places: If they are general descriptions of behavior, they should hold up beyond their original testing ground.” Duvendack, Palmer-Jones, and Reed (2017, p. 47) operationalize replication as “any study whose main purpose is to determine the validity of one or more empirical results from a previously published study.” Duvendack, Palmer-Jones, and Reed (2017, p. 46) further write: “By redoing the original data analysis, by adjusting model specifications, exploring the influence of unusual observations, using different estimation methods, and alternative datasets, replication can identify spurious or fragile results.”

³ Finance academics have long warned against data mining. Lo and MacKinlay (1990) show that few studies are free of data mining, which becomes more severe as the number of studies on a single dataset increases.

for the high percentage of false discoveries. First, it is difficult to publish a nonresult in top academic journals. Second, it is difficult to publish replication studies in finance, while replications routinely appear in top journals in other scientific fields. As a result, finance academics tend to focus on publishing new results, rather than rigorously verifying the reliability of published results.

Harvey (2017) elaborates a complex agency problem behind the publication biases. Editors compete for citation-based impact factors and prefer to publish papers with the most significant results. In response, authors often file away papers with weak or nonresults, instead of submitting them for publication. More disconcertingly, authors sometimes engage in specification search, selecting sample criteria and test procedures until insignificant results become significant (p-hacking). The likely outcome is an embarrassingly large number of false positives that cannot be replicated in the future.

Finance is only the latest field that starts to take replication seriously. In economics, Leamer (1983) exposes the fragility of empirical results to small specification changes and proposes to “take the con out of econometrics” by reporting extensive sensitivity analyses. Dewald, Thursby, and Anderson (1986) attempt to replicate results published in *Journal of Money, Credit, and Banking*, but find that inadvertent errors are so commonplace that the original results often cannot be reproduced.⁴

In a very influential meta-study, Ioannidis (2005) argues that most research findings are false for most designs in most fields. Results are more likely to be false when the studies in a field use smaller samples; when the effect magnitudes are smaller; when there exist many but fewer theoretically predicted relations; when researchers have more degrees of freedom in test designs, variable definitions, and analytical methods; when there exist greater financial and other interest and bias; and when more independent teams are involved in a field. In the almost 15 years since Ioannidis’s (2005) study, replication failures have been widely documented throughout diverse scientific disciplines.⁵

Fama (1998) shows that a number of anomalies weaken or even disappear with value-weighted returns. Conrad, Cooper, and Kaul (2003) argue that data mining can account for up to one half of the relations between characteristics and average returns. Schwert (2003) shows that after anomalies are documented, the patterns often seem to disappear, reverse, or weaken. McLean and Pontiff (2016) report that the average return spreads of 97 anomalies decline out of sample and post publication.

⁴ Other important replication studies in economics include McCullough and Vinod (2003), Brodeur, Lé, Sangnier, and Zylberberg (2016), Camerer et al. (2016), and Chang and Li (2018). In a recent survey of the replication literature in economics, Christensen and Miguel (2018, p. 940) write that “an overall increase in replication research will serve a critical role in establishing the credibility of empirical findings in economics, and in equilibrium, will create stronger incentives for scholars to generate more reliable results.”

⁵ For example, in oncology, Prinz, Schlange, and Asadullah (2011) report that scientists at Bayer fail to reproduce two-thirds of 67 published studies. Begley and Ellis (2012) report that scientists at Amgen attempt to replicate 53 landmark studies but reproduce the original results in only six. In psychology, Open Science Collaboration (2015), which consists of about 270 researchers, conducts replications of 100 studies published in the top three academic journals and reports a success rate of only 36%. Baker (2016) reports that 90% of the respondents in a survey of 1,576 scientists believe that there exists a reproducibility crisis in the published scientific literature. More than 70% of researchers have tried and failed to reproduce other scientists’ experiments, and more than

Most, if not all, of the conditions, against which Ioannidis (2005) warns, apply to the anomalies literature. First, Ioannidis, Stanley, and Doucouliagos (2017) report that the median statistical power is only 18% or less from 64,076 estimates in more than 6,700 studies in economics and finance. Second, the anomalies literature is mostly empirical in nature. Fama and French (1992) reject the classic Capital Asset Pricing Model (CAPM). The consumption CAPM often performs even worse and is rarely used. As a result, empiricists are free to explore hundreds of variables, with little *a priori* hypothesizing as for why a variable should forecast returns.

Third, publication biases are well documented in economics (De Long and Lang 1992; Card and Krueger 1995). Fourth, empiricists have many degrees of freedom in exploiting ambiguities in sample criteria, variable definitions, and test specifications, all of which are tools of *p*-hacking. Fifth, with trillions of dollars invested in factors-based exchange-traded funds and quantitative hedge funds worldwide, the financial interest is overwhelming.⁶ Sixth and finally, armies of academics and investment managers actively engage in searching for significant anomalies, each eager to beat competitors. Consequently, the anomalies literature is one of the biggest areas in finance and accounting.

2. Replicating Procedures

Our replication target consists of 452 anomalies, including 57, 69, 38, 79, 103, and 106 anomalies from the momentum, value versus growth, investment, profitability, intangibles, and trading frictions categories, respectively. Our list encompasses the bulk of the published anomalies literature in finance and accounting. Appendix A details variable definitions and portfolio construction.

Although we vary the methods in forming portfolios and in performing cross-sectional regressions (Section 2.1), we closely follow the variable definitions in the original studies. In addition, when necessary, we perform small perturbations to the original variable definitions, such as changing the scalar of a ratio variable, to evaluate the reliability of its predictive power for returns. For monthly sorted anomalies, we include three different predictive horizons (1, 6, and 12 months). Chan, Jegadeesh, and Lakonishok (1996), for example, highlight the short-lived nature of momentum, by examining how momentum profits vary with the holding period. As such, it seems economically interesting to study how monthly sorted anomalies vary over different horizons.

Our sample criterion is standard. Monthly returns are from the Center for Research in Security Prices (CRSP) and accounting information from the

50% have failed to reproduce their own experiments. Selective reporting, pressure to publish, and poor use of statistics are the three leading causes of the reproducibility crisis.

⁶ ETFGI. 2018. ETFGI reports assets invested in smart beta ETFs and ETPs listed globally reached a new high of \$680 Bn at the end of August 2018 [Press Release]. Retrieved from <https://etfgi.com/news/press-releases/2018/10/etfgi-reports-assets-invested-smart-beta-etfs-and-etps-listed-globally>

Compustat Annual and Quarterly Fundamental Files. The sample period is from January 1967 to December 2016. We exclude financial firms and firms with negative book equity. Some studies exclude stocks with prices per share lower than \$1 or \$5. We do not impose such a screen. In particular, microcaps are included in our sample.

To test whether an anomaly variable predicts returns, we adopt a variety of procedures described in Section 2.1. In Section 2.2, we emphasize the reliability of sorts with NYSE breakpoints and value-weighted returns as well as cross-sectional regressions with weighted least squares. In Section 2.3, we report new evidence on the necessity to control for microcaps.

2.1 A common set of replicating procedures

We adopt a variety of methods to evaluate the reliability of the predictive power of an anomaly variable. For portfolio sorts (into deciles), we vary breakpoints and return weights, including NYSE breakpoints and value-weighted returns (NYSE-VW), NYSE breakpoints and equal-weighted returns (NYSE-EW), NYSE-Amex-NASDAQ breakpoints and value-weighted returns (All-VW), as well as NYSE-Amex-NASDAQ breakpoints and equal-weighted returns (All-EW). For the Fama and MacBeth (1973) cross-sectional regressions, we use both ordinary least squares (FM-OLS) and weighted least squares with the market equity as the weights (FM-WLS).⁷

For annually sorted deciles, we split stocks at the end of June of each year t into deciles on a variable measured at the fiscal year ending in calendar year $t - 1$ and calculate decile returns from July of year t to June of $t + 1$. For monthly sorted portfolios involving the latest earnings data, we use quarterly earnings data in the months immediately after quarterly earnings announcement dates (Compustat quarterly item RDQ). For monthly sorted portfolios involving quarterly accounting data other than earnings, we impose a 4-month lag between the fiscal quarter end and subsequent returns. Unlike earnings, other quarterly items are typically not available on earnings announcement dates. Many firms announce their earnings for a given quarter through a press release and then file SEC reports several weeks later. Easton and Zmijewski (1993) document a median reporting lag of 46 days for NYSE and Amex firms and 52 days for NASDAQ firms. Chen, DeFond, and Park (2002) report that only 37% of quarterly earnings announcements include balance sheet information.

Following Beaver, McNichols, and Price (2007), we adjust monthly returns for delisting by compounding returns in the partial month before delisting with

⁷ As noted, microcaps are included in our sample. In the Internet Appendix, we have also furnished supplementary results from all-but-micro breakpoints and value-weighted returns (ABM-VW), all-but-micro breakpoints and equal-weighted returns (ABM-EW), micro breakpoints and value-weighted returns (Micro-VW), as well as micro breakpoints and equal-weighted returns (Micro-EW). At each portfolio formation date, we form all-but-micro breakpoints using the sample that excludes microcaps and form micro breakpoints using the sample that includes only microcaps. When calculating decile returns, we exclude microcaps in ABM-VW and ABM-EW but include only microcaps in Micro-VW and Micro-EW.

delisting returns from CRSP. If missing, we replace a delisting return with the mean of available delisting returns of the same delisting type and stock exchange in the prior 60 months. Appendix B details our adjustment procedure.

When performing monthly cross-sectional regressions, we winsorize the regressors at the 1%–99% level each month to mitigate the impact of outliers. Also, different anomaly variables often have vastly different units. To make their slopes comparable, we standardize a given winsorized regressor by subtracting its cross-sectional mean and then dividing by its cross-sectional standard deviation. A slope then estimates the change in the average return when the regressor varies by one cross-sectional standard deviation. The slope is also the return to a zero-investment long-short portfolio (Fama 1976).⁸ However, in general, the long and short legs of the slope portfolio do not have total weights that sum to one. As such, the magnitude of the slopes is not directly comparable to the magnitude of the average returns of the high-minus-low deciles.

For anomalies with a multi-month holding period, such as standardized unexpected earnings with a 6-month holding period, denoted by Sue6, at the beginning of month t , we regress the return in month t on Sue6 known at the beginning of month $t-s$, for $s=0, 1, \dots, 5$. We then take the average of the slopes from the six subregressions as the slope for Sue6 in month t and calculate its t -value from the time-series of the average. This procedure is analogous to our portfolio construction of the Sue6 deciles, in which we average across the six subdeciles formed at the beginning of month $t-s$, for $s=0, 1, \dots, 5$, as the return for a given decile (Jegadeesh and Titman 1993). All the t -values are adjusted for heteroskedasticity and autocorrelations (Newey and West 1987).

In addition to their economic magnitudes, we evaluate the statistical significance of the high-minus-low average returns from portfolio sorts and the slopes from cross-sectional regressions. We focus on the traditional, single test absolute t -value, $|t|$, cutoff of 1.96 at the 5% significance level. To adjust for multiple testing, we also adopt two additional $|t|$ -cutoffs of 2.78 and 3.39. Harvey, Liu, and Zhu (2016) propose these two cutoffs, which are relatively

⁸ Let $R_{it+1} = b_{0t} + b_{1t}C_{it} + \epsilon_{it+1}$ be the cross-sectional regression at the beginning of month t , in which R_{it+1} is stock i 's return over month t , and C_{it} is the latest known value of a given characteristic as of month t . Stack the individual returns into an $N_t \times 1$ vector, $\mathbf{R}_{t+1}$, and the individual characteristics into a vector, $\mathbf{C}_t$, in which N_t is the number of stocks in month t . Let $\mathbf{1}_t$ be an $N_t \times 1$ vector of ones, $\mathbf{X}_t \equiv [\mathbf{1}_t \ \mathbf{C}_t]$, and $\mathbf{B}_t \equiv [b_{0t} \ b_{1t}]'$. Then ordinary least squares yield $\mathbf{B}_t = (\mathbf{X}_t' \mathbf{X}_t)^{-1} \mathbf{X}_t' \mathbf{R}_{t+1}$. Rewrite $\mathbf{B}_t = \mathbf{W}_t' \mathbf{R}_{t+1}$, in which $\mathbf{W}_t \equiv [\mathbf{W}_{0t} \ \mathbf{W}_{1t}] = \mathbf{X}_t (\mathbf{X}_t' \mathbf{X}_t)^{-1}$ is an $N_t \times 2$ matrix of portfolio weights, with $\mathbf{W}_{0t}$ the weights for the intercept portfolio, and $\mathbf{W}_{1t}$ the weights for the slope portfolio. In particular,

$$\mathbf{W}_t' \mathbf{X}_t = [\mathbf{W}_{0t} \ \mathbf{W}_{1t}]' [\mathbf{1}_t \ \mathbf{C}_t] = \begin{bmatrix} \mathbf{W}_{0t}' \mathbf{1}_t & \mathbf{W}_{0t}' \mathbf{C}_t \\ \mathbf{W}_{1t}' \mathbf{1}_t & \mathbf{W}_{1t}' \mathbf{C}_t \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

As such, the intercept portfolio is a unit long portfolio ($\mathbf{W}_{0t}' \mathbf{1}_t = 1$), with a zero spread in the characteristic ($\mathbf{W}_{0t}' \mathbf{C}_t = 0$), and the slope portfolio is a zero-investment long-short portfolio ($\mathbf{W}_{1t}' \mathbf{1}_t = 0$), with a unit spread in the characteristic ($\mathbf{W}_{1t}' \mathbf{C}_t = 1$). For weighted least squares, let $\mathbf{M}_t$ be the $N_t \times N_t$ weighting matrix, in which the diagonal element for stock i is given by its value-weight, and the off-diagonal elements are zero. The regression coefficients are given by $\mathbf{B}_t = (\mathbf{X}_t' \mathbf{M}_t \mathbf{X}_t)^{-1} \mathbf{X}_t' \mathbf{M}_t \mathbf{R}_{t+1}$, and the intercept and slope portfolio weights $\mathbf{W}_t = [\mathbf{W}_{0t} \ \mathbf{W}_{1t}] = \mathbf{M}_t' \mathbf{X}_t (\mathbf{X}_t' \mathbf{M}_t \mathbf{X}_t)^{-1}$.

stable over time, based on the Benjamini, Hochberg, and Yekutieli adjustment method at the 5% and 1% levels of the false discovery rate (Benjamini and Hochberg 1995; Benjamini and Yekutieli 2001).⁹

2.2 Reliable procedures that control for microcaps

While reporting results from a variety of different procedures, we emphasize the economic importance and the statistical reliability of sorts with NYSE breakpoints and value-weighted returns as well as cross-sectional regressions with weighted least squares.

When forming portfolios, many studies equal-weight returns. We instead focus on value-weighted returns. First, value-weighting accurately reflects the wealth effect experienced by investors (Fama 1998). Second, microcaps are influential in equal-weighted returns. Microcaps are on average only 3% of the aggregate market capitalization of the NYSE-Amex-NASDAQ universe but account for about 60% of the total number of stocks (Fama and French 2008). Because of high transaction costs, anomalies in microcaps are difficult to exploit in practice (Novy-Marx and Velikov 2016).

Many studies also use NYSE-Amex-NASDAQ breakpoints, as opposed to NYSE breakpoints. We emphasize NYSE breakpoints because the cross-sectional dispersion of anomaly variables is the largest among microcaps. Fama and French (2008) show that microcaps have the highest cross-sectional standard deviations of returns and many anomaly variables among micro, small, and big stocks. With NYSE-Amex-NASDAQ breakpoints, microcaps can account for more than 60% of the stocks in extreme deciles. These microcaps can inflate the magnitude of anomalies, especially when combined with equal-weighted returns. In contrast, NYSE breakpoints assign a fair number of small and big stocks into extreme deciles.

Hundreds of studies use cross-sectional regressions with ordinary least squares. We emphasize univariate regressions with weighted least squares that use the market equity as the weights. First, ordinary least squares can be dominated by microcaps because of their plentifulness. To the extent that the slopes are returns to zero-investment portfolios, cross-sectional regressions are analogous to sorts with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns. However, ordinary least squares can assign even higher weights to microcaps than equal-weights in sorts. Because these regressions minimize the sum of squared errors, while imposing a linear functional form between average returns and anomaly variables, they tend to put more weights on outliers with volatile returns and extreme anomaly variables, which most

⁹ We should acknowledge that the cutoffs of 2.78 and 3.39 are only heuristic in nature. In general, the cutoffs depend on the nature of the underlying test, the correlation structure of the sample, and the specification of the null hypothesis. However, the adopted cutoffs are likely conservative. Directly applying the Benjamini, Hochberg, and Yekutieli adjustment method to our dataset of 452 anomalies yields $|t|$ -cutoffs of 3.47 and 4.27 at the 5% and 1% threshold levels, respectively. Adopting these higher cutoffs would only strengthen our conclusion that most anomalies fail to replicate.

likely belong to microcaps. Using weighted least squares mitigates the impact of microcaps. Harvey and Liu (2018) also argue that value-weighting estimates from cross-sectional regressions better captures their economic importance.

Finally, cross-sectional regressions with many variables are excessively flexible. Leamer and Leonard (1983) show that inferences based on slopes from linear regressions are sensitive to the underlying specification.¹⁰ For example, two individually insignificant variables that are highly correlated can appear significant when used together. Because the set of regressors included in a regression specification is ambiguous, it is common and perhaps even acceptable to explore various specifications, to search for, and then to report a combination that yields “statistical significance” (Simmons, Nelson, and Simonsohn 2011). We avoid this trap by using univariate regressions.

2.3 The economic impact of microcaps

We provide new evidence to show why microcaps must be controlled for.

2.3.1 The extreme nature of microcaps. Table 1 updates Fama and French’s (2008) table I using our 1967–2016 sample. Panel A shows that on average, there are 2,365 microcaps, which account for 60.7% of the total number of firms, 3,896. However, microcaps represent only 3.21% of the total market capitalization, small stocks 6.71%, and big stocks 90.09%. With equal-weights, microcaps earn on average 1.27% per month relative to 1.01% for big stocks. In contrast, the value-weighted market return of 0.91% is close to 0.9% for big stocks. More important, microcaps have the highest cross-sectional standard deviations of monthly returns, 19.26%, followed by small stocks, 11.85%, and then by big stocks, 8.84%. Panel B shows that for the most part, the cross-sectional dispersions in anomaly variables are also the largest for microcaps, followed by small stocks, and then by big stocks.

Figure 1 documents that the economic importance of microcaps has declined in recent decades. Panel A shows that microcaps account for 47.6% of firms at the beginning of the sample. This fraction jumps to 66.6% in 1973 with the addition of NASDAQ, reaches its maximum of 71.6% in 1987, and displays a downward trend afterward. At the end of 2016, microcaps account for 50.1% of firms. In contrast, the numbers of small and big stocks show an upward trend since the mid-1980s and account for 22.8% and 27.2% of firms, respectively, at the end of our sample.

¹⁰ Leamer and Leonard (1983, p. 306) write: “Empirical results reported in economics journals are selected from a large set of estimated models. Journals, through their editorial policies, engage in some selection, which in turn stimulates extensive model searching and prescreening by prospective authors. Since this process is well known to professional readers, the reported results are widely regarded to overstate the precision of the estimates, and probably to distort them as well. As a consequence, statistical analyses are either greatly discounted or completely ignored.”

Table 1
Value- and equal-weighted average monthly returns, and averages and cross-sectional standard deviations of selected anomaly variables, January 1967–December 2016, 600 months

	A. Average monthly values									
	Number	% of total	Value-weighted returns		Equal-weighted returns		Cross-sectional			
	of firms	market cap	Average	Std	Average	Std	std of returns			
Market	3,896	100.00	0.91	4.48	1.17	6.27	16.46			
Micro	2,365	3.21	1.07	6.89	1.27	7.10	19.26			
Small	766	6.71	1.14	6.29	1.15	6.40	11.85			
Big	765	90.09	0.90	4.37	1.01	5.06	8.84			
All-but-micro	1,532	96.79	0.91	4.45	1.08	5.66	10.52			
B. Average monthly cross-sectional standard deviations										
	log(Me)	Bm	Sue	R ⁶	I/A	Roe	Nop	Oa	Rdm	Cop
Market	1.91	0.68	1.74	0.35	0.40	0.11	0.09	0.12	0.10	0.14
Micro	1.07	0.77	1.59	0.38	0.42	0.13	0.11	0.13	0.12	0.15
Small	0.47	0.50	1.77	0.35	0.39	0.08	0.07	0.10	0.06	0.11
Big	0.95	0.43	1.89	0.28	0.31	0.06	0.06	0.08	0.05	0.10
All-but-micro	1.21	0.47	1.84	0.32	0.35	0.07	0.07	0.09	0.06	0.11

Panel A shows averages of monthly value- and equal-weighted average returns, and monthly cross-sectional standard deviations (Std) of returns for all stocks (Market) and microcaps (Micro), small, big, and all-but-micro stocks. Panel A also shows the average number of stocks and the average percentage of the aggregate market capitalization in each size group each month. Panel B shows average monthly cross-sectional standard deviations of selected anomaly variables. Micro stocks are below the 20th percentile of NYSE market equity, small stocks are between the 20th and 50th percentiles, and big stocks are above the NYSE median. The anomaly variables are the log market equity (log(Me)), book-to-market (Bm), standardized unexpected earnings (Sue), prior 6-month returns (R^6), investment-to-assets (I/A), return on equity (Roe), net payout yield (Nop), operating accruals (Oa), R&D-to-market (Rdm), and cash-based operating profitability (Cop). Panel B winsorizes all the variables at the 1%–99% level. Appendix A details the variable definitions.

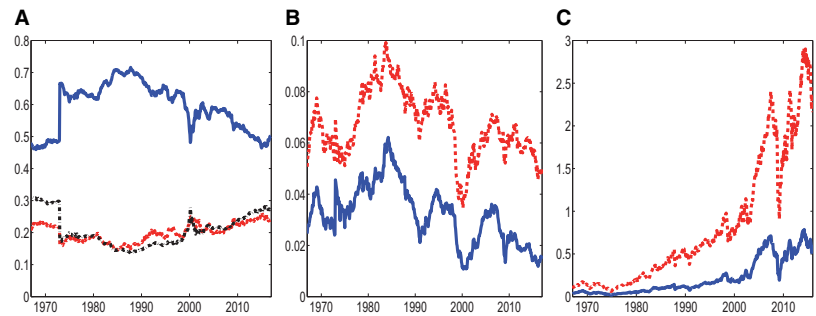


Figure 1
Time-series properties of microcaps, January 1967–December 2016, 600 months

Microcaps are smaller than the 20th percentile of market equity for NYSE stocks; small stocks are bigger than the 20th percentile but smaller than the NYSE median; and big stocks are bigger than the NYSE median. Panel A shows the time-series of the number of microcaps (solid-blue line), small stocks (red-dashed line), and big stocks (black-dashdot line) as a fraction of the total number of stocks. Panel B plots the time-series of the total market cap of microcaps (solid-blue line) and small stocks (red-dashed line) as a percentage of the aggregate market cap. Finally, panel C plots the breakpoints for the 20th percentile of NYSE market equity (solid-blue line) and the NYSE median (red-dashed line) in billions of dollars. Panel A: The number of stocks for a size group as a fraction of the total number of stocks in the market. Panel B: Total market equity for a size group as a fraction of the aggregate market cap. Panel C: Size breakpoints, billion.

Panel B shows that microcaps represent 2.5% of the total market cap in 1967. This fraction increases to 4.6% with the addition of NASDAQ, reaches its maximum of 6.2% in 1984, and exhibits a downward trend afterward. At the end of 2016, microcaps represent only 1.6% of the aggregate market cap, in contrast to 5.1% for small stocks and 93.3% for big stocks. Panel C shows that the breakpoints of microcaps and small stocks have increased over time. At the end of 2016, the 20th percentile of NYSE market equity is 724 million dollars, and the median 2.6 billion dollars.¹¹

2.3.2 Portfolio weights and investment capacity. Table 2 shows that anomalies driven by microcaps might be illusionary. Panel A reports average portfolio weights on microcaps for the extreme portfolios of anomalies. The sorts with NYSE breakpoints and value-weighted returns assign a modest amount of weights to microcaps, while NYSE-Amex-NASDAQ breakpoints and equal-weighted returns invest a disproportionately large amount. For example, in the momentum category, the low decile assigns on average 8% to microcaps under the former but 63.9% under the latter. In the value versus growth category, the high decile assigns on average 7.4% to microcaps under the former but 64.2% under the latter.

Similarly, cross-sectional regressions with weighted least squares assign a modest amount of weights to microcaps, while ordinary least squares invest a disproportionately large amount. We separate each zero-investment slope portfolio into two. The short portfolio consists of individual stocks with negative weights, and the long portfolio positive weights.¹² In the investment category, for example, the short portfolio assigns on average only 3.6% to microcaps with weighted least squares but 62.1% with ordinary least squares. In the profitability category, the long portfolio assigns 3.5% to microcaps under the former procedure but 53.5% under the latter.

From panels B and C, the investment capacity on microcaps is extremely limited. We measure a portfolio's investment capacity as $\min_i \{Me_i/|w_i|\}$, in which i is the index of the stocks in the portfolio, Me_i the market equity of stock i , and w_i its portfolio weight. If $w_i > 0$, $Me_i/|w_i|$ is the maximum amount from buying up all the shares of stock i , without considering the availability of shares of other stocks in the portfolio. If $w_i < 0$, $Me_i/|w_i|$ is the maximum amount from short-selling all its shares. We must take the minimum $Me_i/|w_i|$ across the index i because buying or selling all the shares of any stock would exhaust the investment capacity of the portfolio.

For an equal-weighted portfolio, $w_i = \pm 1/n$, in which n is the number of stocks in the portfolio. As such, the investment capacity is $\min_i \{Me_i/|w_i|\} =$

¹¹ Our evidence that the economic weight of microcaps has declined in recent decades is consistent with Kahle and Stulz (2017). Kahle and Stulz document that the percentage of public firms having market equity less than \$100 million in 2015 dollars has dramatically dropped, from 61.5% in 1975 to 43.9% in 1995 and to 22.6% in 2015.

¹² As noted, the long and short portfolios do not have total weights that sum to one. To ease comparison with sorts, we scale the long and short portfolios from regressions to make their total weights equal 1 and -1 , respectively.

Table 2
Portfolio weights on microcaps and investment capacity, January 1967–December 2016, 600 months

	Low						High							
	All	Mom	VvG	Inv	Prof	Intan	Fric	All	Mom	VvG	Inv	Prof	Intan	Fric
A. Portfolio weights allocated to microcaps (in %)														
NYSE-VW	7.19	8.00	3.89	7.36	9.46	4.00	10.24	10.12	3.87	7.38	5.54	5.84	10.29	19.96
NYSE-EW	55.24	62.23	48.58	68.47	65.48	46.56	51.88	59.53	47.66	62.87	58.26	53.52	59.00	69.21
All-VW	10.40	10.76	5.53	9.99	15.04	4.93	15.38	14.97	4.41	8.93	7.11	8.18	15.98	31.46
All-EW	57.56	63.86	51.05	71.21	68.86	47.51	54.86	62.12	48.65	64.16	60.53	54.89	62.41	73.72
FM-WLS	3.88	4.82	2.37	3.59	5.60	2.43	4.57	8.22	2.73	4.79	4.93	3.51	7.79	18.52
FM-OLS	54.31	62.64	46.20	62.08	63.52	46.45	53.11	60.14	47.72	63.00	59.20	53.47	59.79	70.59
B. Investment capacity as a fraction of the aggregate market capitalization (in %)														
NYSE-VW	10.70	5.95	13.12	5.96	8.52	9.88	15.80	8.56	8.93	5.64	7.50	11.10	8.59	8.74
NYSE-EW	0.12	0.02	0.04	0.01	0.03	0.19	0.27	0.08	0.04	0.02	0.02	0.03	0.06	0.24
All-VW	10.16	5.00	8.52	4.39	6.29	11.24	17.91	7.19	8.60	4.47	4.71	9.57	7.17	7.35
All-EW	0.11	0.02	0.03	0.01	0.03	0.19	0.23	0.07	0.04	0.02	0.02	0.02	0.05	0.19
FM-WLS	17.94	10.71	22.00	9.03	10.70	19.60	26.14	6.87	11.28	4.22	3.91	7.19	7.08	6.84
FM-OLS	0.09	0.09	0.07	0.02	0.05	0.14	0.13	0.09	0.06	0.02	0.02	0.03	0.07	0.24
C. Investment capacity at the end of December 2016 (in \$billion)														
NYSE-VW	2,015.65	1,201.65	2,052.98	1,332.69	1,748.85	2,095.75	2,827.82	1,569.06	1,320.97	900.09	1,702.07	2,402.44	1,499.78	1,533.15
NYSE-EW	26.64	1.33	3.00	0.72	4.87	52.32	58.10	11.91	1.97	1.31	1.34	2.15	3.52	44.35
All-VW	1,892.68	1,067.00	1,306.85	1,081.72	1,162.29	2,284.57	3,234.00	1,331.77	1,297.23	779.22	964.12	2,140.04	1,225.08	1,340.40
All-EW	21.86	1.19	2.73	0.61	4.50	37.65	52.24	9.74	1.85	1.19	0.96	2.02	3.73	35.09
FM-WLS	3,367.74	1,975.25	3,939.30	1,659.65	2,066.62	4,018.27	4,765.58	1,219.84	1,821.52	625.06	697.68	1,560.16	1,267.34	1,170.33
FM-OLS	7.69	1.37	5.57	1.05	7.01	13.69	9.80	8.25	2.41	1.35	0.95	3.51	3.54	27.21

The six categories of anomalies, momentum, value versus growth, investment, profitability, intangibles, and trading frictions, are denoted by “Mom,” “VvG,” “Inv,” “Prof,” “Intan,” and “Fric,” respectively. In portfolio sorts, “Low” denotes the low decile, and “High” the high decile. “NYSE-VW” denotes NYSE breakpoints and value-weighted returns, “NYSE-EW” NYSE breakpoints and equal-weighted returns, “All-VW” NYSE-Amex-NASDAQ breakpoints and value-weighted returns, and “All-EW” NYSE-Amex-NASDAQ breakpoints and equal-weighted returns. In univariate cross-sectional regressions, “FM-WLS” denotes weighted least squares with the market equity as the weights, and “FM-OLS” ordinary least squares. We separate each zero-investment slope portfolio into two: “Low,” is the short portfolio with negative weights on individual stocks, and “High” the long portfolio with positive weights on individual stocks. To ease comparison with the results from sorts, we scale the long and short portfolios to make their total weights sum to 1 and -1 , respectively. Panel A calculates the time-series average of weights on microcaps for the low and high portfolios of each anomaly and reports the average across all the anomalies in a given category. In panel B, the investment capacity of a portfolio is $\min_i (Me_i / w_i)$, in which Me_i is stock i ’s market equity, and w_i its weight. For the low and high portfolios of each anomaly, we compute the investment capacity as a fraction of aggregate market cap at each month, take its time-series average, and report the average across all the anomalies in a given category. Panel C calculates the investment capacity in billions of dollars for the low and high portfolios of each anomaly at the end of December 2016 and report the average across all the anomalies in a given category.

$n \times \min_i \{Me_i\}$. Intuitively, if an equal amount of wealth is invested in each stock in the portfolio, its investment capacity is restricted by the stock with the smallest market equity. For a value-weighted portfolio, $w_i = \pm Me_i / \sum_i Me_i$. The investment capacity is $\min_i \{Me_i / |w_i|\} = \min_i \{\sum_i Me_i\} = \sum_i Me_i$, the total market equity of all stocks in the portfolio, which is much higher than the investment capacity under the equal-weights. Finally, the investment capacity for the long and short portfolios from cross-sectional regressions is also $\min_i \{Me_i / |w_i|\}$, in which w_i is defined over all stocks (footnote 8).

The investment capacity with NYSE breakpoints and value-weighted returns is orders of magnitude larger than that with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns. In the momentum category, the investment capacity of the low decile is on average 5.95% of the aggregate market cap with the former procedure but only 0.02% with the latter (panel B). In terms of dollar values at the end of December 2016, the contrast is between \$1.2 trillion and \$1.19 billion (panel C). In the value versus growth category, the investment capacity of the high decile is on average 5.64% of the aggregate market cap under the former procedure but only 0.02% with the latter. In dollar values, the contrast is between \$900.1 billion and \$1.19 billion.

Similarly, for cross-sectional regressions, the investment capacity of the short portfolio with weighted least squares in the investment category is 9.03% of the aggregate market cap but only 0.02% with ordinary least squares (panel B). In terms of dollar values at the end of December 2016, the contrast is between \$1.66 trillion and \$1.05 billion (panel C). In the profitability category, the investment capacity of the long portfolio is 7.19% with the former procedure but only 0.03% with the latter. In terms of dollar values, the contrast is between \$1.56 trillion and \$3.51 billion.

2.3.3 Limitations. We have argued that because of their limited investment capacity and high transaction costs, microcaps are not economically important for investment management. However, we should clarify that microcaps play an important role in the real economy. In particular, using data collected by the U.S. Small Business Administration and the Census Bureau, Luttmer (2010) shows that the employment size distribution appears stationary and closely resembles a Pareto distribution with a long left tail. A large fraction of aggregate employment resides in microcaps. For example, firms with less than 50 employees contribute to more than 30% of total employment. Most of these firms are not publicly traded. Axtell (2001) shows that in 1997 the mean employment size in Compustat is 4,605 employees but only 19 in the Census data, implying that Compustat is heavily censored with respect to microcaps. Finally, a long literature in economics has established that microcaps have contributed more than larger firms to aggregate employment growth (Birch 1987; Moscarini and Postel-Vinay 2012) and to economic growth more generally (Evans 1987; Hall 1987).

3. Replication Results

Section 3.1 provides a bird's eye view of our results. Section 3.2 details the results for individual anomalies. Section 3.3 examines the commonality among the replicated anomalies.

3.1 The big picture

As noted, we treat an anomaly as a replication success if the average return of its high-minus-low decile with NYSE breakpoints and value-weighted returns is significant at the 5% threshold with $|t| \geq 1.96$. For cross-sectional regressions, an anomaly is treated as a replication success if its slope from weighted least squares with the market equity as the weights is significant with $|t| \geq 1.96$.

3.1.1 Replication rates across the 452 anomalies. Despite our lax criterion without adjusting for multiple testing, most anomalies fail to replicate. Panel A of Figure 2 shows that only 158 anomalies are significant in sorts with NYSE breakpoints and value-weighted returns, implying a low replication success rate of 35%. Cross-sectional regressions with weighted least squares yield largely similar results, with a low replication rate of 33.6%.

Although controlling for microcaps, we emphasize that microcaps are still in our sample. We have experimented with dropping microcaps when calculating value-weighted returns (but after forming deciles with NYSE breakpoints) and when performing cross-sectional regressions. The replication rates are reduced to 30.5% and 27.2%, respectively (the Internet Appendix).

Adjusting for multiple testing further reduces the replication rates. With the $|t|$ -cutoff of 2.78, the replication rates drop to 17.9% in sorts with NYSE breakpoints and value-weighted returns and to 13.3% in cross-sectional regressions with weighted least squares (panel A).

The low replication rates are not due to our extended samples through December 2016. We repeat our replication tests but stop the sample of a given anomaly at the end of its original study. If the start of its original sample is later than January 1967, we begin our sample at the same date. Otherwise, we start in January 1967, which is the earliest date in our sample. The results from the shorter, original samples are quantitatively similar to those from the extended samples. From panel B of Figure 2, with $|t| \geq 1.96$, the replication rate in sorts is 34.7%, which is close to 35% from the extended samples. Sampling variation plays a limited role. Once the samples are extended through December 2016, 31 anomalies that are significant in the original samples become insignificant, but 32 insignificant anomalies in the original samples become significant in the extended samples.

Cross-sectional regressions with weighted least squares yield a replication rate of 31.2% in the original samples. As the shorter samples make it more difficult to clear the $|t| \geq 1.96$ hurdle, the replication rate is somewhat lower than 33.6% in the extended samples. Sampling variation again plays a limited

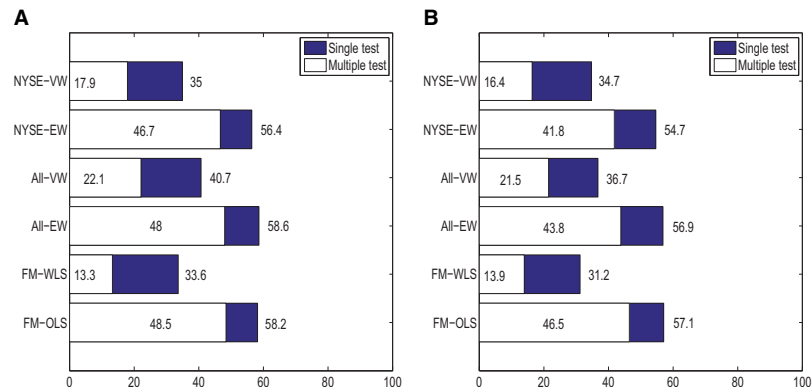


Figure 2
Replication rates (as a percentage), single and multiple tests, January 1967–December 2016, 600 months
“NYSE-VW” and “NYSE-EW” denote NYSE breakpoints with value- and equal-weighted returns; and “All-VW” and “All-EW” denote NYSE-Amex-NASDAQ breakpoints with value- and equal-weighted returns, respectively, in portfolio sorts. “FM-WLS” denotes weighted least squares with the market equity as the weights, and “FM-OLS” denotes ordinary least squares, in univariate cross-sectional regressions. We winsorize the regressors at the 1%–99% level each month and standardize them in the regressions. Standardizing a variable means subtracting its cross-sectional mean and then dividing by its cross-sectional standard deviation. We apply the absolute t -cutoff of 1.96 for single tests and 2.78 for multiple tests, both at the 5% threshold level. The bars indicate the fractions (as a percentage) of anomalies that are successfully replicated (significant at the 5% level) in the set of 452. Panel A shows the results from the extended samples through December 2016, and panel B from the shorter samples in the original studies. The multiple testing bars (in white) are overlaid on the single testing bars (in blue). Panel A: The extended samples through December 2016. Panel B: The shorter samples in original studies.

role. In total, 22 anomalies that are significant in the original samples lose their significance in the extended samples, but 33 insignificant anomalies in the original samples gain their significance in the extended samples. On net, the longer samples yield 11 more replicated anomalies. Finally, the results adjusted for multiple testing from the original samples are also similar to those from the extended samples. In all, our evidence on the low replication rates is robust in the shorter, original samples.

Controlling for microcaps in our robust procedures goes a long way in explaining the low replication rates, but not completely. Panel A of Figure 2 also reports results from sorts with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns (All-EW) as well as cross-sectional regressions with ordinary least squares (FM-OLS). Both assign maximum weights to microcaps in their respective setting. The replication rates are 58.6% with All-EW and 58.2% with FM-OLS. Both are far lower than 80%, which is generally viewed as an ideal replication rate (Ioannidis 2005).

Equal- versus value-weighting is more effective than NYSE-Amex-NASDAQ versus NYSE breakpoints in overweighting microcaps. With NYSE-Amex-NASDAQ breakpoints and value-weighted returns (All-VW), the replication rate is 40.7% under $|t| \geq 1.96$, rising modestly from 35% with NYSE-VW. The increment is more substantial with NYSE breakpoints and

equal-weighted returns (NYSE-EW), yielding a replication rate of 56.4%, which is close to 58.6% with All-EW.

More important, with multiple testing adjusted with $|t| \geq 2.78$, most anomalies fail to replicate regardless of microcaps. In the extended samples, panel A reports low replication rates of 48% for NYSE-Amex-NASDAQ breakpoints and equal-weighted returns as well as 48.5% for cross-sectional regressions with ordinary least squares, despite their maximum weights to microcaps. In the shorter, original samples, the replication rates are 43.8% and 46.5%, respectively (panel B).

3.1.2 Replication rates for each category of anomalies. Figure 3 shows the replication rates for each of the six categories of anomalies. For example, with NYSE breakpoints and value-weighted returns, the replication rates are acceptable in the momentum and investment categories, 63.2% and 73.7%, moderate in the value versus growth and profitability categories, 42% and 44.3%, but poor in the intangibles and trading frictions categories, 25.2% and

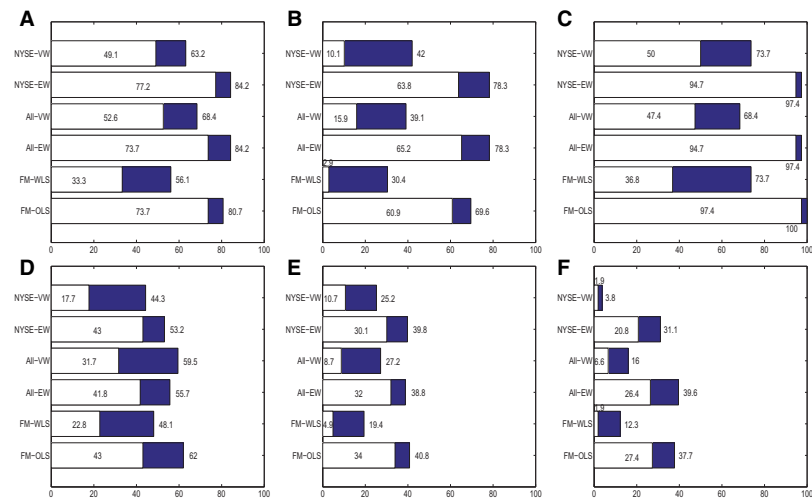


Figure 3
Replication rates (as a percentage) for each category of anomalies, single and multiple tests, January 1967–December 2016, 600 months
“NYSE-VW” and “NYSE-EW” denote NYSE breakpoints with value- and equal-weighted returns; and “All-VW” and “All-EW” NYSE-Amex-NASDAQ breakpoints with value- and equal-weighted returns, respectively, in portfolio sorts. “FM-WLS” denotes weighted least squares with the market equity as the weights, and “FM-OLS” ordinary least squares, in univariate cross-sectional regressions. We winsorize the regressors at the 1%–99% level each month before standardizing them. Standardizing means subtracting a variable’s cross-sectional mean and then dividing by its cross-sectional standard deviation. We apply the absolute t -cutoff of 1.96 for single tests and 2.78 for multiple tests, both at the 5% significance level. For each category, the bars report the fractions (as a percentage) of anomalies that are successfully replicated (significant at the 5% level). The multiple testing bars (in white) are overlaid on the single testing bars (in blue). Panel A: Momentum (57 anomalies). Panel B: Value versus growth (69 anomalies). Panel C: Investment (38 anomalies). Panel D: Intangibles (103 anomalies). Panel E: Profitability (79 anomalies). Panel F: Trading frictions (106 anomalies).

3.8%, respectively. Most strikingly, 96.2% of the trading frictions variables fail to replicate in single tests!

Cross-sectional regressions with weighted least squares yield largely similar results. The replication rates are acceptable in the momentum and investment categories, 56.1% and 73.7%, moderate in the value versus growth and profitability categories, 30.4% and 48.1%, and poor in the intangibles and trading frictions categories, 19.4% and 12.3%, respectively. Still, the vast majority, 87.7%, of the trading frictions variables fail to replicate in single tests.

As noted, microcaps are in our sample. We have experimented with dropping microcaps when calculating value-weighted returns (but after forming deciles with NYSE breakpoints) and when performing cross-sectional regressions with weighted least squares. The replication rates are generally lower, 56.1%, 68.4%, 27.5%, 34.2%, 25.2%, and 7.55% in the sorts and 47.4%, 68.4%, 26.1%, 34.2%, 17.5%, and 6.6% in the regressions across the momentum, investment, value versus growth, profitability, intangibles, and trading frictions categories, respectively (the Internet Appendix).

Overweighting microcaps is more effective in increasing the replication rates for the momentum, value versus growth, investment, and profitability categories than for the intangibles and trading frictions categories. Across the six categories, sorts with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns yield the replication rates of 84.2%, 78.3%, 97.4%, 55.7%, 38.8%, and 39.6%, respectively. In addition, cross-sectional regressions with ordinary least squares yield 80.7%, 69.6%, 100%, 62%, 40.8%, and 37.7%, respectively. In particular, even with maximum weights to microcaps with the two respective procedures, 60.4% and 62.3% of the trading frictions anomalies still fail to replicate.¹³ The replication rates from the original samples are largely similar (the Internet Appendix).

3.2 Individual anomalies

In this subsection, we detail the replication results for individual anomalies. We compare our estimates with those from the original studies in terms of economic importance and statistical significance. Many prominent anomalies fail to replicate. Also, even for replicated anomalies, their economic magnitudes are much lower than originally reported. We discuss possible procedural sources for the differences.

For each of the 452 anomalies, Table 3 reports the average returns of the high-minus-low deciles from different sorts, including NYSE-VW, NYSE-EW, All-VW, and All-EW, the univariate cross-sectional slopes with

¹³ The evidence is also striking with microcap breakpoints and equal-weighted returns (Micro-EW, the Internet Appendix). With $|t| \geq 1.96$, the replication rates are 62.6% across the 452 anomalies and 87.7%, 75.4%, 94.7%, 69.6%, 44.7%, and 41.5% across the momentum, value versus growth, investment, profitability, intangibles, and trading frictions categories, respectively. As such, even with only microcaps, most of the trading frictions variables, 58.5%, still fail to replicate.

weighted least squares (FM-WLS) and with ordinary least squares (FM-OLS), as well as their absolute t -values adjusted for heteroskedasticity and autocorrelations. For NYSE-VW and FM-WLS, we also report results from the shorter, original samples (NYSE-VW-SS and FM-WLS-SS, respectively). Due to data limitations, some anomalies start their samples later than January 1967 and occasionally end earlier than December 2016. Table 3 indicates these start and end dates. We proceed category by category.

3.2.1 Momentum. Panel A of Table 3 reports the replication results for the 57 momentum anomalies. With NYSE-VW, the high-minus-low earnings surprise (Sue) deciles at the 1-, 6-, and 12-month horizons earn on average 0.46%, 0.16%, and 0.08% per month ($t=3.48, 1.44, \text{ and } 0.73$), respectively. These estimates are lower than those in Chan, Jegadeesh, and Lakonishok (1996), who report 6- and 12-month buy-and-hold returns of 6.8% and 7.5% (1.13% and 0.63% per month), respectively. Overweighting microcaps via equal-weighting partially explains the differences, as our All-EW estimates are 1.34%, 0.64%, and 0.24% ($t=10.33, 5.3, \text{ and } 2.15$) across the three horizons, respectively.

The high-minus-low deciles on abnormal returns around earnings announcements (Abr) earn on average 0.7%, 0.33%, and 0.23% per month at the 1-, 6-, and 12-month horizons ($t=5.45, 3.41, \text{ and } 2.99$), respectively, with NYSE-VW. The 6- and 12-month estimates are smaller than the buy-and-hold returns of 5.9% and 8.3% (0.98% and 0.69% per month), respectively, over the same horizons in Chan, Jegadeesh, and Lakonishok (1996). The high-minus-low deciles on revisions in analysts' earnings forecasts (Re) earn on average 0.75%, 0.47%, and 0.24% ($t=3.18, 2.24, \text{ and } 1.3$) at the 1-, 6- and 12-month horizons, respectively. The 6- and 12-month estimates are again smaller than the buy-and-hold returns of 7.7% and 9.7% (1.28% and 0.81% per month) in Chan, Jegadeesh, and Lakonishok (1996), respectively.

Price momentum fares well in our replication. In particular, the high-minus-low decile on the prior 6-month return with the 6-month horizon (R^6_6) earns on average 0.82% per month ($t=3.5$) with NYSE-VW. This estimate is smaller than 1.1% ($t=3.61$) in Jegadeesh and Titman (1993). We reproduce their estimate with All-EW (closest to their procedure) in their original sample and obtain 1.18% ($t=4.22$) (untabulated). However, this estimate falls to 0.7% ($t=2.63$) in the extended sample. The estimate is 1.06% ($t=3.82$) in the original sample with NYSE-VW.

The high-minus-low tax expense surprise (Tes) deciles at the 1-, 6-, and 12-month horizons earn average returns of 0.23%, 0.24%, and 0.16% per month ($t=1.41, 1.68, \text{ and } 1.19$), respectively, with NYSE-VW. The estimates are lower than the 3-month buy-and-hold return of 3.9% (1.3% per month) in Thomas and Zhang (2011) based on All-EW. Our All-EW estimates are at most 0.85%

Table 3
Average returns of the high-minus-low deciles, the univariate cross-sectional regression slopes, and their absolute *t*-values, 452 anomalies, January 1967–December 2016, 600 months

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	<i>t</i>	$\bar{R}$	<i>t</i>	$\bar{R}$	<i>t</i>	$\bar{R}$	<i>t</i>	$\bar{R}$	<i>t</i>	$\bar{R}$	<i>t</i>	$\bar{R}$	<i>t</i>	$\bar{R}$	<i>t</i>
A. Momentum (57 anomalies)																
Sue: Earnings surprise, 1-, 6-, and 12-month, Foster, Olsen, and Shevlin (1984)																
Sue1	0.46	3.48 ^c	1.33	9.96 ^c	0.48	3.68 ^c	1.34	10.33 ^c	0.12	3.67 ^c	0.38	10.58 ^c	0.50	1.14	0.15	1.38
Sue6	0.16	1.44	0.63	5.10 ^c	0.17	1.55	0.64	5.30 ^c	0.06	1.84	0.19	5.55 ^c	0.37	0.85	0.09	0.81
Sue12	0.08	0.73	0.23	1.98 ^a	0.08	0.74	0.24	2.15 ^a	0.02	0.69	0.07	2.26 ^a	0.01	0.01	-0.01	0.14
Abr★: Cumulative abnormal stock returns around earnings announcements, 1-, 6-, and 12-month, Chan, Jegadeesh, and Lakonishok (1996), 1972/1																
Abr1	0.70	5.45 ^c	1.26	14.44 ^c	1.01	5.98 ^c	1.52	14.60 ^c	0.26	5.53 ^c	0.41	14.41 ^c	0.96	5.36 ^c	0.35	5.49 ^c
Abr6	0.33	3.41 ^c	0.74	10.89 ^c	0.44	3.72 ^c	0.84	10.89 ^c	0.12	3.14 ^b	0.23	10.59 ^c	0.41	3.39 ^c	0.17	3.64 ^c
Abr12	0.23	2.99 ^b	0.46	8.13 ^c	0.32	3.20 ^b	0.53	8.06 ^c	0.10	2.91 ^b	0.15	8.56 ^c	0.30	2.70 ^a	0.13	3.11 ^b
Re★: Revisions in analysts' earnings forecasts, 1-, 6-, and 12-month, Chan, Jegadeesh, and Lakonishok (1996), 1976/7																
Re1	0.75	3.18 ^b	1.21	6.55 ^c	0.77	3.01 ^b	1.40	7.27 ^c	0.26	2.01 ^a	0.31	4.88 ^c	1.06	3.09 ^b	0.46	2.96 ^b
Re6	0.47	2.24 ^a	0.73	4.43 ^c	0.53	2.25 ^a	0.85	4.92 ^c	0.17	1.48	0.19	3.20 ^b	0.59	1.76	0.31	2.09 ^a
Re12	0.24	1.30	0.41	2.92 ^b	0.27	1.28	0.48	3.15 ^b	0.08	0.79	0.10	1.79	0.26	0.84	0.14	1.02
R ⁶ : Price momentum, prior 6-month returns, 1-, 6-, and 12-month, Jegadeesh and Titman (1993)																
R ⁶ 1	0.60	2.08 ^a	0.67	2.72 ^a	1.41	4.01 ^c	0.69	2.31 ^a	0.20	2.00 ^a	0.20	2.57 ^a	0.90	2.95 ^b	0.22	1.86
R ⁶ 6	0.82	3.50 ^c	0.64	2.90 ^b	1.29	4.63 ^c	0.70	2.63 ^a	0.28	3.18 ^b	0.20	2.85 ^b	1.06	3.82 ^c	0.30	2.71 ^a
R ⁶ 12	0.55	2.91 ^b	0.26	1.46	0.85	3.81 ^c	0.26	1.15	0.20	2.48 ^a	0.07	1.16	0.85	3.66 ^c	0.24	2.51 ^a
R ¹¹ : Price momentum, prior 11-month returns, 1-, 6-, and 12-month, Fama and French (1996)																
R ¹¹ 1	1.16	3.99 ^c	0.89	3.35 ^b	1.62	4.29 ^c	0.87	2.69 ^a	0.36	3.39 ^c	0.28	3.30 ^b	1.58	4.94 ^c	0.45	3.73 ^c
R ¹¹ 6	0.80	3.13 ^b	0.42	1.76	1.10	3.51 ^c	0.42	1.43	0.27	2.69 ^a	0.12	1.53	1.26	4.48 ^c	0.37	3.26 ^b
R ¹¹ 12	0.43	1.93	-0.03	0.14	0.54	2.01 ^a	-0.11	0.41	0.15	1.70	-0.03	0.53	0.79	3.17 ^b	0.22	2.24 ^a
Im: Industry momentum, 1-, 6-, and 12-month, Moskowitz and Grinblatt (1999)																
Im1	0.68	2.86 ^b	1.46	7.21 ^c	0.68	2.86 ^b	1.46	7.21 ^c	0.13	1.92	0.41	6.86 ^c	0.76	2.68 ^a	0.15	2.03 ^a
Im6	0.60	3.01 ^b	0.93	5.55 ^c	0.60	3.01 ^b	0.93	5.55 ^c	0.09	1.68	0.26	4.99 ^c	0.69	2.91 ^b	0.08	1.22
Im12	0.63	3.57 ^c	0.80	5.13 ^c	0.63	3.57 ^c	0.80	5.13 ^c	0.09	1.68	0.20	4.28 ^c	0.75	3.71 ^c	0.11	1.89
Rs: Revenue surprise, 1-, 6-, and 12-month, Jegadeesh and Livnat (2006)																
Rs1	0.32	2.28 ^a	0.88	5.82 ^c	0.30	2.05 ^a	0.92	5.97 ^c	0.07	1.96	0.23	5.16 ^c	0.31	1.15	0.07	1.00
Rs6	0.15	1.12	0.41	2.84 ^b	0.15	1.11	0.43	2.92 ^b	0.04	1.19	0.10	2.30 ^a	0.24	0.93	0.06	0.80
Rs12	0.07	0.52	0.10	0.68	0.06	0.51	0.11	0.75	0.02	0.64	0.01	0.19	0.24	0.99	0.06	0.86

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $
Tes★: Tax expense surprise, 1-, 6-, and 12-month, Thomas and Zhang (2011), 1976/1																
Tes1	0.23	1.41	0.78	8.31 ^c	0.30	1.55	0.85	8.03 ^c	0.10	1.98 ^a	0.20	7.15 ^c	0.26	1.40	0.13	1.96 ^a
Tes6	0.24	1.68	0.40	4.89 ^c	0.32	1.91	0.43	4.81 ^c	0.10	1.88	0.10	4.17 ^c	0.37	2.20 ^a	0.14	2.15 ^a
Tes12	0.16	1.19	0.15	2.02 ^a	0.15	1.02	0.17	2.08 ^a	0.06	1.15	0.04	1.78	0.29	1.83	0.10	1.63
dEF★: Change in analysts' earnings forecasts, 1-, 6-, and 12-month, Hawkins, Chamberlin, and Daniel (1984), 1976/3																
dEF1	0.94	4.33 ^c	1.51	9.87 ^c	1.25	5.29 ^c	1.66	10.53 ^c	0.34	4.25 ^c	0.34	7.94 ^c	1.91	3.68 ^c	0.62	3.58 ^c
dEF6	0.56	3.19 ^b	0.82	6.81 ^c	0.71	4.04 ^c	0.92	7.31 ^c	0.20	3.17 ^b	0.19	5.07 ^c	1.47	2.89 ^b	0.48	2.95 ^b
dEF12	0.33	2.38 ^a	0.49	4.80 ^c	0.41	2.92 ^b	0.54	5.08 ^c	0.12	2.23 ^a	0.11	3.17 ^b	0.84	1.80	0.31	2.07 ^a
NEI★: Number of consecutive quarters with earnings increases, 1-, 6-, and 12-month, Barth, Elliott, and Finn (1999), 1969/1																
Nei1	0.33	3.04 ^b	0.76	5.58 ^c	0.33	3.04 ^b	0.76	5.58 ^c	0.08	2.27 ^a	0.27	6.74 ^c	0.42	1.89	0.13	1.74
Nei6	0.19	1.78	0.39	2.93 ^b	0.19	1.78	0.39	2.93 ^b	0.04	1.16	0.13	3.28 ^b	0.26	1.20	0.08	1.18
Nei12	0.12	1.11	0.16	1.19	0.12	1.11	0.16	1.19	0.02	0.53	0.05	1.34	0.19	0.90	0.06	0.84
52w: 52-week high, 1-, 6-, and 12-month, George and Hwang (2004)																
52w1	0.13	0.38	-0.16	0.46	0.30	0.74	-0.72	1.72	0.13	1.08	-0.05	0.44	0.35	1.09	0.19	1.50
52w6	0.56	2.01 ^a	0.39	1.29	0.92	2.57 ^a	0.30	0.81	0.26	2.50 ^a	0.16	1.45	0.87	3.16 ^b	0.34	3.04 ^b
52w12	0.45	1.88	0.22	0.82	0.62	1.96 ^a	0.14	0.40	0.21	2.34 ^a	0.08	0.83	0.65	2.66 ^a	0.29	2.87 ^b
ε ⁶ : Residual momentum, prior 6-month returns, 1-, 6-, and 12-month, Blitz, Huij, and Martens (2011)																
ε ⁶ 1	0.18	1.15	0.68	4.90 ^c	0.23	1.37	0.69	4.88 ^c	0.04	0.88	0.20	5.09 ^c	0.28	1.54	0.06	1.22
ε ⁶ 6	0.45	3.74 ^c	0.62	5.28 ^c	0.48	3.89 ^c	0.64	5.42 ^c	0.12	3.55 ^c	0.18	5.42 ^c	0.57	4.20 ^c	0.15	3.92 ^c
ε ⁶ 12	0.37	3.85 ^c	0.39	4.14 ^c	0.38	3.91 ^c	0.40	4.18 ^c	0.10	3.44 ^c	0.12	4.28 ^c	0.46	4.28 ^c	0.12	3.74 ^c
ε ¹¹ : Residual momentum, prior 11-month returns, 1-, 6-, and 12-month, Blitz, Huij, and Martens (2011)																
ε ¹¹ 1	0.61	3.72 ^c	1.18	8.22 ^c	0.62	3.71 ^c	1.20	8.15 ^c	0.19	4.11 ^c	0.36	8.88 ^c	0.76	4.18 ^c	0.24	4.60 ^c
ε ¹¹ 6	0.50	3.82 ^c	0.66	5.31 ^c	0.52	3.88 ^c	0.68	5.34 ^c	0.14	3.50 ^c	0.57	5.38 ^c	0.62	4.21 ^c	0.18	3.92 ^c
ε ¹¹ 12	0.33	2.88 ^b	0.30	2.95 ^b	0.33	2.80 ^b	0.31	2.95 ^b	0.09	2.46 ^a	0.26	2.94 ^b	0.40	3.09 ^b	0.10	2.67 ^a
Sm★: Segment momentum, 1-, 6-, and 12-month, Cohen and Lou (2012), 1977/7																
Sm1	0.53	2.36 ^b	1.22	6.62 ^c	0.56	2.40 ^a	1.29	6.86 ^c	0.16	2.46 ^a	0.37	7.65 ^c	0.63	2.45 ^a	0.20	2.64 ^a
Sm6	0.09	0.94	0.45	4.58 ^c	0.10	1.06	0.47	4.80 ^c	0.02	0.50	0.13	4.86 ^c	0.14	1.30	0.03	0.75
Sm12	0.14	1.94	0.39	5.81 ^c	0.13	1.69	0.41	5.80 ^c	0.04	1.95	0.11	5.86 ^c	0.17	1.92	0.05	1.94

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		AIL-VW		AIL-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
IIR: Industry lead-lag effect in prior returns, 1-, 6-, and 12-month, Hou (2007)																
IIR1	0.69	3.33 ^b	1.29	7.43 ^c	0.69	3.33 ^b	1.29	7.43 ^c	0.21	3.59 ^c	0.42	8.03 ^c	0.83	3.57 ^c	0.33	4.53 ^c
IIR6	0.34	3.36 ^b	0.64	6.88 ^c	0.34	3.35 ^b	0.64	6.88 ^c	0.04	1.41	0.17	5.99 ^c	0.36	2.95 ^b	0.07	1.83
IIR12	0.35	4.27 ^c	0.49	6.50 ^c	0.35	4.27 ^c	0.49	6.50 ^c	0.06	2.95 ^b	0.14	6.33 ^c	0.41	4.27 ^c	0.09	3.27 ^b
IIE: Industry lead-lag effect in earnings surprises, 1-, 6-, and 12-month, Hou (2007)																
IIE1	0.58	3.48 ^c	0.70	4.37 ^c	0.58	3.48 ^c	0.70	4.37 ^c	0.06	1.32	0.19	4.81 ^c	0.67	3.33 ^b	0.08	1.27
IIE6	0.23	1.55	0.35	2.60 ^a	0.23	1.55	0.35	2.60 ^a	0.03	0.62	0.09	2.44 ^a	0.30	1.63	0.04	0.59
IIE12	0.09	0.64	0.14	1.15	0.09	0.63	0.14	1.15	0.00	0.02	0.03	0.87	0.16	0.94	0.00	0.03
Cm★: Customer momentum, 1-, 6-, and 12-month, Cohen and Frazzini (2008), 1979/7																
Cm1	0.78	3.85 ^c	0.79	5.47 ^c	0.78	3.92 ^c	0.83	5.68 ^c	0.25	3.20 ^b	0.26	5.04 ^c	0.89	3.34 ^b	0.35	3.31 ^b
Cm6	0.16	1.72	0.37	6.19 ^c	0.21	1.98 ^a	0.41	6.29 ^c	0.08	1.88	0.14	6.01 ^c	0.15	1.19	0.10	1.73
Cm12	0.15	2.23 ^a	0.30	5.79 ^c	0.17	2.34 ^a	0.33	6.01 ^c	0.07	2.41 ^a	0.12	6.29 ^c	0.14	1.61	0.07	2.09 ^a
Sim: Supplier industries momentum, 1-, 6-, and 12-month, Menzly and Ozbas (2010)																
Sim1	0.79	3.65 ^c	1.29	6.00 ^c	0.79	3.65 ^c	1.29	6.00 ^c	0.21	4.15 ^c	0.36	6.43 ^c	0.90	3.44 ^c	0.25	3.87 ^c
Sim6	0.10	0.96	0.46	3.70 ^c	0.10	0.96	0.46	3.70 ^c	0.03	1.12	0.14	4.07 ^c	0.14	1.11	0.04	1.11
Sim12	0.11	1.42	0.31	3.69 ^c	0.11	1.42	0.31	3.69 ^c	0.04	2.06 ^a	0.10	4.30 ^c	0.14	1.46	0.04	1.77
Cim: Customer industries momentum, 1-, 6-, and 12-month, Menzly and Ozbas (2010)																
Cim1	0.75	3.35 ^b	1.36	6.50 ^c	0.75	3.35 ^b	1.36	6.50 ^c	0.20	3.70 ^c	0.34	6.51 ^c	0.83	3.15 ^b	0.23	3.71 ^c
Cim6	0.29	2.76 ^a	0.58	5.48 ^c	0.29	2.76 ^a	0.58	5.48 ^c	0.07	2.88 ^b	0.14	5.24 ^c	0.33	2.70 ^a	0.09	2.79 ^b
Cim12	0.27	3.41 ^c	0.46	6.17 ^c	0.27	3.41 ^c	0.46	6.17 ^c	0.08	4.08 ^c	0.11	6.03 ^c	0.31	3.44 ^c	0.09	4.16 ^c
B. Value versus growth (69 anomalies)																
Bm: Book-to-market equity, Rosenberg, Reid, and Lanstein (1985)																
Bm	0.54	2.61 ^a	1.19	6.08 ^c	0.64	2.80 ^b	1.39	6.43 ^c	0.20	2.06 ^a	0.34	5.95 ^c	1.41	3.10 ^b	0.57	3.10 ^b
Bmj: Book-to-June-end market equity, Asness and Frazzini (2013)																
Bmj	0.46	2.12 ^a	1.12	5.44 ^c	0.52	2.01 ^a	1.25	5.46 ^c	0.17	1.49	0.33	5.20 ^c	0.53	2.32 ^a	0.20	1.60
Bm ^d : Quarterly book-to-market equity, 1-, 6-, and 12-month																
Bm ^d 1	0.43	1.70	1.53	5.83 ^c	0.38	1.31	1.78	6.03 ^c	0.21	1.59	0.53	6.65 ^c	0.54	1.97 ^a	0.25	1.71
Bm ^d 6	0.42	1.78	1.12	4.75 ^c	0.39	1.46	1.27	4.91 ^c	0.18	1.43	0.35	4.99 ^c	0.50	2.00 ^a	0.21	1.59
Bm ^d 12	0.48	2.21 ^a	1.13	5.19 ^c	0.50	2.05 ^a	1.28	5.39 ^c	0.19	1.66	0.34	5.20 ^c	0.55	2.42 ^a	0.22	1.81

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Dm: Debt-to-market, Bhandari (1988)																
Dm	0.31	1.61	0.40	1.97 ^a	0.33	1.46	0.48	2.18 ^a	0.13	1.55	0.09	1.49	0.51	1.38	0.21	1.31
Dm ^d ★: Quarterly debt-to-market, 1-, 6-, and 12-month, 1972/1																
Dm ^d 1	0.29	1.25	0.54	1.99 ^a	0.32	1.09	0.61	2.15 ^a	0.17	1.13	0.09	1.15	0.94	1.56	0.38	1.32
Dm ^d 6	0.28	1.25	0.30	1.19	0.34	1.27	0.36	1.36	0.18	1.26	0.01	0.16	0.92	1.69	0.36	1.38
Dm ^d 12	0.33	1.57	0.35	1.53	0.43	1.71	0.41	1.67	0.19	1.51	0.04	0.56	0.87	1.70	0.37	1.44
Am: Assets-to-market, Fama and French (1992)																
Am	0.31	1.52	0.83	3.73 ^c	0.44	1.93	1.06	4.35 ^c	0.16	1.69	0.19	3.20 ^b	0.38	1.43	0.14	1.43
Am ^d ★: Quarterly assets-to-market, 1-, 6-, and 12-month, 1972/1																
Am ^d 1	0.33	1.20	1.19	3.83 ^c	0.34	1.18	1.36	4.03 ^c	0.20	1.39	0.29	3.52 ^c	0.27	0.89	0.18	1.52
Am ^d 6	0.38	1.48	0.82	2.94 ^b	0.38	1.37	0.91	3.07 ^b	0.18	1.33	0.15	1.93	0.27	0.94	0.15	1.35
Am ^d 12	0.37	1.59	0.81	3.15 ^b	0.40	1.56	0.94	3.43 ^c	0.17	1.37	0.16	2.13 ^a	0.30	1.09	0.14	1.24
Rev: Long-term reversal, 1-, 6-, and 12-month, De Bondt and Thaler (1985)																
Rev1	-0.43	1.92	-0.95	4.41 ^c	-0.37	1.35	-1.13	4.23 ^c	-0.05	0.83	-0.25	4.50 ^c	-0.87	2.11 ^a	-0.19	1.66
Rev6	-0.42	2.01 ^a	-0.85	4.25 ^c	-0.37	1.45	-1.01	3.98 ^c	-0.05	0.74	-0.23	4.50 ^c	-0.81	2.06 ^a	-0.18	1.55
Rev12	-0.39	1.99 ^a	-0.80	4.33 ^c	-0.33	1.36	-0.95	4.00 ^c	-0.05	0.86	-0.22	4.62 ^c	-0.76	2.07 ^a	-0.18	1.63
Ep: Earnings-to-price, Basu (1983)																
Ep	0.44	2.26 ^a	0.65	4.55 ^c	0.64	3.01 ^b	0.75	4.86 ^c	0.20	2.38 ^a	0.19	4.52 ^c	0.77	1.97 ^a	0.32	2.01 ^a
Ep ^d : Quarterly earnings-to-price, 1-, 6-, and 12-month																
Ep ^d 1	0.93	4.94 ^c	1.70	10.58 ^c	0.96	4.46 ^c	1.80	10.38 ^c	0.38	3.78 ^c	0.48	10.25 ^c	1.18	2.94 ^b	0.47	2.89 ^b
Ep ^d 6	0.59	3.42 ^c	1.04	7.72 ^c	0.66	3.46 ^c	1.11	7.75 ^c	0.26	2.89 ^b	0.29	7.39 ^c	0.81	2.25 ^a	0.38	2.45 ^a
Ep ^d 12	0.43	2.60 ^a	0.66	5.47 ^c	0.49	2.69 ^a	0.72	5.70 ^c	0.20	2.31 ^a	0.19	5.46 ^c	0.64	1.83	0.32	2.13 ^a
Ep ^f ★: Analysts' earnings forecasts-to-price, 1-, 6-, and 12-month, Elgers, Lo, and Pfeiffer (2001), 1976/2																
Ep ^f 1	0.42	1.80	0.59	2.59 ^a	0.43	1.50	0.64	2.52 ^a	0.13	1.06	0.18	2.66 ^a	-0.01	0.02	-0.07	0.61
Ep ^f 6	0.38	1.62	0.28	1.26	0.40	1.46	0.31	1.28	0.11	0.95	0.09	1.33	-0.07	0.26	-0.10	0.95
Ep ^f 12	0.34	1.52	0.28	1.37	0.37	1.40	0.30	1.32	0.11	0.98	0.09	1.46	-0.03	0.12	-0.08	0.85
Cp: Cash flow-to-price, Lakonishok, Shleifer, and Vishny (1994)																
Cp	0.43	2.14 ^a	0.87	5.07 ^c	0.37	1.59	0.97	5.07 ^c	0.20	2.20 ^a	0.25	5.03 ^c	0.75	2.60 ^a	0.29	2.71 ^a

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Cp ^q : Quarterly cash flow-to-price, 1-, 6-, and 12-month																
Cp ^q 1	0.62	2.93 ^b	1.27	6.39 ^c	0.72	3.18 ^b	1.46	6.70 ^c	0.31	2.75 ^a	0.36	6.31 ^c	0.83	2.85 ^b	0.37	2.93 ^b
Cp ^q 6	0.48	2.42 ^a	0.85	4.64 ^c	0.57	2.59 ^a	0.98	4.96 ^c	0.25	2.39 ^a	0.24	4.63 ^c	0.63	2.34 ^a	0.31	2.87 ^b
Cp ^q 12	0.40	2.12 ^a	0.76	4.47 ^c	0.50	2.41 ^a	0.86	4.71 ^c	0.22	2.22 ^a	0.21	4.35 ^c	0.63	2.43 ^a	0.30	2.82 ^b
Dp: Dividend yield, Litzenger and Ramaswamy (1979)																
Dp	0.25	1.03	0.16	0.89	0.21	0.87	0.19	1.03	0.10	1.29	0.05	1.00	0.49	1.10	0.18	1.25
Dp ^q : Quarterly dividend yield, 1-, 6-, and 12-month																
Dp ^q 1	0.28	1.12	0.38	2.12 ^a	0.26	1.05	0.37	2.02 ^a	0.13	1.74	0.12	2.58 ^a	0.50	1.10	0.22	1.47
Dp ^q 6	0.19	0.80	0.20	1.11	0.18	0.78	0.19	1.06	0.11	1.42	0.06	1.18	0.46	1.04	0.19	1.32
Dp ^q 12	0.21	0.90	0.25	1.45	0.21	0.89	0.25	1.42	0.10	1.40	0.07	1.50	0.44	0.99	0.19	1.28
Op [★] : Payout yield, Boudoukh et al. (2007), 1972/7																
Op	0.38	1.86	0.51	3.54 ^c	0.43	1.87	0.58	3.50 ^c	0.17	2.16 ^a	0.15	4.01 ^c	0.56	1.84	0.15	1.13
Op ^q ★: Quarterly payout yield, 1-, 6-, and 12-month, 1985/1																
Op ^q 1	0.11	0.49	0.39	2.90 ^b	0.12	0.52	0.42	2.77 ^a	0.18	1.97 ^a	0.13	3.18 ^b	0.23	0.75	0.24	1.77
Op ^q 6	0.11	0.61	0.33	2.66 ^a	0.16	0.78	0.39	2.96 ^b	0.11	1.34	0.09	2.62 ^a	0.16	0.60	0.14	1.16
Op ^q 12	0.17	0.91	0.31	2.76 ^a	0.23	1.19	0.36	2.96 ^b	0.11	1.35	0.09	2.98 ^b	0.28	1.06	0.17	1.38
Nop [★] : Net payout yield, Boudoukh et al. (2007), 1972/7																
Nop	0.64	3.40 ^c	0.69	4.15 ^c	0.61	3.01 ^b	0.81	4.16 ^c	0.21	2.11 ^a	0.19	3.47 ^c	0.97	3.02 ^b	0.26	1.34
Nop ^q ★: Quarterly net payout yield, 1-, 6-, and 12-month, 1985/1																
Nop ^q 1	0.19	0.86	0.49	2.40 ^a	0.27	1.07	0.61	2.38 ^a	0.14	1.36	0.13	2.15 ^a	0.41	1.19	0.20	1.34
Nop ^q 6	0.25	1.19	0.61	3.15 ^b	0.38	1.62	0.82	3.48 ^c	0.14	1.35	0.19	3.56 ^c	0.47	1.47	0.20	1.31
Nop ^q 12	0.31	1.55	0.54	2.95 ^b	0.47	2.14 ^a	0.78	3.52 ^c	0.14	1.43	0.19	3.76 ^c	0.53	1.72	0.23	1.58
Sr: 5-year sales growth rank, Lakonishok, Shleifer, and Vishny (1994)																
Sr	-0.19	1.08	-0.47	3.52 ^c	-0.13	0.68	-0.52	3.65 ^c	-0.02	0.37	-0.12	3.26 ^b	-0.45	1.97 ^a	-0.11	1.56
Sg: Annual sales growth, Lakonishok, Shleifer, and Vishny (1994)																
Sg	-0.03	0.19	-0.66	5.56 ^c	-0.17	0.96	-0.87	6.49 ^c	-0.15	1.94	-0.23	6.45 ^c	-0.20	0.90	-0.16	1.59
Em: Enterprise multiple, Loughran and Wellman (2011)																
Em	-0.54	2.86 ^b	-0.85	5.92 ^c	-0.71	3.21 ^b	-0.98	5.74 ^c	-0.17	2.16 ^a	-0.18	4.30 ^c	-0.67	3.27 ^b	-0.21	2.33 ^a

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $
Enr ¹ ★: Quarterly enterprise multiple, 1-, 6-, and 12-month, 1976/1																
Enr ¹ 1	-0.71	3.21 ^b	-1.63	8.97 ^c	-1.02	3.83 ^c	-1.87	8.93 ^c	-0.22	2.64 ^a	-0.32	6.64 ^c	-0.86	3.43 ^c	-0.25	2.52 ^a
Enr ¹ 6	-0.43	2.05 ^a	-0.99	6.49 ^c	-0.64	2.62 ^a	-1.13	6.58 ^c	-0.14	1.78	-0.22	5.25 ^c	-0.58	2.46 ^a	-0.18	1.87
Enr ¹ 12	-0.43	2.15 ^a	-0.83	5.80 ^c	-0.65	2.75 ^a	-0.96	6.00 ^c	-0.14	1.71	-0.18	4.68 ^c	-0.57	2.51 ^a	-0.17	1.82
Sp: Sales-to-price, Barbee, Mukherji, and Raines (1996)																
Sp	0.50	2.37 ^a	0.89	4.07 ^c	0.74	2.87 ^b	1.14	4.38 ^c	0.29	2.30 ^a	0.22	3.99 ^c	0.41	1.30	0.12	0.69
Sp ¹ : Quarterly sales-to-price, 1-, 6-, and 12-month																
Sp ¹ 1	0.59	2.39 ^a	1.33	4.91 ^c	0.72	2.55 ^a	1.58	5.30 ^c	0.40	2.51 ^a	0.35	4.90 ^c	0.24	0.68	0.18	0.95
Sp ¹ 6	0.56	2.43 ^a	1.00	4.00 ^c	0.61	2.30 ^a	1.19	4.32 ^c	0.35	2.31 ^a	0.25	3.71 ^c	0.21	0.64	0.12	0.69
Sp ¹ 12	0.53	2.47 ^a	0.91	3.85 ^c	0.60	2.39 ^a	1.09	4.14 ^c	0.31	2.22 ^a	0.22	3.64 ^c	0.30	0.97	0.10	0.59
Ocp★: Operating cash flow-to-price, Desai, Rajgopal, and Venkatachalam (2004), 1972/7																
Ocp	0.70	3.14 ^b	0.81	4.31 ^c	0.83	3.26 ^b	0.89	4.34 ^c	0.27	2.63 ^a	0.19	3.56 ^c	0.57	2.11 ^a	0.21	1.88
Ocp ¹ ★: Quarterly operating cash flow-to-price, 1-, 6-, and 12-month, 1985/1																
Ocp ¹ 1	0.64	2.28 ^a	0.93	3.44 ^c	0.65	2.19 ^a	0.94	3.17 ^b	0.39	1.89	0.23	2.78 ^a	0.27	0.81	0.07	0.32
Ocp ¹ 6	0.47	1.82	0.62	2.39 ^a	0.63	2.28 ^a	0.66	2.42 ^a	0.28	1.50	0.14	1.80	0.22	0.75	-0.01	0.05
Ocp ¹ 12	0.37	1.56	0.59	2.54 ^a	0.54	2.11 ^a	0.63	2.57 ^a	0.24	1.41	0.13	1.79	0.18	0.65	0.01	0.04
Ir: Intangible return, Daniel and Titman (2006)																
Ir	-0.47	2.22 ^a	-0.97	5.09 ^c	-0.57	2.32 ^a	-1.09	5.06 ^c	-0.14	1.87	-0.30	5.04 ^c	-0.63	2.39 ^a	-0.17	1.74
Vhp: Intrinsic value-to-market, Frankel and Lee (1998)																
Vhp	0.38	2.05 ^a	0.46	3.51 ^c	0.55	2.58 ^a	0.59	4.02 ^c	0.20	2.51 ^a	0.15	4.01 ^c	0.21	0.70	0.13	1.24
Vfp★: Analysts' forecasts-based intrinsic value-to-market, Frankel and Lee (1998), 1976/7																
Vfp	0.47	2.18 ^a	0.27	1.39	0.45	1.69	0.33	1.54	0.13	1.19	0.08	1.43	0.33	1.15	0.07	0.71
Efp: Enterprise book-to-price, Penman, Richardson, and Tuna (2007)																
Efp	0.41	2.00 ^a	0.90	4.74 ^c	0.41	1.68	1.11	5.20 ^c	0.16	1.56	0.30	5.53 ^c	0.44	1.90	0.15	1.20
Ebp ¹ ★: Quarterly enterprise book-to-price, 1-, 6-, and 12-month, 1976/1																
Ebp ¹ 1	0.25	0.90	1.30	5.08 ^c	0.34	1.04	1.72	5.87 ^c	0.13	0.90	0.48	6.32 ^c	0.25	0.81	0.09	0.48
Ebp ¹ 6	0.21	0.84	0.93	3.92 ^c	0.25	0.83	1.21	4.52 ^c	0.11	0.80	0.34	5.01 ^c	0.22	0.74	0.05	0.28
Ebp ¹ 12	0.31	1.28	0.95	4.23 ^c	0.38	1.31	1.19	4.75 ^c	0.13	1.04	0.33	5.15 ^c	0.36	1.30	0.08	0.51

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Ndp: Net debt-to-price, Penman, Richardson, and Tuna (2007)	0.28	1.52	0.42	2.25 ^a	0.35	1.68	0.44	2.17 ^a	0.15	1.80	0.09	1.57	0.26	1.21	0.13	1.46
Ndp ^q ★: Quarterly net debt-to-price, 1-, 6-, and 12-month, 1976/1																
Ndp ^q 1	0.16	0.66	0.40	1.42	0.36	1.35	0.46	1.49	0.13	0.82	0.05	0.51	-0.06	0.20	0.10	0.58
Ndp ^q 6	0.17	0.75	0.25	0.93	0.35	1.27	0.27	0.91	0.15	0.99	-0.02	0.27	0.00	0.01	0.06	0.39
Ndp ^q 12	0.26	1.25	0.33	1.39	0.43	1.79	0.34	1.31	0.18	1.36	0.02	0.23	0.15	0.59	0.11	0.79
Dur: Equity duration, Dechow, Sloan, and Soliman (2004)																
Dur	-0.42	2.19 ^a	-0.76	3.97 ^c	-1.06	4.06 ^c	-0.86	3.61 ^c	-0.23	2.23 ^a	-0.25	3.76 ^c	-0.62	2.62 ^a	-0.23	2.21 ^a
Lig★: Long-term growth forecasts of analysts, La Porta (1996), 1982/1																
Lig	0.13	0.38	-0.18	0.57	-0.06	0.14	-0.38	0.99	-0.04	0.30	-0.13	1.23	0.16	0.25	-0.06	0.29
Lig ^m ★: Long-term growth forecasts of analysts, monthly sorts, 1-, 6-, and 12-month, 1982/1																
Lig ^m 1	0.02	0.05	-0.32	1.01	-0.14	0.30	-0.52	1.34	-0.06	0.40	-0.16	1.40	0.05	0.07	-0.01	0.02
Lig ^m 6	0.01	0.02	-0.34	1.09	-0.19	0.40	-0.57	1.51	-0.07	0.49	-0.18	1.62	-0.05	0.07	-0.03	0.13
Lig ^m 12	0.02	0.07	-0.32	1.06	-0.18	0.39	-0.55	1.47	-0.07	0.46	-0.17	1.60	0.02	0.02	-0.03	0.16
C. Investment (38 anomalies)																
Aci: Abnormal corporate investment, Titman, Wei, and Xie (2004)																
Aci	-0.30	2.13 ^a	-0.30	4.30 ^c	-0.28	2.03 ^a	-0.36	4.45 ^c	-0.09	2.12 ^a	-0.09	5.31 ^c	-0.39	2.50 ^a	-0.11	2.13 ^a
I/A: Investment-to-assets, Cooper, Gulen, and Schill (2008)																
I/A	-0.44	2.89 ^b	-1.02	6.91 ^c	-0.56	3.40 ^c	-1.27	6.99 ^c	-0.18	2.91 ^b	-0.36	9.29 ^c	-0.57	3.03 ^b	-0.26	3.27 ^b
Ia ^q ★: Quarterly investment-to-assets, 1-, 6-, and 12-month, 1973/1																
Ia ^q 1	-0.31	1.74	-1.08	5.84 ^c	-0.34	1.54	-1.30	5.72 ^c	-0.14	2.12 ^a	-0.39	8.02 ^c	-0.44	1.89	-0.18	1.99 ^a
Ia ^q 6	-0.50	3.00 ^b	-1.18	6.46 ^c	-0.66	3.42 ^c	-1.43	6.45 ^c	-0.19	2.95 ^b	-0.41	8.86 ^c	-0.69	3.26 ^b	-0.25	2.89 ^b
Ia ^q 12	-0.48	3.11 ^b	-1.16	6.79 ^c	-0.61	3.50 ^c	-1.40	6.70 ^c	-0.20	3.20 ^b	-0.40	9.46 ^c	-0.66	3.40 ^c	-0.26	3.21 ^b
dPia: Change in property, plant, and equipment and inventory scaled by book assets, Lyandres, Sun, and Zhang (2008)																
dPia	-0.48	3.64 ^c	-0.98	7.58 ^c	-0.61	4.43 ^c	-1.08	8.06 ^c	-0.13	2.84 ^b	-0.30	8.85 ^c	-0.55	3.60 ^c	-0.15	2.77 ^a
Noa: Net operating assets, Hishleifer et al. (2004)																
Noa	-0.44	3.25 ^b	-0.88	5.64 ^c	-0.49	3.06 ^b	-1.01	5.23 ^c	-0.18	4.87 ^c	-0.29	5.94 ^c	-0.48	2.88 ^b	-0.20	4.38 ^c

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
dNoa: Change in net operating assets, Hirshleifer et al. (2004)																
dNoa	-0.55	4.14 ^c	-1.04	8.05 ^c	-0.68	4.47 ^c	-1.20	8.03 ^c	-0.22	4.36 ^c	-0.35	9.58 ^c	-0.68	4.14 ^c	-0.26	4.04 ^c
dLno: Change in long-term net operating assets, Fairfield, Whisenant, and Yohn (2003)																
dLno	-0.39	2.99 ^b	-0.84	6.64 ^c	-0.47	3.66 ^c	-0.91	6.76 ^c	-0.12	3.36 ^b	-0.27	7.83 ^c	-0.39	2.34 ^a	-0.11	2.16 ^a
Ig: Investment growth, Xie (2001)																
Ig	-0.46	3.76 ^c	-0.56	6.52 ^c	-0.49	3.19 ^b	-0.71	6.67 ^c	-0.17	2.43 ^a	-0.17	6.54 ^c	-0.55	3.91 ^c	-0.20	2.33 ^a
2lg: 2-year investment growth, Anderson and Garcia-Feijoo (2006)																
2lg	-0.33	2.52 ^a	-0.46	5.17 ^c	-0.27	1.61	-0.57	5.20 ^c	-0.16	2.28 ^a	-0.16	5.53 ^c	-0.34	2.12 ^a	-0.16	1.85
3lg: 3-year investment growth, Anderson and Garcia-Feijoo (2006)																
3lg	-0.16	1.15	-0.47	5.09 ^c	-0.28	1.58	-0.54	4.71 ^c	-0.12	1.80	-0.15	5.79 ^c	-0.15	0.79	-0.10	1.13
Nsi: Net stock issues, Pontiff and Woodgate (2008)																
Nsi	-0.64	4.47 ^c	-0.99	6.32 ^c	-0.78	5.19 ^c	-1.13	6.70 ^c	-0.17	4.32 ^c	-0.30	6.83 ^c	-0.86	4.77 ^c	-0.21	4.38 ^c
dli: Percentage change in investment relative to industry, Abarbanell and Bushee (1998)																
dli	-0.29	2.61 ^a	-0.45	6.32 ^c	-0.45	3.36 ^b	-0.57	7.13 ^c	-0.14	2.30 ^a	-0.14	5.50 ^c	-0.32	2.03 ^a	-0.10	1.34
Cei: Composite equity issuance, Daniel and Titman (2006)																
Cei	-0.57	3.32 ^b	-0.67	3.67 ^c	-0.63	3.41 ^c	-0.81	3.89 ^c	-0.19	3.09 ^b	-0.24	3.82 ^c	-0.69	3.13 ^b	-0.24	3.23 ^b
Cdi: Composite debt issuance, Lyandres, Sun, and Zhang (2008)																
Cdi	0.05	0.43	-0.37	5.42 ^c	-0.02	0.15	-0.42	5.32 ^c	-0.03	0.72	-0.11	5.09 ^c	0.01	0.06	-0.03	0.63
Ivg: Inventory growth, Belo and Lin (2011)																
Ivg	-0.33	2.44 ^a	-0.72	6.79 ^c	-0.44	2.90 ^b	-0.82	6.37 ^c	-0.13	2.56 ^a	-0.23	8.23 ^c	-0.45	3.16 ^b	-0.17	3.13 ^b
Ivc: Inventory changes, Thomas and Zhang (2002)																
Ivc	-0.44	3.33 ^b	-0.66	6.36 ^c	-0.45	2.74 ^a	-0.78	6.52 ^c	-0.13	2.29 ^a	-0.21	6.64 ^c	-0.49	2.80 ^b	-0.15	1.86
Oa: Operating accruals, Sloan (1996)																
Oa	-0.27	2.19 ^a	-0.38	3.61 ^c	-0.28	1.71	-0.44	3.40 ^c	-0.10	1.66	-0.11	2.79 ^b	-0.33	1.98 ^a	-0.13	1.96 ^a
Ta: Total accruals, Richardson et al. (2005)																
Ta	-0.22	1.63	-0.45	3.54 ^c	-0.22	1.24	-0.53	3.29 ^b	-0.10	1.73	-0.17	4.06 ^c	-0.31	1.83	-0.13	1.77
dWc: Change in net noncash working capital, Richardson et al. (2005)																
dWc	-0.42	3.25 ^b	-0.44	5.08 ^c	-0.43	2.46 ^a	-0.63	5.52 ^c	-0.16	2.86 ^b	-0.18	6.17 ^c	-0.53	3.38 ^b	-0.21	2.94 ^b

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
dCoa: Change in current operating assets, Richardson et al. (2005)	-0.31	2.28 ^a	-0.83	7.22 ^c	-0.47	2.93 ^b	-1.01	7.82 ^c	-0.15	2.18 ^a	-0.29	8.49 ^c	-0.41	2.55 ^a	-0.21	2.43 ^a
dCol: Change in current operating liabilities, Richardson et al. (2005)	-0.12	0.81	-0.64	6.90 ^c	-0.18	1.02	-0.79	7.37 ^c	-0.03	0.48	-0.24	7.96 ^c	-0.12	0.69	-0.06	0.74
dNco: Change in net noncurrent operating assets, Richardson et al. (2005)	-0.41	3.52 ^c	-0.97	7.51 ^c	-0.55	4.16 ^c	-1.07	7.86 ^c	-0.16	4.12 ^c	-0.32	9.77 ^c	-0.48	3.40 ^c	-0.18	3.75 ^c
dNco: Change in noncurrent operating assets, Richardson et al. (2005)	-0.42	3.47 ^c	-0.96	7.14 ^c	-0.50	4.01 ^c	-1.05	7.42 ^c	-0.15	3.88 ^c	-0.32	9.54 ^c	-0.51	3.48 ^c	-0.17	3.62 ^c
dNcl: Change in noncurrent operating liabilities, Richardson et al. (2005)	-0.08	0.64	-0.24	2.66 ^a	-0.11	1.00	-0.22	2.57 ^a	-0.03	0.97	-0.09	3.91 ^c	-0.23	1.65	-0.04	1.21
dFin: Change in net financial assets, Richardson et al. (2005)	0.28	2.39 ^a	0.58	7.81 ^c	0.28	1.93	0.56	5.86 ^c	0.11	2.45 ^a	0.17	6.72 ^c	0.35	2.44 ^a	0.12	2.31 ^a
dSit★: Change in short-term investments, Richardson et al. (2005), 1971/77	0.18	1.22	-0.10	1.03	-0.19	0.92	-0.23	1.70	0.02	0.34	-0.08	2.38 ^a	0.40	2.20 ^a	0.04	0.54
dLit: Change in long-term investments, Richardson et al. (2005)	-0.23	1.59	-0.42	4.62 ^c	-0.04	0.22	-0.50	4.48 ^c	-0.05	0.90	-0.14	4.62 ^c	-0.29	1.59	-0.08	1.17
dFnl: Change in financial liabilities, Richardson et al. (2005)	-0.32	3.09 ^b	-0.81	10.52 ^c	-0.29	2.53 ^a	-0.89	10.61 ^c	-0.15	4.09 ^c	-0.27	10.39 ^c	-0.37	2.89 ^b	-0.18	4.08 ^c
dBe: Change in common equity, Richardson et al. (2005)	-0.32	2.03 ^a	-0.63	4.39 ^c	-0.69	3.57 ^c	-0.83	4.79 ^c	-0.17	2.45 ^a	-0.25	6.36 ^c	-0.38	1.89	-0.20	2.28 ^a
Dac: Discretionary accruals, Xie (2001)	-0.39	2.95 ^b	-0.37	4.47 ^c	-0.42	2.39 ^a	-0.39	3.60 ^c	-0.14	2.45 ^a	-0.12	3.67 ^c	-0.39	2.46 ^a	-0.15	2.96 ^b
Poa: Percent operating accruals, Hafzalla, Lundholm, and Van Winkle (2011)	-0.39	2.89 ^b	-0.55	6.30 ^c	-0.46	2.68 ^a	-0.64	6.69 ^c	-0.07	1.51	-0.11	5.02 ^c	-0.41	1.70	0.01	0.18
Pia: Percent total accruals, Hafzalla, Lundholm, and Van Winkle (2011)	-0.42	3.14 ^b	-0.36	5.22 ^c	-0.56	3.72 ^c	-0.45	5.90 ^c	-0.15	3.50 ^c	-0.11	6.02 ^c	-0.23	0.92	-0.07	1.26
Pda: Percent discretionary accruals	-0.48	3.91 ^c	-0.37	5.66 ^c	-0.40	2.97 ^b	-0.35	4.52 ^c	-0.10	2.77 ^a	-0.08	4.40 ^c	-0.55	2.88 ^b	-0.02	0.31

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Nxf★: Net external finance, Bradshaw, Richardson, and Sloan (2006), 1972/7																
Nxf	-0.29	1.58	-0.79	5.84 ^c	-0.51	2.44 ^a	-0.99	5.85 ^c	-0.15	2.13 ^a	-0.31	6.38 ^c	-0.45	2.01 ^a	-0.22	2.54 ^a
Nef★: Net equity finance, Bradshaw, Richardson, and Sloan (2006), 1972/7																
Nef	-0.18	0.96	-0.53	2.84 ^b	-0.44	1.97 ^a	-0.78	3.57 ^c	-0.12	1.50	-0.20	3.54 ^c	-0.34	1.43	-0.21	2.11 ^a
Ndf★: Net debt finance, Bradshaw, Richardson, and Sloan (2006), 1972/7																
Ndf	-0.30	2.45 ^a	-0.69	8.82 ^c	-0.23	1.70	-0.78	8.82 ^c	-0.08	2.34 ^a	-0.23	9.54 ^c	-0.37	2.31 ^a	-0.10	2.14 ^a
D. Profitability (79 anomalies)																
Roe: Return on equity, 1-, 6-, and 12-month, Hou, Xue, and Zhang (2015)																
Roe1	0.68	3.12 ^b	1.23	6.32 ^c	0.77	3.03 ^b	1.36	5.47 ^c	0.24	2.69 ^a	0.35	4.90 ^c	0.82	3.22 ^b	0.28	2.62 ^a
Roe6	0.42	1.98 ^a	0.72	3.73 ^c	0.38	1.58	0.89	3.58 ^c	0.18	2.17 ^a	0.25	3.44 ^c	0.48	1.92	0.21	2.13 ^a
Roe12	0.23	1.18	0.27	1.45	0.26	1.14	0.41	1.71	0.12	1.49	0.14	1.88	0.28	1.23	0.14	1.45
dRoe: 4-quarter change in return on equity, 1-, 6-, and 12-month																
dRoe1	0.75	5.53 ^c	1.46	12.26 ^c	0.88	5.04 ^c	1.56	12.04 ^c	0.26	4.75 ^c	0.35	9.38 ^c	0.83	5.35 ^c	0.28	4.43 ^c
dRoe6	0.36	3.16 ^b	0.78	7.94 ^c	0.41	3.00 ^b	0.85	7.83 ^c	0.14	2.92 ^b	0.20	6.23 ^c	0.42	3.17 ^b	0.16	2.98 ^b
dRoe12	0.24	2.39 ^a	0.37	4.28 ^c	0.27	2.45 ^a	0.38	4.02 ^c	0.10	2.43 ^a	0.10	3.57 ^c	0.31	2.67 ^a	0.13	2.58 ^a
RoA★: Return on assets, 1-, 6-, and 12-month, Balakrishnan, Bartov, and Faurel (2010), 1972/1																
RoA1	0.57	2.63 ^a	1.06	4.97 ^c	0.77	2.90 ^b	1.12	4.04 ^c	0.25	2.64 ^a	0.31	3.53 ^c	0.68	2.50 ^a	0.31	2.59 ^a
RoA6	0.39	1.82	0.64	3.02 ^b	0.53	1.98 ^a	0.71	2.52 ^a	0.20	2.17 ^a	0.21	2.33 ^a	0.35	1.34	0.24	2.15 ^a
RoA12	0.25	1.27	0.24	1.16	0.40	1.60	0.30	1.09	0.13	1.49	0.09	1.07	0.27	1.11	0.19	1.76
dRoA★: 4-quarter change in return on assets, 1-, 6-, and 12-month, 1973/1																
dRoA1	0.57	3.76 ^c	1.44	11.90 ^c	0.72	3.98 ^c	1.38	10.20 ^c	0.26	4.74 ^c	0.34	8.86 ^c	0.70	4.21 ^c	0.28	4.15 ^c
dRoA6	0.27	1.99 ^a	0.75	7.38 ^c	0.41	2.71 ^a	0.70	6.03 ^c	0.15	2.97 ^b	0.17	5.13 ^c	0.39	2.53 ^a	0.18	3.05 ^b
dRoA12	0.17	1.45	0.33	3.82 ^c	0.31	2.44 ^a	0.30	3.04 ^b	0.10	2.07 ^a	0.07	2.44 ^a	0.30	2.24 ^a	0.15	2.63 ^a
Rna: Return on net operating assets, Soliman (2008)																
Rna	0.15	0.81	-0.13	0.69	0.34	1.25	-0.01	0.05	0.09	1.38	0.01	0.13	0.37	1.18	0.25	1.99 ^a
Rna ^q ★: Quarterly return on net operating assets, 1-, 6-, and 12-month, 1976/1																
Rna ^q 1	0.64	2.77 ^a	0.79	3.64 ^c	0.93	2.86 ^b	0.90	3.01 ^b	0.36	4.04 ^c	0.17	2.32 ^a	0.92	2.45 ^a	0.62	3.93 ^c
Rna ^q 6	0.43	2.01 ^a	0.44	2.06 ^a	0.86	2.78 ^b	0.61	2.08 ^a	0.28	3.27 ^b	0.13	1.79	0.71	2.06 ^a	0.54	3.69 ^c
Rna ^q 12	0.35	1.68	0.20	0.95	0.69	2.25 ^a	0.34	1.18	0.24	2.86 ^b	0.08	1.14	0.67	2.00 ^a	0.48	3.31 ^b

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Pm: Profit margin, Soliman (2008)																
Pm	0.03	0.14	-0.17	0.73	0.44	1.48	0.04	0.14	0.12	1.47	0.03	0.50	0.40	0.98	0.29	1.76
Pm ^q : Quarterly profit margin, 1-, 6-, and 12-month																
Pm ^q 1	0.36	1.68	0.53	2.35 ^a	0.82	2.94 ^b	0.82	2.90 ^b	0.21	2.66 ^a	0.13	2.43 ^a	0.70	1.64	0.44	2.63 ^a
Pm ^q 6	0.17	0.87	0.24	1.12	0.66	2.44 ^a	0.51	1.83	0.15	2.00 ^a	0.10	1.93	0.50	1.24	0.39	2.42 ^a
Pm ^q 12	0.17	0.90	0.04	0.18	0.57	2.16 ^a	0.27	0.96	0.14	1.82	0.07	1.37	0.47	1.20	0.36	2.20 ^a
Atot: Assets turnover, Soliman (2008)																
Atot	0.33	1.89	0.25	1.60	0.33	1.92	0.25	1.57	0.05	1.03	0.01	0.44	0.48	1.48	0.10	0.93
Atot ^q ★: Quarterly assets turnover, 1-, 6-, and 12-month, 1972/1																
Atot ^q 1	0.62	3.44 ^c	1.02	6.53 ^c	0.81	4.06 ^c	1.05	6.58 ^c	0.18	3.73 ^c	0.21	6.18 ^c	0.94	3.24 ^b	0.27	2.96 ^b
Atot ^q 6	0.53	3.07 ^b	0.79	5.09 ^c	0.62	3.34 ^b	0.81	5.09 ^c	0.16	3.27 ^b	0.16	4.96 ^c	0.80	2.90 ^b	0.26	2.77 ^a
Atot ^q 12	0.42	2.56 ^a	0.60	3.86 ^c	0.51	2.85 ^b	0.62	3.87 ^c	0.14	2.85 ^b	0.12	3.61 ^c	0.64	2.32 ^a	0.21	2.24 ^a
Cto: Capital turnover, Haugen and Baker (1996)																
Cto	0.28	1.67	0.25	1.29	0.32	1.73	0.23	1.08	0.06	1.29	0.02	0.39	0.44	1.27	0.11	1.21
Cto ^q ★: Quarterly capital turnover, 1-, 6-, and 12-month, 1972/1																
Cto ^q 1	0.44	2.44 ^a	0.99	5.35 ^c	0.73	3.13 ^b	1.03	4.95 ^c	0.14	2.89 ^b	0.26	5.35 ^c	0.69	2.05 ^a	0.19	2.08 ^a
Cto ^q 6	0.40	2.34 ^a	0.77	4.15 ^c	0.61	2.78 ^a	0.82	3.97 ^c	0.12	2.61 ^a	0.20	4.25 ^c	0.61	1.79	0.16	1.84
Cto ^q 12	0.36	2.14 ^a	0.58	3.12 ^b	0.50	2.39 ^a	0.62	3.00 ^b	0.10	2.19 ^a	0.13	2.88 ^b	0.55	1.61	0.13	1.49
Gpa: Gross profits-to-assets, Novy-Marx (2013)																
Gpa	0.37	2.63 ^a	0.67	4.35 ^c	0.58	2.83 ^b	0.64	3.29 ^b	0.12	2.05 ^a	0.18	3.97 ^c	0.41	2.64 ^a	0.12	1.96 ^a
Gla: Gross profits-to-lagged assets																
Gla	0.17	1.13	0.24	1.54	0.31	1.67	0.24	1.24	0.06	0.96	0.04	0.92	0.16	1.03	0.06	0.85
Gla ^q ★: Quarterly gross profits-to-assets, 1-, 6-, and 12-month, 1976/1																
Gla ^q 1	0.51	3.48 ^c	0.96	5.48 ^c	0.87	4.42 ^c	0.96	4.40 ^c	0.18	2.93 ^b	0.29	5.84 ^c	0.53	3.28 ^b	0.19	2.79 ^b
Gla ^q 6	0.33	2.46 ^a	0.69	3.96 ^c	0.65	3.36 ^b	0.66	2.94 ^b	0.14	2.33 ^a	0.21	4.29 ^c	0.38	2.53 ^a	0.15	2.31 ^a
Gla ^q 12	0.29	2.18 ^a	0.50	2.89 ^b	0.58	3.23 ^b	0.46	2.04 ^a	0.12	2.06 ^a	0.15	3.16 ^b	0.32	2.25 ^a	0.13	2.05 ^a
Ope: Operating profits-to-book equity, Fama and French (2015)																
Ope	0.27	1.34	0.13	0.66	0.57	1.88	0.29	1.03	0.17	2.08 ^a	0.11	1.38	0.26	1.20	0.18	2.06 ^a

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $
Ole: Operating profits-to-lagged book equity																
Ole	0.11	0.58	-0.13	0.70	0.68	2.40 ^a	0.14	0.52	0.07	1.22	0.04	0.51	0.07	0.38	0.07	1.17
Ole4 [★] : Quarterly operating profits-to-lagged book equity, 1-, 6-, and 12-month, 1972/1																
Ole4 ¹	0.71	3.40 ^c	0.93	4.49 ^c	0.78	3.17 ^b	1.19	4.20 ^c	0.24	3.63 ^c	0.32	4.18 ^c	0.69	3.15 ^b	0.25	3.49 ^c
Ole4 ⁶	0.48	2.40 ^a	0.50	2.54 ^a	0.55	2.36 ^a	0.75	2.77 ^a	0.18	2.93 ^b	0.23	3.00 ^b	0.47	2.25 ^a	0.19	2.91 ^b
Ole4 ¹²	0.36	1.90	0.21	1.06	0.50	2.20 ^a	0.44	1.63	0.14	2.47 ^a	0.14	1.83	0.36	1.81	0.15	2.44 ^a
Opa: Operating profits-to-book assets, Ball et al. (2016)																
Opa	0.41	2.09 ^a	0.43	2.43 ^a	0.81	2.58 ^a	0.59	2.49 ^a	0.13	1.86	0.17	2.39 ^a	0.34	1.70	0.12	1.59
Ola: Operating profits-to-lagged book assets																
Ola	0.20	1.11	0.03	0.14	0.62	2.18 ^a	0.20	0.86	0.03	0.42	0.05	0.85	0.17	0.91	0.01	0.17
Ola4 [★] : Quarterly operating profits-to-lagged book assets, 1-, 6-, and 12-month, 1976/1																
Ola4 ¹	0.75	3.53 ^c	1.05	5.34 ^c	1.00	3.66 ^c	1.34	5.20 ^c	0.25	3.45 ^c	0.38	5.13 ^c	0.70	3.19 ^b	0.24	3.17 ^b
Ola4 ⁶	0.52	2.59 ^a	0.67	3.50 ^c	0.75	3.00 ^b	0.91	3.64 ^c	0.16	2.31 ^a	0.27	3.82 ^c	0.49	2.34 ^a	0.16	2.09 ^a
Ola4 ¹²	0.46	2.46 ^a	0.41	2.22 ^a	0.67	2.82 ^b	0.59	2.45 ^a	0.15	2.13 ^a	0.19	2.74 ^a	0.45	2.29 ^a	0.14	1.92
Cop: Cash-based operating profits-to-book assets, Ball et al. (2016)																
Cop	0.63	3.57 ^c	0.69	4.81 ^c	0.82	3.34 ^b	0.81	4.15 ^c	0.16	2.42 ^a	0.23	4.13 ^c	0.62	3.41 ^c	0.16	2.33 ^a
Cla: Cash-based operating profits-to-lagged book assets																
Cla	0.55	3.23 ^b	0.46	3.03 ^b	0.93	3.97 ^c	0.72	3.68 ^c	0.10	1.57	0.20	3.67 ^c	0.52	3.01 ^b	0.10	1.45
Cla4 [★] : Quarterly cash-based operating profits-to-lagged book assets, 1-, 6-, and 12-month, 1976/1																
Cla4 ¹	0.52	3.26 ^b	0.90	6.01 ^c	0.54	3.09 ^b	1.06	5.82 ^c	0.20	3.58 ^c	0.33	6.16 ^c	0.50	3.04 ^b	0.21	3.55 ^c
Cla4 ⁶	0.49	3.60 ^c	0.73	5.50 ^c	0.60	3.95 ^c	0.92	5.52 ^c	0.17	3.22 ^b	0.27	5.62 ^c	0.48	3.48 ^c	0.17	3.13 ^b
Cla4 ¹²	0.46	3.63 ^c	0.62	4.96 ^c	0.56	4.09 ^c	0.80	5.30 ^c	0.17	3.44 ^c	0.23	5.21 ^c	0.47	3.59 ^c	0.17	3.38 ^b
F [★] : Fundamental score, Piotroski (2000), 1972/7																
F	0.29	1.11	0.46	2.06 ^a	0.29	1.11	0.46	2.06 ^a	0.07	1.54	0.12	2.15 ^a	0.65	2.19 ^a	0.09	1.55
F4 [★] : Quarterly fundamental score, 1-, 6-, and 12-month, 1985/1																
F4 ¹	0.52	2.32 ^a	1.42	5.61 ^c	0.52	2.32 ^a	1.42	5.61 ^c	0.12	2.46 ^a	0.37	5.08 ^c	0.65	2.38 ^a	0.17	2.69 ^a
F4 ⁶	0.48	2.39 ^a	0.98	4.13 ^c	0.48	2.39 ^a	0.98	4.13 ^c	0.09	2.06 ^a	0.26	3.71 ^c	0.61	2.44 ^a	0.15	2.57 ^a
F4 ¹²	0.38	2.05 ^a	0.65	2.81 ^b	0.38	2.05 ^a	0.65	2.81 ^b	0.06	1.57	0.17	2.50 ^a	0.55	2.54 ^a	0.12	2.50 ^a

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Fp [★] : Failure probability, Campbell, Hilscher, and Szilagyi (2008), 1976/7																
Fp	-0.39	1.35	-0.29	1.12	-0.55	1.50	-0.34	1.03	-0.19	1.62	-0.09	0.84	-0.82	2.09 ^a	-0.40	2.27 ^a
Fp ^m [★] : Failure probability, monthly sorts, 1-, 6-, and 12-month, Campbell, Hilscher, and Szilagyi (2008), 1976/1																
Fp ^m 1	-0.45	1.38	-0.57	2.06 ^a	-0.85	2.29 ^a	-0.78	2.42 ^a	-0.20	1.46	-0.23	2.42 ^a	-0.97	2.31 ^a	-0.39	1.99 ^a
Fp ^m 6	-0.62	1.99 ^a	-0.53	1.96	-0.89	2.46 ^a	-0.71	2.23 ^a	-0.23	1.86	-0.21	2.08 ^a	-1.15	2.92 ^b	-0.43	2.40 ^a
Fp ^m 12	-0.36	1.26	-0.28	1.07	-0.53	1.55	-0.35	1.11	-0.17	1.45	-0.10	1.00	-0.74	2.04 ^a	-0.32	1.95
O: Ohlson's (1980) O-score, Dichev (1998)																
O	-0.09	0.48	-0.01	0.04	-0.52	1.82	-0.10	0.40	-0.04	0.63	-0.04	0.51	-0.60	2.05 ^a	-0.11	1.09
O4 [★] : Quarterly O-score, 1-, 6-, and 12-month, 1976/1																
O41	-0.37	1.66	-0.14	0.65	-0.94	2.62 ^a	-0.24	0.74	-0.15	1.70	-0.08	0.88	-0.83	2.71 ^a	-0.19	1.56
O46	-0.23	1.06	-0.04	0.20	-0.80	2.29 ^a	-0.11	0.34	-0.13	1.52	-0.04	0.47	-0.75	2.48 ^a	-0.17	1.44
O412	-0.16	0.76	0.01	0.05	-0.71	2.11 ^a	-0.06	0.18	-0.12	1.41	-0.03	0.31	-0.68	2.27 ^a	-0.17	1.43
Z: Altman's (1968) Z-score, Dichev (1998)																
Z	0.01	0.06	-0.21	1.26	-0.13	0.61	-0.43	2.26 ^a	-0.05	0.64	-0.15	3.52 ^c	-0.04	0.13	-0.06	0.40
Z4 [★] : Quarterly Z-score, 1-, 6-, and 12-month, 1976/1																
Z41	0.00	0.00	-0.17	0.78	-0.10	0.37	-0.38	1.41	-0.04	0.54	-0.14	2.39 ^a	0.08	0.23	-0.03	0.20
Z46	-0.03	0.17	-0.28	1.34	-0.21	0.82	-0.49	1.95	-0.06	0.73	-0.16	3.02 ^b	0.06	0.18	-0.04	0.32
Z412	-0.09	0.46	-0.29	1.48	-0.20	0.85	-0.51	2.15 ^a	-0.05	0.73	-0.15	3.17 ^b	-0.01	0.02	-0.06	0.45
C4 [★] : Credit rating, 1-, 6-, and 12-month, Avramov et al. (2009), 1986/1																
C41	0.03	0.09	-0.28	0.82	-0.10	0.26	-0.63	1.53	-0.03	0.31	-0.15	1.18	-0.08	0.20	-0.05	0.44
C46	-0.01	0.03	-0.26	0.77	-0.12	0.31	-0.57	1.41	-0.02	0.27	-0.13	1.01	-0.11	0.26	-0.04	0.38
C412	-0.01	0.02	-0.26	0.78	-0.13	0.36	-0.54	1.38	-0.02	0.19	-0.12	0.93	-0.08	0.19	-0.03	0.30
Tbi: Taxable income-to-book income, Lev and Nissim (2004)																
Tbi	0.14	1.06	0.13	1.40	0.04	0.29	0.06	0.60	0.05	1.37	0.05	2.34 ^a	0.14	0.67	0.06	1.14
Tbi4 [★] : Quarterly taxable income-to-book income, 1-, 6-, and 12-month																
Tbi41	0.18	1.41	-0.04	0.43	0.23	1.58	-0.02	0.16	0.02	0.58	0.01	0.39	0.18	0.80	0.06	1.12
Tbi46	0.21	1.90	0.05	0.61	0.18	1.59	0.04	0.43	0.04	1.22	0.02	1.17	0.24	1.16	0.06	1.22
Tbi412	0.20	1.88	0.09	1.17	0.11	1.00	0.07	0.73	0.04	1.28	0.03	1.72	0.27	1.34	0.07	1.27

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
G★: Growth score, Mohanram (2005), 1976/7																
G	0.24	1.22	0.18	0.81	0.24	1.22	0.18	0.81	0.06	1.10	0.06	0.88	0.71	2.73 ^a	0.17	2.83 ^b
BI: Book leverage, Fama and French (1992)																
BI	-0.02	0.12	-0.30	2.36 ^a	0.17	0.94	-0.20	1.36	0.00	0.11	-0.09	2.97 ^b	-0.02	0.10	-0.06	1.10
BIq★: Quarterly book leverage, 1-, 6-, and 12-month, 1972/1																
BIq1	0.10	0.58	-0.25	1.61	0.29	1.32	-0.13	0.74	0.05	1.12	-0.09	2.30 ^a	0.15	0.69	0.03	0.45
BIq6	0.12	0.71	-0.28	1.91	0.31	1.46	-0.18	1.12	0.04	0.90	-0.10	2.69 ^a	0.19	0.93	0.01	0.15
BIq12	0.09	0.52	-0.27	1.93	0.26	1.24	-0.19	1.18	0.03	0.66	-0.10	2.77 ^a	0.13	0.62	-0.01	0.13
SGq1: Quarterly sales growth, 1-, 6-, and 12-month																
SGq1	0.30	1.76	0.56	4.28 ^c	0.30	1.61	0.48	3.29 ^b	0.08	1.09	0.06	1.55	0.30	1.19	0.12	1.29
SGq6	0.12	0.72	0.04	0.33	0.07	0.41	-0.06	0.45	-0.03	0.36	-0.07	1.92	0.06	0.26	-0.01	0.16
SGq12	-0.09	0.63	-0.33	2.83 ^b	-0.09	0.58	-0.44	3.53 ^c	-0.08	1.28	-0.16	4.80 ^c	-0.23	1.08	-0.08	1.01
E. Intangibles (103 anomalies)																
Oca: Organizational capital-to-book assets, Eisfeldt and Papanikolaou (2013)																
Oca	0.54	2.67 ^a	0.78	3.77 ^c	0.43	2.26 ^a	0.78	3.57 ^c	0.14	1.71	0.20	3.90 ^c	0.51	2.15 ^a	0.12	1.26
Ioca: Industry-adjusted organizational capital-to-book assets, Eisfeldt and Papanikolaou (2013)																
Ioca	0.53	4.31 ^c	0.67	5.33 ^c	0.50	3.93 ^c	0.72	5.23 ^c	0.14	2.48 ^a	0.18	4.62 ^c	0.59	4.16 ^c	0.18	2.70 ^a
Adm★: Advertising expense-to-market, Chan, Lakonishok, and Sougiannis (2001), 1973/7																
Adm	0.66	2.71 ^a	0.72	3.14 ^b	0.67	2.48 ^a	0.90	3.45 ^c	0.30	2.01 ^a	0.13	2.38 ^a	0.82	2.73 ^a	0.46	3.23 ^b
gAd★: Growth in advertising expense, Lou (2014), 1974/7																
gAd	-0.07	0.38	-0.59	5.02 ^c	-0.13	0.59	-0.66	4.45 ^c	-0.06	0.82	-0.19	4.92 ^c	-0.14	0.66	-0.08	0.95
Rdm★: R&D expense-to-market, Chan, Lakonishok, and Sougiannis (2001), 1976/7																
Rdm	0.70	2.75 ^a	1.56	6.43 ^c	0.79	2.23 ^a	1.77	5.44 ^c	0.37	2.32 ^a	0.44	4.98 ^c	0.47	1.38	0.11	0.69
Rdmq★: Quarterly R&D expense-to-market, 1-, 6-, and 12-month, 1990/1																
Rdmq1	1.12	2.91 ^b	2.36	5.77 ^c	1.78	3.34 ^b	3.24	5.97 ^c	0.90	3.03 ^b	0.98	6.38 ^c	1.12	1.76	0.41	1.09
Rdmq6	0.80	2.18 ^a	2.01	5.38 ^c	1.13	2.42 ^a	2.52	5.21 ^c	0.66	2.59 ^a	0.70	5.03 ^c	0.79	1.21	0.33	0.91
Rdmq12	0.82	2.43 ^a	2.03	5.88 ^c	1.06	2.54 ^a	2.44	5.59 ^c	0.61	2.66 ^a	0.66	5.18 ^c	0.76	1.43	0.25	0.74
Rds★: R&D expense-to-sales, Chan, Lakonishok, and Sougiannis (2001), 1976/7																
Rds	0.09	0.35	0.30	1.09	-0.29	0.64	-0.05	0.13	-0.08	0.60	-0.04	0.56	0.08	0.25	-0.17	1.03

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Rds ^q ★: Quarterly R&D expense-to-sales, 1-, 6-, and 12-month, 1990/1																
Rds ^q 1	0.35	1.21	0.33	0.89	-0.40	0.71	-0.15	0.31	-0.04	0.31	-0.13	1.75	0.46	0.98	-0.07	0.30
Rds ^q 6	0.45	1.71	0.42	1.16	-0.33	0.61	0.01	0.01	-0.04	0.36	-0.10	1.36	0.69	1.66	-0.06	0.23
Rds ^q 12	0.45	1.70	0.44	1.19	-0.25	0.46	0.03	0.05	-0.04	0.27	-0.09	1.21	0.73	1.87	-0.02	0.06
OI: Operating leverage, Novy-Marx (2011)																
OI	0.44	2.63 ^a	0.52	3.05 ^b	0.49	2.77 ^a	0.58	3.26 ^b	0.11	2.30 ^a	0.14	3.54 ^c	0.50	2.65 ^a	0.13	2.36 ^a
OI ^q ★: Quarterly operating leverage, 1-, 6-, and 12-month, 1973/1																
OI ^q 1	0.49	2.60 ^a	0.62	3.37 ^b	0.48	2.39 ^a	0.65	3.41 ^c	0.13	2.53 ^a	0.17	4.22 ^c	0.53	2.42 ^a	0.15	2.50 ^a
OI ^q 6	0.48	2.62 ^a	0.56	3.10 ^b	0.49	2.58 ^a	0.62	3.27 ^b	0.12	2.40 ^a	0.16	3.99 ^c	0.54	2.53 ^a	0.14	2.43 ^a
OI ^q 12	0.48	2.77 ^a	0.52	2.89 ^b	0.48	2.62 ^a	0.58	3.08 ^b	0.12	2.41 ^a	0.15	3.67 ^c	0.55	2.74 ^a	0.14	2.47 ^a
Hn: Hiring rate, Belo, Lin, and Bazzdresch (2014)																
Hn	-0.25	1.63	-0.68	5.84 ^c	-0.19	1.08	-0.85	6.34 ^c	-0.08	1.21	-0.22	5.96 ^c	-0.29	1.80	-0.09	1.09
Rca★: R&D capital-to-book assets, Li (2011), 1980/7																
Rca	0.35	1.49	0.69	2.46 ^a	0.34	0.76	0.80	1.91	0.20	1.18	0.17	1.46	0.31	1.08	0.17	0.78
Bca★: Brand capital-to-book assets, Belo, Lin, and Vitorino (2014), 1973/7																
Bca	0.15	0.65	0.25	1.61	0.19	0.81	0.33	1.94	0.07	0.90	0.05	1.23	0.27	1.01	0.11	1.14
Aop★: Analysts optimism, Frankel and Lee (1998), 1976/7																
Aop	-0.16	0.92	-0.05	0.44	-0.16	0.92	-0.13	1.06	-0.04	0.72	-0.03	0.90	-0.17	0.63	-0.02	0.17
Pafe★: Predicted analysts forecast error, Frankel and Lee (1998), 1985/7																
Pafe	0.25	0.73	0.00	0.00	0.08	0.22	-0.05	0.15	0.07	0.61	-0.02	0.22	0.15	0.25	0.03	0.16
Parc★: Patent-to-R&D capital, Hirschleifer, Hsu, and Li (2013), 1982/7-2008/6																
Parc	0.09	0.38	-0.04	0.34	0.82	2.33 ^a	-0.10	0.68	0.12	1.16	-0.04	1.10	0.09	0.38	0.12	1.16
Crd★: Citations-to-R&D expense, Hirschleifer, Hsu, and Li (2013), 1983/7-2008/6																
Crd	0.16	0.64	0.07	0.61	0.40	1.03	0.22	1.51	0.07	0.53	0.01	0.35	0.16	0.64	0.07	0.53
Hs: Industry concentration in sales, Hou and Robinson (2006)																
Hs	-0.31	2.12 ^a	-0.20	1.53	-0.32	2.18 ^a	-0.17	1.41	-0.04	1.18	-0.04	1.07	-0.37	2.30 ^a	-0.06	1.40
Ha: Industry concentration in total assets, Hou and Robinson (2006)																
Ha	-0.24	1.71	-0.01	0.05	-0.29	1.81	-0.15	0.92	-0.05	1.26	-0.04	0.82	-0.26	1.55	-0.05	1.21

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
He: Industry concentration in book equity, Hou and Robinson (2006)																
He	-0.22	1.58	0.02	0.10	-0.24	1.53	-0.14	0.80	-0.05	1.21	-0.04	0.87	-0.29	1.66	-0.06	1.37
Age: Firm age, 1-, 6-, and 12-month, Jiang, Lee, and Zhang (2005)																
Age1	0.02	0.09	0.07	0.32	-0.02	0.09	0.31	1.43	-0.01	0.23	0.05	0.69	0.07	0.28	0.00	0.06
Age6	0.02	0.13	0.07	0.35	0.03	0.15	0.36	1.70	-0.01	0.20	0.05	0.81	0.07	0.29	0.00	0.04
Age12	0.01	0.07	0.04	0.22	0.05	0.23	0.34	1.61	-0.01	0.24	0.05	0.72	0.05	0.24	0.00	0.08
D1: Price delay based on R^2 , Hou and Moskowitz (2005)																
D1	0.20	0.98	0.40	1.92	0.26	0.96	0.98	3.74 ^c	0.17	0.97	0.24	2.99 ^b	0.18	0.71	0.12	0.60
D2: Price delay based on slopes, Hou and Moskowitz (2005)																
D2	0.23	1.10	0.41	1.88	0.25	0.88	1.04	3.73 ^c	0.21	1.12	0.26	3.09 ^b	0.23	0.88	0.17	0.75
D3: Price delay based on adjusted slopes, Hou and Moskowitz (2005)																
D3	0.24	1.13	0.39	1.79	0.28	0.98	1.03	3.69 ^c	0.20	1.09	0.26	3.08 ^b	0.24	0.92	0.16	0.71
dSi: Percentage change in sales minus percentage change in inventory, Abarbanell and Bushee (1998)																
dSi	0.10	0.74	0.13	2.26 ^a	0.10	0.67	0.13	1.76	0.06	1.63	0.07	3.59 ^c	0.05	0.27	0.13	2.23 ^a
dSa: Percentage change in sales minus percentage change in accounts receivable, Abarbanell and Bushee (1998)																
dSa	0.18	1.47	0.13	2.03 ^a	0.00	0.00	0.12	1.63	0.09	2.43 ^a	0.06	3.24 ^b	0.29	2.01 ^a	0.13	2.38 ^a
dGs: Percentage change in gross margin minus percentage change in sales, Abarbanell and Bushee (1998)																
dGs	0.09	0.70	0.13	1.61	0.08	0.57	0.16	1.86	0.01	0.13	0.03	1.69	0.10	0.52	0.02	0.32
dSs: Percentage change in sales minus percentage change in SG&A, Abarbanell and Bushee (1998)																
dSs	-0.02	0.10	-0.14	1.51	0.09	0.52	-0.23	2.45 ^a	-0.02	0.32	-0.07	3.04 ^b	-0.10	0.54	-0.04	0.51
Etr: Effective tax rate, Abarbanell and Bushee (1998)																
Etr	0.24	2.29 ^a	0.01	0.21	0.19	1.67	0.00	0.04	0.05	1.54	-0.01	0.52	0.20	1.47	0.08	1.74
Lfe: Labor force efficiency, Abarbanell and Bushee (1998)																
Lfe	0.17	1.40	-0.06	1.03	0.24	1.72	-0.10	1.28	-0.04	0.87	-0.06	2.33 ^a	0.12	0.69	-0.06	0.77
Ana [★] : Analysts coverage, 1-, 6-, and 12-month, Elgers, Lo, and Pfeiffer (2001), 1976/2																
Ana1	-0.13	0.82	-0.14	0.77	-0.13	0.81	-0.08	0.44	-0.04	0.97	-0.04	0.70	0.33	1.32	0.09	1.35
Ana6	-0.12	0.72	-0.14	0.79	-0.07	0.46	-0.10	0.59	-0.04	0.86	-0.04	0.68	0.38	1.53	0.09	1.35
Ana12	-0.11	0.66	-0.14	0.78	-0.09	0.56	-0.12	0.68	-0.04	0.85	-0.04	0.68	0.39	1.59	0.09	1.30

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Tan: Tangibility of assets, Hahn and Lee (2009)																
Tan	0.02	0.16	0.49	3.25 ^b	-0.11	0.71	0.46	2.72 ^a	0.01	0.33	0.14	3.13 ^b	-0.12	0.66	-0.03	0.61
Tan4★: Quarterly tangibility of assets, 1-, 6-, and 12-month, 1972/1																
Tan41	0.20	1.08	0.73	4.74 ^c	0.18	0.94	0.75	4.26 ^c	0.07	1.31	0.21	4.50 ^c	0.06	0.25	0.02	0.30
Tan46	0.19	1.14	0.67	4.24 ^c	0.12	0.66	0.69	3.99 ^c	0.06	1.19	0.21	4.37 ^c	-0.01	0.05	0.00	0.06
Tan412	0.12	0.78	0.58	3.82 ^c	0.02	0.14	0.58	3.43 ^c	0.04	0.85	0.18	3.83 ^c	-0.08	0.45	-0.02	0.39
Re★: Real estate ratio, Tuzel (2010), 1970/7																
Re	0.34	2.44 ^a	0.22	2.50 ^a	0.27	1.72	0.26	2.38 ^a	0.07	1.76	0.06	2.34 ^a	0.43	2.58 ^a	0.08	1.70
Kz: The Kaplan-Zingales (1997) index of financing constraints, Lamont, Polk, and Saa-Requejo (2001)																
Kz	-0.11	0.62	-0.20	1.30	-0.14	0.63	-0.18	1.02	0.03	0.54	0.03	1.02	-0.20	0.86	0.01	0.13
Kz4★: Quarterly Kaplan-Zingales index, 1-, 6-, and 12-month, 1977/1																
Kz41	-0.13	0.65	-0.44	2.16 ^a	-0.33	1.41	-0.57	2.44 ^a	-0.02	0.38	-0.02	0.62	-0.24	0.91	-0.06	0.85
Kz46	-0.14	0.75	-0.33	1.75	-0.27	1.18	-0.38	1.73	-0.02	0.34	0.01	0.31	-0.26	0.99	-0.06	0.87
Kz412	-0.13	0.69	-0.30	1.72	-0.25	1.12	-0.34	1.68	-0.04	0.65	0.01	0.19	-0.26	1.01	-0.07	1.05
Ww: The Whited-Wu index of financing constraints, Whited and Wu (2006)																
Ww	0.17	0.70	0.37	1.54	-0.52	1.66	0.54	1.71	0.08	0.91	0.14	1.54	0.24	0.66	0.09	0.66
Ww4★: Quarterly Whited-Wu index of financing constraints, 1-, 6-, and 12-month, 1972/1																
Ww41	0.03	0.10	0.21	0.82	-0.68	1.97 ^a	0.26	0.75	0.06	0.62	0.06	0.56	0.08	0.20	0.04	0.31
Ww46	0.07	0.27	0.26	1.02	-0.60	1.75	0.34	1.00	0.07	0.73	0.08	0.77	0.13	0.34	0.05	0.37
Ww412	0.08	0.29	0.32	1.25	-0.55	1.64	0.40	1.20	0.06	0.67	0.10	0.98	0.14	0.35	0.06	0.41
Sdd★: Secured debt-to-total debt, Valtia (2016), 1982/7																
Sdd	0.08	0.32	0.19	1.57	-0.21	0.90	0.07	0.71	0.02	0.23	0.01	0.30	0.18	0.64	0.04	0.40
Cdd★: Convertible debt-to-total debt, Valtia (2016), 1970/7																
Cdd	0.10	0.41	0.01	0.06	-0.05	0.20	-0.02	0.11	-0.01	0.14	0.04	0.62	0.14	0.43	-0.02	0.19
Vef★: Cash flow volatility, 1-, 6-, and 12-month, Huang (2009), 1978/1																
Vef1	-0.36	1.69	-0.42	1.60	-0.54	1.82	-0.62	1.89	-0.14	1.88	-0.09	1.47	-0.60	2.01 ^a	-0.24	2.30 ^a
Vef6	-0.33	1.63	-0.41	1.60	-0.52	1.79	-0.61	1.88	-0.13	1.74	-0.09	1.47	-0.57	2.01 ^a	-0.23	2.21 ^a
Vef12	-0.29	1.41	-0.38	1.49	-0.47	1.68	-0.56	1.76	-0.12	1.59	-0.08	1.33	-0.49	1.78	-0.22	2.10 ^a

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Cia★: Cash-to-assets, 1-, 6-, and 12-month, Palazzo (2012), 1972/1																
Cia1	0.27	1.36	0.27	1.35	0.32	1.14	0.20	0.82	0.10	1.09	0.05	0.63	0.22	0.99	0.10	0.88
Cia6	0.14	0.69	0.15	0.76	0.18	0.66	0.09	0.34	0.07	0.74	0.02	0.24	0.06	0.28	0.05	0.48
Cia12	0.11	0.58	0.13	0.67	0.18	0.70	0.07	0.29	0.05	0.60	0.01	0.14	0.04	0.20	0.04	0.33
Gind★: Corporate governance, Gompers, Ishii, and Metrick (2003), 1990/9-2006/12																
Gind	0.02	0.06	-0.20	0.81	0.02	0.06	-0.20	0.81	-0.04	0.52	-0.05	0.69	-0.73	2.04 ^a	-0.16	1.92
Acq: Accrual quality, Francis et al. (2005)																
Acq	-0.12	0.60	0.10	0.45	-0.22	0.93	0.03	0.14	-0.01	0.15	0.00	0.01	-0.19	0.63	-0.04	0.40
Acq ^m : Accrual quality, monthly sorts, 1-, 6-, and 12-month																
Acq ^m 1	-0.07	0.34	0.03	0.14	-0.19	0.85	0.01	0.03	-0.03	0.41	-0.01	0.07	-0.19	0.67	-0.06	0.63
Acq ^m 6	-0.04	0.19	0.07	0.33	-0.16	0.73	0.03	0.14	-0.01	0.17	0.00	0.05	-0.18	0.64	-0.05	0.49
Acq ^m 12	-0.01	0.04	0.11	0.52	-0.12	0.54	0.08	0.34	0.00	0.03	0.01	0.17	-0.14	0.50	-0.04	0.35
Ob★: Order backlog, Rajgopal, Shevlin, and Venkatachalam (2003), 1971/7																
Ob	0.18	0.77	0.03	0.18	0.19	0.76	0.03	0.19	0.02	0.31	-0.02	0.50	0.08	0.23	-0.05	0.53
Eper: Earnings persistence, Francis et al. (2004)																
Eper	-0.02	0.11	-0.21	1.89	-0.07	0.53	-0.19	1.74	-0.01	0.15	-0.09	2.60 ^a	-0.05	0.23	0.01	0.10
Eprd: Earnings predictability, Francis et al. (2004)																
Eprd	-0.53	2.96 ^b	-0.75	4.05 ^c	-0.51	2.79 ^b	-0.75	3.98 ^c	-0.19	2.75 ^a	-0.12	2.98 ^b	-0.61	2.50 ^a	-0.22	2.21 ^a
Esm: Earnings smoothness, Francis et al. (2004)																
Esm	-0.04	0.31	0.07	0.51	-0.04	0.31	0.09	0.66	0.05	1.30	0.03	0.68	-0.06	0.31	0.06	1.07
Evr: Value relevance of earnings, Francis et al. (2004)																
Evr	0.18	1.31	0.07	0.87	0.25	1.78	0.04	0.52	0.05	1.48	0.02	0.77	0.18	0.86	0.03	0.56
Etl: Earnings timeliness, Francis et al. (2004)																
Etl	0.34	2.79 ^b	0.19	2.17 ^a	0.35	2.74 ^a	0.14	1.63	0.06	2.03 ^a	0.04	1.76	0.32	1.77	0.05	1.13
Ecs: Earnings conservatism, Francis et al. (2004)																
Ecs	0.03	0.25	0.08	1.20	0.05	0.49	0.07	1.06	-0.01	0.41	0.02	1.41	0.09	0.63	-0.02	0.55
Frm★: Pension funding rate scaled by market equity, Franzoni and Marin (2006), 1981/7																
Frm	0.06	0.30	0.01	0.05	0.11	0.55	0.02	0.13	0.01	0.26	0.01	0.22	0.31	1.23	0.12	1.82

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		AII-VW		AII-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Fra★: Pension funding rate scaled by book assets, Franzoni and Marin (2006), 1981/7																
Fra	-0.14	0.96	0.05	0.36	-0.11	0.82	0.04	0.31	-0.02	0.57	0.02	0.57	-0.13	0.64	-0.01	0.12
Ala: Liquidity of book assets, Ortiz-Molina and Phillips (2014)																
Ala	-0.08	0.42	-0.39	3.15 ^b	-0.13	0.57	-0.66	4.48 ^c	-0.10	1.38	-0.24	5.60 ^c	-0.10	0.35	-0.13	1.02
Ala★: Quarterly liquidity of book assets, 1-, 6-, and 12-month, 1976/1																
Ala★1	0.40	1.67	0.37	2.17 ^a	0.46	1.60	0.29	1.38	0.08	0.96	0.06	1.07	0.36	0.98	0.11	0.85
Ala★6	0.26	1.11	0.24	1.38	0.36	1.29	0.06	0.30	0.04	0.56	-0.01	0.10	0.16	0.43	0.06	0.47
Ala★12	0.19	0.81	0.09	0.51	0.22	0.83	-0.14	0.70	0.01	0.10	-0.06	1.00	0.14	0.40	0.02	0.15
Alm: Liquidity of market assets, Ortiz-Molina and Phillips (2014)																
Alm	0.13	0.71	0.34	2.46 ^a	-0.07	0.34	0.35	2.26 ^a	0.06	0.84	0.08	2.02 ^a	-0.05	0.20	0.02	0.16
Alm★: Quarterly liquidity of market assets, 1-, 6-, and 12-month, 1976/1																
Alm★1	0.58	2.75 ^a	1.21	6.34 ^c	0.65	2.61 ^a	1.52	6.99 ^c	0.17	1.72	0.42	6.88 ^c	0.56	1.89	0.16	1.07
Alm★6	0.60	3.05 ^b	1.18	6.47 ^c	0.65	3.00 ^b	1.42	6.93 ^c	0.20	2.09 ^a	0.40	6.99 ^c	0.59	2.17 ^a	0.21	1.43
Alm★12	0.54	2.84 ^b	1.04	5.91 ^c	0.50	2.44 ^a	1.24	6.31 ^c	0.18	1.93	0.34	6.20 ^c	0.47	1.77	0.18	1.29
Dis★: Disparity between long- and short-term earnings growth forecasts, 1-, 6-, and 12-month, Da and Warachka (2011), 1982/1																
Dis1	-0.29	1.48	-0.57	3.64 ^c	-0.48	2.15 ^a	-0.64	3.39 ^b	-0.08	1.25	-0.06	1.55	-0.19	0.81	-0.09	1.29
Dis6	-0.01	0.07	-0.17	1.22	-0.21	1.10	-0.22	1.24	-0.01	0.24	0.02	0.43	0.05	0.27	-0.01	0.16
Dis12	0.06	0.46	-0.10	0.83	-0.06	0.38	-0.13	0.87	0.00	0.06	0.02	0.82	0.08	0.44	-0.01	0.13
Dis★: Dispersion of analysts' earnings forecasts, 1-, 6-, and 12-month, Diether, Malloy, and Scherbina (2002), 1976/2																
Dis1	-0.19	0.72	0.64	3.04 ^b	-0.23	0.86	-0.69	3.14 ^b	-0.06	0.86	-0.16	3.93 ^c	-0.62	1.47	-0.15	1.38
Dis6	-0.18	0.71	-0.44	2.15 ^a	-0.14	0.58	-0.49	2.31 ^a	-0.02	0.37	-0.10	2.51 ^a	-0.60	1.47	-0.11	1.14
Dis12	-0.08	0.32	-0.27	1.39	-0.03	0.11	-0.32	1.58	0.00	0.01	-0.06	1.39	-0.47	1.22	-0.08	0.80
Dig★: Dispersion in analysts' long-term growth forecasts, 1-, 6-, and 12-month, Anderson, Ghysels, and Juergens (2005), 1982/1																
Dig1	-0.11	0.45	-0.16	0.81	0.02	0.06	-0.19	0.86	-0.02	0.17	-0.07	0.91	-0.26	0.67	-0.04	0.23
Dig6	-0.08	0.32	-0.13	0.71	0.03	0.13	-0.14	0.69	0.00	0.01	-0.06	0.80	-0.26	0.67	-0.03	0.17
Dig12	-0.06	0.26	-0.11	0.62	0.04	0.15	-0.12	0.58	0.02	0.18	-0.04	0.60	-0.21	0.57	-0.01	0.08
R ₁ ¹ : Year 1-lagged return, annual, Heston and Sadka (2008)																
R ₁ ¹	0.67	3.43 ^c	0.70	5.46 ^c	0.72	3.34 ^b	0.81	5.36 ^c	0.26	2.92 ^b	0.24	5.06 ^c	0.93	3.90 ^c	0.38	3.49 ^c

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $
R_1 : Year 1-lagged return, nonannual, Heston and Sadka (2008)																
$R_1^{[2,5]}$	0.54	1.76	-0.01	0.05	0.76	2.04 ^a	-0.37	1.11	0.24	1.90	-0.05	0.51	0.77	2.26 ^a	0.34	2.19 ^a
$R_a^{[2,5]}$: Years 2–5 lagged returns, annual, Heston and Sadka (2008)																
$R_a^{[2,5]}$	0.69	4.11 ^c	0.69	6.07 ^c	0.80	3.66 ^c	0.87	6.19 ^c	0.21	3.23 ^b	0.25	5.93 ^c	0.97	4.98 ^c	0.32	4.16 ^c
$R_n^{[2,5]}$: Years 2–5 lagged returns, nonannual, Heston and Sadka (2008)																
$R_n^{[2,5]}$	-0.50	2.22	-1.08	5.70 ^c	-0.74	2.70 ^a	-1.35	6.07 ^c	-0.17	1.81	-0.35	5.57 ^c	-0.81	2.92 ^b	-0.28	2.36 ^a
$R_a^{[6,10]}$: Years 6–10 lagged returns, annual, Heston and Sadka (2008)																
$R_a^{[6,10]}$	0.83	5.06 ^c	0.76	6.96 ^c	0.90	4.59 ^c	0.82	6.22 ^c	0.30	4.91 ^c	0.26	6.78 ^c	0.94	4.76 ^c	0.33	4.35 ^c
$R_n^{[6,10]}$: Years 6–10 lagged returns, nonannual, Heston and Sadka (2008)																
$R_n^{[6,10]}$	-0.46	2.38 ^a	-0.57	4.70 ^c	-0.50	2.36 ^a	-0.66	4.72 ^c	-0.14	1.97 ^a	-0.20	4.72 ^c	-0.41	1.65	-0.15	1.55
$R_1^{[11,15]}$: Years 11–15 lagged returns, annual, Heston and Sadka (2008)																
$R_1^{[11,15]}$	0.62	4.46 ^c	0.61	6.36 ^c	0.72	4.46 ^c	0.69	6.25 ^c	0.19	3.94 ^c	0.21	6.74 ^c	0.84	4.96 ^c	0.23	3.96 ^c
$R_a^{[11,15]}$: Years 11–15 lagged returns, nonannual, Heston and Sadka (2008)																
$R_a^{[11,15]}$	-0.30	1.88	-0.19	1.55	-0.14	0.80	-0.21	1.58	-0.08	1.61	-0.06	1.42	-0.29	1.46	-0.09	1.44
$R_n^{[16,20]}$: Years 16–20 lagged returns, annual, Heston and Sadka (2008)																
$R_n^{[16,20]}$	0.54	3.26 ^b	0.51	4.66 ^c	0.65	3.56 ^c	0.59	5.10 ^c	0.14	2.54 ^a	0.18	5.57 ^c	0.67	3.35 ^b	0.18	2.62 ^a
$R_a^{[16,20]}$: Years 16–20 lagged returns, nonannual, Heston and Sadka (2008)																
$R_a^{[16,20]}$	-0.26	1.60	-0.29	2.59 ^a	-0.28	1.56	-0.33	2.77 ^a	-0.04	1.00	-0.08	2.57 ^a	-0.27	1.42	-0.02	0.46

F. Trading frictions (106 anomalies)

Me: Size (the market equity), Banz (1981)																
Me	-0.25	1.02	-0.46	1.83	-0.31	1.06	-1.14	3.91 ^c	-0.11	1.32	-0.23	2.59 ^a	-0.26	0.41	-0.05	0.25
Ivff: Idiosyncratic volatility from the Fama-French 3-factor model, 1-, 6-, and 12-month, Ang et al. (2006)																
Ivff1	-0.52	1.71	-0.07	0.22	-1.22	3.38 ^b	-0.31	0.89	-0.18	1.02	-0.15	1.43	-0.72	2.04 ^a	-0.26	1.28
Ivff6	-0.32	1.12	-0.10	0.34	-0.81	2.50 ^a	-0.10	0.28	-0.12	0.70	-0.05	0.46	-0.43	1.28	-0.17	0.87
Ivff12	-0.18	0.67	-0.03	0.10	-0.53	1.69	0.08	0.24	-0.05	0.31	0.02	0.14	-0.25	0.75	-0.08	0.41

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Iv: Idiosyncratic volatility, Ali, Hwang, and Trombley (2003)																
Iv	-0.25	0.77	0.09	0.26	-0.83	2.07 ^a	0.23	0.56	-0.03	0.14	0.06	0.50	-0.14	0.31	0.06	0.21
Ivc: Idiosyncratic volatility from the CAPM, 1-, 6-, and 12-month																
Ivc1	-0.48	1.52	-0.08	0.27	-1.15	3.19 ^b	-0.33	0.94	-0.17	1.00	-0.16	1.51	-0.69	1.89	-0.26	1.27
Ivc6	-0.31	1.07	-0.10	0.34	-0.80	2.41 ^a	-0.10	0.28	-0.12	0.72	-0.05	0.49	-0.42	1.24	-0.18	0.90
Ivc12	-0.20	0.72	-0.03	0.11	-0.56	1.75	0.08	0.24	-0.06	0.36	0.01	0.11	-0.28	0.82	-0.09	0.44
Ivq★: Idiosyncratic volatility from the q-factor model, 1-, 6-, and 12-month, 1967/2																
Ivq1	-0.48	1.59	-0.09	0.31	-1.14	3.19 ^b	-0.33	0.95	-0.17	1.02	-0.16	1.52	-0.72	1.98 ^a	-0.27	1.38
Ivq6	-0.31	1.10	-0.12	0.41	-0.80	2.47 ^a	-0.11	0.31	-0.12	0.73	-0.06	0.54	-0.42	1.28	-0.18	0.93
Ivq12	-0.20	0.75	-0.05	0.18	-0.53	1.72	0.07	0.21	-0.06	0.36	0.01	0.07	-0.29	0.88	-0.10	0.50
Tv: Total volatility, 1-, 6-, and 12-month, Ang et al. (2006)																
Tv1	-0.39	1.18	-0.08	0.26	-1.18	3.02 ^b	-0.33	0.92	-0.14	0.78	-0.16	1.48	-0.55	1.43	-0.20	0.96
Tv6	-0.24	0.77	-0.10	0.33	-0.79	2.16 ^a	-0.10	0.28	-0.11	0.62	-0.06	0.54	-0.33	0.88	-0.15	0.73
Tv12	-0.20	0.65	-0.04	0.15	-0.60	1.72	0.07	0.19	-0.07	0.38	0.00	0.03	-0.24	0.66	-0.08	0.40
Sv★: Systematic volatility, 1-, 6-, and 12-month, Ang et al. (2006), 1986/2																
Sv1	-0.49	2.24 ^a	-0.27	2.32 ^a	-0.42	1.56	-0.33	2.13 ^a	-0.25	2.32 ^a	-0.10	2.25 ^a	-1.06	3.57 ^c	-0.52	3.50 ^c
Sv6	-0.18	1.27	-0.07	1.24	-0.21	1.25	-0.08	1.11	-0.09	1.42	-0.03	1.18	-0.30	1.53	-0.19	2.19 ^a
Sv12	-0.14	1.22	-0.04	0.75	-0.17	1.25	-0.05	0.86	-0.07	1.18	-0.02	0.95	-0.28	1.77	-0.17	2.36 ^a
β: The market beta, 1-, 6-, and 12-month, Fama and MacBeth (1973)																
β1	0.02	0.06	-0.07	0.26	0.00	0.00	-0.11	0.39	-0.01	0.10	-0.06	0.62	2.34	1.85	0.79	1.71
β6	0.02	0.05	-0.03	0.13	-0.04	0.10	-0.09	0.29	0.00	0.03	-0.05	0.51	1.95	1.57	0.77	1.72
β12	-0.02	0.08	-0.09	0.34	-0.09	0.24	-0.13	0.46	0.00	0.00	-0.06	0.63	1.82	1.43	0.80	1.71
βFP: The Frazzini-Pedersen beta, 1-, 6-, and 12-month, Frazzini and Pedersen (2014)																
βFP1	-0.22	0.66	-0.47	1.53	-0.31	0.88	-0.50	1.52	-0.10	0.85	-0.15	1.51	-0.17	0.48	-0.10	0.76
βFP6	-0.23	0.72	-0.40	1.34	-0.27	0.80	-0.40	1.27	-0.09	0.78	-0.12	1.24	-0.19	0.58	-0.09	0.74
βFP12	-0.18	0.57	-0.34	1.18	-0.22	0.67	-0.36	1.16	-0.07	0.65	-0.10	1.09	-0.14	0.43	-0.08	0.63
βD: The Dimson beta, 1-, 6-, and 12-month, Dimson (1979)																
βD1	0.15	0.71	-0.17	1.21	0.04	0.18	-0.27	1.80	0.01	0.12	-0.08	1.74	0.07	0.17	-0.02	0.13
βD6	0.07	0.39	-0.11	0.98	0.02	0.14	-0.20	1.71	0.05	0.54	-0.05	1.25	-0.22	0.79	-0.08	0.73

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
β_{D12}	0.01	0.06	-0.16	1.59	-0.05	0.33	-0.24	2.32 ^a	0.01	0.17	-0.06	1.78	-0.24	0.96	-0.10	1.00
Tur: Share turnover, 1-, 6-, and 12-month, Datar, Naik, and Radcliffe (1998)																
Tur1	-0.15	0.61	-0.91	4.22 ^c	-0.15	0.54	-0.86	3.53 ^c	-0.04	0.42	-0.30	4.72 ^c	-0.32	0.99	-0.12	1.08
Tur6	-0.16	0.62	-0.99	4.71 ^c	-0.18	0.69	-0.98	4.19 ^c	-0.03	0.31	-0.31	5.04 ^c	-0.27	0.85	-0.11	1.01
Tur12	-0.11	0.46	-0.96	4.78 ^c	-0.17	0.65	-0.97	4.26 ^c	-0.03	0.31	-0.31	5.15 ^c	-0.24	0.76	-0.11	1.01
Cvt: Coefficient of variation for share turnover, 1-, 6-, and 12-month, Chordia, Subrahmanyam, and Anshuman (2001)																
Cvt1	0.12	0.82	0.33	1.81	0.04	0.25	0.40	2.47 ^a	0.06	0.86	0.13	3.03 ^b	0.14	0.89	0.07	0.94
Cvt6	0.09	0.64	0.39	2.18 ^a	0.09	0.58	0.46	2.86 ^b	0.08	1.05	0.15	3.55 ^c	0.16	1.06	0.08	1.08
Cvt12	0.15	1.10	0.43	2.44 ^a	0.12	0.89	0.50	3.26 ^b	0.09	1.19	0.15	3.82 ^c	0.26	1.73	0.11	1.66
Div: Dollar trading volume, 1-, 6-, and 12-month, Brennan, Chordia, and Subrahmanyam (1998)																
Div1	-0.25	1.37	-0.57	2.62 ^a	-0.26	1.27	-1.16	4.75 ^c	-0.03	1.63	-0.11	2.47 ^a	-0.20	0.89	-0.03	1.42
Div6	-0.34	1.92	-0.61	2.86 ^b	-0.34	1.71	-1.17	4.95 ^c	-0.03	1.68	-0.11	2.57 ^a	-0.29	1.30	-0.04	1.47
Div12	-0.40	2.23 ^a	-0.65	3.09 ^b	-0.37	1.95	-1.18	5.09 ^c	-0.04	1.83	-0.12	2.80 ^b	-0.36	1.66	-0.04	1.62
Cvd: Coefficient of variation for dollar trading volume, 1-, 6-, and 12-month, Chordia, Subrahmanyam, and Anshuman (2001)																
Cvd1	0.08	0.57	0.30	1.58	0.03	0.20	0.32	1.77	0.08	1.02	0.09	2.07 ^a	0.15	0.94	0.07	0.96
Cvd6	0.11	0.75	0.38	1.98 ^a	0.09	0.61	0.43	2.48 ^a	0.09	1.11	0.13	2.98 ^b	0.17	1.08	0.08	1.08
Cvd12	0.15	1.10	0.42	2.23 ^a	0.14	0.99	0.48	2.87 ^b	0.09	1.16	0.14	3.30 ^b	0.27	1.79	0.12	1.67
Pps: Price per share, 1-, 6-, and 12-month, Miller and Scholes (1982)																
Pps1	-0.02	0.07	-0.28	0.99	-0.27	0.60	-1.36	3.45 ^c	-0.02	0.35	-0.09	1.07	-0.69	1.12	-0.07	0.87
Pps6	0.05	0.16	-0.21	0.76	0.22	0.53	-0.93	2.45 ^a	0.00	0.10	-0.07	0.85	-0.44	0.75	-0.06	0.73
Pps12	-0.04	0.14	-0.30	1.14	0.18	0.46	-0.94	2.59 ^a	-0.01	0.32	-0.10	1.25	-0.48	0.84	-0.06	0.78
Ami: Absolute return-to-volume, 1-, 6-, and 12-month, Amihud (2002)																
Ami1	0.25	1.20	0.44	1.81	0.09	0.33	1.01	3.49 ^c	0.11	0.81	0.33	4.97 ^c	0.18	0.69	0.09	0.45
Ami6	0.34	1.64	0.49	2.07 ^a	0.28	1.07	1.13	3.90 ^c	0.20	1.47	0.36	5.43 ^c	0.27	1.06	0.17	0.90
Ami12	0.39	1.91	0.54	2.29 ^a	0.41	1.56	1.16	4.09 ^c	0.26	1.99 ^a	0.37	5.68 ^c	0.34	1.33	0.26	1.37
Lm ¹ : Prior 1-month turnover-adjusted number of zero daily trading volume, 1-, 6-, and 12-month, Liu (2006)																
Lm ¹ 1	-0.07	0.32	0.44	2.33 ^a	0.04	0.17	0.47	2.08 ^a	-0.03	0.66	0.06	1.33	-0.07	0.27	-0.05	0.80
Lm ¹ 6	0.21	0.96	0.85	4.90 ^c	0.31	1.29	1.00	4.67 ^c	0.08	1.59	0.18	4.14 ^c	0.32	1.28	0.08	1.23
Lm ¹ 12	0.21	0.99	0.89	5.30 ^c	0.34	1.46	1.05	5.09 ^c	0.10	2.00 ^a	0.20	4.79 ^c	0.31	1.29	0.11	1.64

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Lm ⁶ : Prior 6-month turnover-adjusted number of zero daily trading volume, 1-, 6-, and 12-month, Liu (2006)																
Lm ⁶ 1	0.38	1.85	1.00	5.55 ^c	0.33	1.35	1.19	5.15 ^c	0.09	1.63	0.19	4.26 ^c	0.49	2.09 ^a	0.09	1.28
Lm ⁶ 6	0.36	1.74	1.02	5.77 ^c	0.42	1.74	1.24	5.56 ^c	0.12	2.22 ^a	0.23	5.10 ^c	0.49	2.08 ^a	0.13	1.84
Lm ⁶ 12	0.31	1.48	1.00	5.92 ^c	0.38	1.66	1.17	5.50 ^c	0.13	2.34 ^a	0.22	5.18 ^c	0.43	1.84	0.14	2.01 ^a
Lm ¹² : Prior 12-month turnover-adjusted number of zero daily trading volume, 1-, 6-, and 12-month, Liu (2006)																
Lm ¹² 1	0.39	1.88	1.00	5.52 ^c	0.44	1.86	1.21	5.30 ^c	0.11	2.00 ^a	0.22	4.92 ^c	0.49	2.03 ^a	0.12	1.64
Lm ¹² 6	0.34	1.65	0.99	5.68 ^c	0.39	1.67	1.17	5.37 ^c	0.12	2.29 ^a	0.23	5.24 ^c	0.45	1.90	0.14	1.94
Lm ¹² 12	0.25	1.19	0.94	5.59 ^c	0.31	1.35	1.10	5.22 ^c	0.12	2.30 ^a	0.22	5.09 ^c	0.39	1.60	0.14	2.01 ^a
Mdr: Maximum daily return, 1-, 6-, and 12-month, Bali, Cakici, and Whitelaw (2011)																
Mdr1	-0.36	1.27	-0.44	1.63	-0.80	2.40 ^a	-0.77	2.48 ^a	-0.14	0.87	-0.30	3.34 ^b	-0.39	1.19	-0.16	0.87
Mdr6	-0.17	0.65	-0.15	0.58	-0.45	1.47	-0.22	0.74	-0.09	0.47	-0.07	0.80	-0.22	0.69	-0.12	0.66
Mdr12	-0.07	0.27	-0.05	0.18	-0.28	0.94	-0.03	0.10	-0.03	0.18	0.00	0.03	-0.10	0.33	-0.05	0.25
Ts: Total skewness, 1-, 6-, and 12-month, Bali, Engle, and Murray (2016)																
Ts1	0.20	1.90	-0.52	4.80 ^c	0.31	2.59 ^a	-0.46	4.06 ^c	0.09	2.53 ^a	-0.12	4.03 ^c	0.22	1.93	0.10	2.60 ^a
Ts6	0.03	0.45	-0.05	0.69	0.07	1.05	-0.04	0.52	0.00	0.05	-0.02	0.75	0.02	0.28	-0.01	0.26
Ts12	0.03	0.53	0.00	0.04	0.07	1.32	0.01	0.19	0.00	0.29	0.00	0.13	0.02	0.43	0.00	0.20
Isc: Idiosyncratic skewness from the CAPM, 1-, 6-, and 12-month																
Isc1	0.15	1.55	-0.44	4.23 ^c	0.35	2.91 ^b	-0.38	3.57 ^c	0.09	3.09 ^b	-0.11	3.87 ^c	0.17	1.57	0.10	3.16 ^b
Isc6	-0.02	0.43	-0.02	0.30	0.04	0.62	-0.03	0.39	0.01	0.58	-0.01	0.52	-0.03	0.45	0.01	0.51
Isc12	0.05	0.93	0.03	0.51	0.09	1.65	0.02	0.31	0.02	1.54	0.00	0.21	0.05	0.92	0.03	1.52
Isf1: Idiosyncratic skewness from the Fama-French 3-factor model, 1-, 6-, and 12-month, Bali, Engle, and Murray (2016)																
Isf1	0.28	3.11 ^b	-0.29	3.03 ^b	0.33	3.08 ^b	-0.23	2.38 ^a	0.09	3.05 ^b	-0.07	2.85 ^b	0.30	3.08 ^b	0.10	3.22 ^b
Isf6	0.05	0.92	-0.01	0.10	0.07	1.31	-0.02	0.22	0.01	0.94	0.00	0.18	0.05	0.83	0.02	0.92
Isf12	0.08	1.72	0.03	0.52	0.11	2.05 ^a	0.03	0.45	0.03	1.74	0.01	0.38	0.08	1.57	0.03	1.66
Isq1★: Idiosyncratic skewness from the q-factor model, 1-, 6-, and 12-month, 1967/72																
Isq1	0.25	2.80 ^b	-0.28	3.12 ^b	0.33	3.32 ^b	-0.23	2.52 ^a	0.07	2.46 ^a	-0.07	2.99 ^b	0.28	2.94 ^b	0.08	2.65 ^a
Isq6	0.07	1.23	-0.01	0.16	0.11	1.82	-0.01	0.10	0.02	1.45	-0.01	0.30	0.07	1.09	0.02	1.41
Isq12	0.09	1.77	0.03	0.50	0.11	2.27 ^a	0.03	0.45	0.03	2.04 ^a	0.01	0.32	0.09	1.74	0.03	1.97 ^a

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $	$\bar{R}$	$ t $
Cs: Cskewness, 1-, 6-, and 12-month, Harvey and Siddique (2000)																
Cs1	-0.11	0.93	-0.01	0.15	-0.18	1.46	-0.03	0.44	-0.01	0.36	0.01	0.65	0.08	0.54	0.04	0.99
Cs6	-0.02	0.32	0.01	0.35	-0.05	0.74	0.01	0.36	0.00	0.25	0.00	0.16	-0.05	0.79	-0.01	0.80
Cs12	-0.02	0.34	0.01	0.39	-0.04	0.78	0.01	0.44	-0.01	0.46	0.00	0.23	-0.08	1.49	-0.03	1.88
Tail: Tail risk, 1-, 6-, and 12-month, Kelly and Jiang (2014)																
Tail1	0.11	0.54	0.22	1.58	0.12	0.51	0.21	1.20	0.09	1.16	0.07	1.29	0.11	0.50	0.10	1.09
Tail6	0.14	0.79	0.28	2.34 ^a	0.10	0.47	0.30	2.01 ^a	0.07	0.96	0.09	1.79	0.14	0.69	0.08	0.91
Tail12	0.17	1.05	0.30	2.78 ^b	0.18	0.94	0.33	2.40 ^a	0.09	1.25	0.10	2.17 ^a	0.19	1.04	0.10	1.20
β ret: The Acharya-Pedersen liquidity beta (return-return), 1-, 6-, and 12-month, Acharya and Pedersen (2005)																
β ret1	0.00	0.01	-0.07	0.29	-0.04	0.12	-0.15	0.50	-0.02	0.12	-0.07	0.75	0.19	0.51	0.05	0.39
β ret6	0.00	0.01	-0.07	0.28	-0.07	0.19	-0.13	0.46	-0.01	0.05	-0.06	0.67	0.22	0.59	0.07	0.52
β ret12	-0.03	0.09	-0.11	0.46	-0.12	0.34	-0.17	0.62	0.00	0.04	-0.07	0.77	0.20	0.55	0.08	0.58
β lec: The Acharya-Pedersen liquidity beta (illiquidity-illiquidity), 1-, 6-, and 12-month, Acharya and Pedersen (2005)																
β lec1	0.31	1.48	0.26	1.80	0.31	1.40	0.26	1.48	0.11	0.93	0.09	1.61	0.25	0.94	0.12	0.77
β lec6	0.30	1.42	0.25	1.75	0.34	1.62	0.26	1.49	0.12	1.01	0.09	1.59	0.27	1.03	0.13	0.83
β lec12	0.29	1.46	0.21	1.55	0.31	1.48	0.24	1.42	0.12	1.01	0.08	1.45	0.29	1.15	0.13	0.89
β lec: The Acharya-Pedersen liquidity beta (return-illiquidity), 1-, 6-, and 12-month, Acharya and Pedersen (2005)																
β lec1	0.07	0.24	-0.04	0.20	0.08	0.24	0.00	0.01	0.01	0.05	0.01	0.13	0.07	0.23	-0.02	0.16
β lec6	0.05	0.18	-0.10	0.50	0.11	0.35	-0.03	0.14	0.00	0.04	-0.02	0.19	0.10	0.31	0.00	0.02
β lec12	0.07	0.28	-0.04	0.22	0.17	0.55	0.01	0.06	0.01	0.06	-0.01	0.10	0.08	0.24	0.00	0.00
β ler: The Acharya-Pedersen liquidity beta (illiquidity-return), 1-, 6-, and 12-month, Acharya and Pedersen (2005)																
β ler1	0.06	0.49	0.05	0.62	0.16	1.15	0.05	0.45	0.00	0.04	0.00	0.07	0.06	0.37	0.02	0.21
β ler6	-0.02	0.13	0.00	0.05	0.20	1.52	0.02	0.20	-0.01	0.06	-0.01	0.23	0.03	0.23	0.02	0.23
β ler12	-0.04	0.32	-0.02	0.30	0.25	2.11 ^a	0.03	0.28	0.00	0.04	0.00	0.08	-0.02	0.13	0.02	0.19
β net: The Acharya-Pedersen net liquidity beta, 1-, 6-, and 12-month, Acharya and Pedersen (2005)																
β net1	0.09	0.27	0.04	0.18	0.14	0.39	0.09	0.33	0.01	0.04	0.01	0.09	0.34	0.93	0.10	0.38
β net6	0.11	0.33	0.03	0.15	0.02	0.04	0.02	0.06	0.01	0.06	0.00	0.05	0.39	1.06	0.11	0.43
β net12	0.06	0.20	0.00	0.00	-0.09	0.27	-0.04	0.17	0.01	0.05	-0.01	0.07	0.34	0.95	0.11	0.45

(continued)

Table 3
Continued

	NYSE-VW		NYSE-EW		All-VW		All-EW		FM-WLS		FM-OLS		NYSE-VW-SS		FM-WLS-SS	
	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t	$\bar{R}$	t
Srev: Short-term reversal, Jegadeesh (1990)	-0.27	1.40	-1.86	8.90 ^c	-0.36	1.42	-2.57	9.22 ^c	-0.14	1.67	-0.63	8.24 ^c	-0.65	2.39 ^d	-0.27	2.61 ^a
β^- : Downside beta, 1-, 6-, and 12-month, Ang, Chen, and Xing (2006)																
β^-1	-0.15	0.51	-0.72	2.86 ^b	-0.42	1.28	-0.82	2.94 ^b	-0.08	0.69	-0.24	2.83 ^b	-0.09	0.29	-0.06	0.46
β^-6	-0.19	0.66	-0.83	3.45 ^c	-0.33	1.02	-0.96	3.61 ^c	-0.10	0.83	-0.27	3.34 ^b	-0.11	0.36	-0.08	0.63
β^-12	-0.13	0.47	-0.70	3.01 ^b	-0.16	0.55	-0.80	3.15 ^b	-0.08	0.68	-0.23	2.95 ^b	-0.04	0.15	-0.06	0.48
Shl: The high-low bid-ask spread, 1-, 6-, and 12-month, Corwin and Schultz (2012)																
Shl1	-0.16	0.55	0.02	0.09	-0.43	1.13	0.53	1.54	-0.06	0.22	0.20	2.04 ^a	-0.15	0.48	-0.06	0.19
Shl6	-0.16	0.61	0.05	0.20	-0.43	1.26	0.54	1.65	-0.12	0.45	0.20	2.12 ^a	-0.20	0.68	-0.14	0.46
Shl12	-0.13	0.50	0.13	0.53	-0.27	0.84	0.66	2.09 ^a	-0.03	0.14	0.22	2.45 ^a	-0.14	0.52	-0.05	0.17
Sba★: The bid-ask spread, 1-, 6-, and 12-month, Hou and Loh (2016), 1984/2–2015/1, missing 1986/2–1987/1																
Sba1	-0.18	0.68	0.08	0.27	-0.67	1.94	0.49	1.24	-0.05	0.19	0.16	1.47	-0.23	0.80	-0.07	0.24
Sba6	-0.07	0.28	0.09	0.31	-0.41	1.28	0.56	1.51	0.04	0.14	0.19	1.78	-0.10	0.36	0.03	0.12
Sba12	-0.01	0.04	0.15	0.53	-0.26	0.84	0.62	1.71	0.10	0.39	0.21	1.99 ^a	-0.03	0.12	0.10	0.35
$\beta^{\text{lev}} \star$: The financial intermediary leverage beta, 1-, 6-, and 12-month, Adrian, Etula, and Muir (2014), 1973/1																
$\beta^{\text{lev}}1$	0.39	1.90	0.38	2.37 ^a	0.35	1.48	0.40	2.29 ^a	0.12	1.42	0.13	2.32 ^a	0.43	1.79	0.13	1.35
$\beta^{\text{lev}}6$	0.26	1.31	0.28	1.95	0.27	1.17	0.28	1.71	0.08	1.01	0.10	1.92	0.29	1.27	0.09	1.00
$\beta^{\text{lev}}12$	0.25	1.30	0.23	1.65	0.28	1.27	0.22	1.44	0.07	0.91	0.08	1.69	0.28	1.27	0.08	0.89
β^{PS} : The Pastor-Stambaugh liquidity beta, 1-, 6-, and 12-month																
$\beta^{\text{PS}}1$	0.08	0.47	0.02	0.18	0.02	0.10	0.03	0.26	0.05	0.77	0.01	0.42	0.16	0.98	0.04	0.53
$\beta^{\text{PS}}6$	0.11	0.74	0.02	0.23	0.06	0.34	0.04	0.41	0.04	0.70	0.02	0.67	0.10	0.60	0.02	0.21
$\beta^{\text{PS}}12$	0.17	1.24	0.04	0.47	0.08	0.50	0.05	0.48	0.06	1.00	0.03	0.99	0.17	1.17	0.04	0.58
Pin★: Probability of information-based trading, Easley, Hvidkjaer, and O'Hara (2002), 1984/1–2002/12																
Pin	-0.23	0.91	0.21	0.71	-0.24	0.92	0.40	1.35	0.01	0.04	0.10	1.10	-0.23	0.91	0.01	0.04

For portfolio sorts, "NYSE-VW" and "NYSE-EW" denote NYSE breakpoints with value- and equal-weighted returns, and "All-VW" and "All-EW" NYSE-Amex-NASDAQ breakpoints with value- and equal-weighted returns, respectively. "NYSE-VW-SS" denote NYSE breakpoints and value-weighted returns in the shorter samples in the original studies. For each procedure, we show the average return, $\bar{R}$, and the absolute t -value, $|t|$, for the high-minus-low decile formed on each of the 452 anomaly variables. For univariate cross-sectional regressions, "FM-WLS" and "FM-OLS" denote NYSE breakpoints with value- and equal-weighted returns, and "FM-WLS-SS" weighted least squares in the shorter, original samples. We winsorize the regressors at the 1%-99% level each month and then standardize them before performing cross-sectional regressions. Standardizing a variable means subtracting its cross-sectional mean and then dividing by its cross-sectional standard deviation. For each anomaly variable, we report the slope and its absolute t -value. ^a, ^b, and ^c indicate absolute t -values exceeding the thresholds of 1.96, 2.78, and 3.39, respectively. All the t -values are adjusted for heteroskedasticity and autocorrelations. Anomalies with sample starting dates later than January 1967 (and occasionally ending dates earlier than December 2016) are denoted by ★. For example, Abr starts from January 1972 (1972/1).

($t=8.03$). The difference likely arises from the time lag between the fiscal quarter end and subsequent returns. While Thomas and Zhang (2011) impose a 3-month lag, we adopt a 4-month lag. However, the large difference between the value- and equal-weighted estimates is mostly driven by microcaps.

The high-minus-low deciles on the industry lead-lag effect in earnings surprise (Ile) at the 1-, 6-, and 12-month horizons earn on average 0.58%, 0.23%, and 0.09% ($t=3.48$, 1.55, and 0.64), respectively, with NYSE-VW. In contrast, Hou (2007) reports highly significant effects at short horizons via weekly cross-sectional regressions with ordinary least squares.

Finally, the high-minus-low customer momentum (Cm) quintiles earn 0.78%, 0.16%, and 0.15% per month ($t=3.85$, 1.72, and 2.23) at the 1-, 6-, and 12-month horizons, respectively, with NYSE-VW. Following Cohen and Frazzini (2008), we form quintiles because many firms have the same Cm values, yielding fewer than ten portfolios in some months. The 0.78% estimate is substantially lower than their estimate of 1.58% ($t=3.79$) with NYSE-Amex-NASDAQ breakpoints and a price screen of \$5. Using their breakpoints minimally affects our estimate (All-VW). As such, the difference mainly arises from the \$5 price screen. The average return is also sensitive to the holding period.

3.2.2 Value versus growth. Panel B of Table 3 reports the replication results for the 68 value versus growth anomalies. Several high-profile value versus growth anomalies fail to replicate. The high-minus-low 5-year sales growth (Sr) decile with NYSE-VW earns an average return of only -0.19% per month ($t=-1.08$). This estimate is much smaller in magnitude than -7.3% per annum (-0.61% per month) in Lakonishok, Shleifer, and Vishny (1994) based on NYSE-Amex breakpoints and equal-weighted returns (no NASDAQ stocks). Sampling variation plays an important role, as our NYSE-VW estimate is -0.45% ($t=-1.97$) in their 1963–1990 original sample. Overweighting microcaps plays an equally important role, as the All-EW estimate in our extended sample is -0.52% ($t=-3.65$).

The high-minus-low decile on long-term analysts' forecasts (Ltg) yields 0.13% per month ($t=0.38$) with NYSE-VW. This estimate differs dramatically from -20.9% per annum (-1.74% per month) in La Porta (1996) with NYSE-Amex breakpoints and equal-weighted returns (no NASDAQ stocks). With All-EW, our estimates are -0.38% ($t=-0.99$) in the extended sample and -0.47% ($t=-0.72$) in La Porta's original 1982–1991 sample. A potential reason why we fail to replicate the original estimate is that IBES has implemented large-scale and nonrandom revisions to its data (Ljungqvist, Malloy, and Marston 2009). Still, based on the latest available data, Ltg fails to predict returns regardless of sorting frequency, breakpoints, and return weights.

Turning to the replicated anomalies, panel B shows that the annually sorted high-minus-low book-to-market (Bm) decile earns on average 0.54% per month ($t=2.61$) with NYSE-VW. The estimate is 1.41% ($t=3.1$) in Rosenberg, Reid, and Lanstein's (1985) original sample from January 1973

to September 1984, which is likely too short to be representative. The estimate is 0.69% ($t=2.39$) from January 1967 to December 1990, which is the sample in Fama and French (1992).

Many other value versus growth anomaly variables predict returns reliably, although the average returns are lower in magnitude than original estimates, which often overweight microcaps. For example, the high-minus-low cash flow-to-price (Cp) decile with NYSE-VW earns on average 0.43% per month ($t=2.14$). This estimate is much lower than 9.9% per annum (0.83% per month) in Lakonishok, Shleifer, and Vishny (1994) based on NYSE-Amex breakpoints and equal-weighted returns (no NASDAQ stocks). For comparison, our All-EW estimate is 0.97% ($t=5.07$).

Asness and Frazzini (2013) show that using the more updated market equity strengthens the predictive power of the value versus growth variables. To incorporate their insight, we form monthly sorted value versus growth deciles with the most recent market equity and quarterly accounting variables. Consistent with their work, the monthly sorted deciles in general yield higher average return spreads, especially at the 1-month horizon, than annually sorted deciles. The monthly sorted high-minus-low deciles on earnings-to-price (Ep^q), cash flow-to-price (Cp^q), enterprise multiple (Em^q), and sales-to-price (Sp^q) at the 1-month horizon earn on average 0.93%, 0.62%, -0.71% , and 0.59% per month ($t=4.94, 2.93, -3.21, \text{ and } 2.39$), which are higher in magnitude than 0.44%, 0.43%, -0.54% , and 0.5% ($t=2.26, 2.14, -2.86, \text{ and } 2.37$), respectively, for the annually sorted deciles.

3.2.3 Investment. Panel C of Table 3 details the replication results for the 38 investment anomalies. The high-minus-low investment-to-assets (I/A) decile with NYSE-VW earns on average -0.44% per month ($t=-2.89$). This estimate is much lower in magnitude than -1.05% ($t=-5.04$) with value-weighted returns and -1.73% ($t=-8.45$) with equal-weighted returns in Cooper, Gulen, and Schill (2008), who use NYSE-Amex-NASDAQ breakpoints. For comparison, our All-VW estimate is -0.56% ($t=-3.4$), and the All-EW estimate -1.27% ($t=-6.99$) in the extended sample. As such, overweighting microcaps via equal-weighting greatly exaggerates the investment premium.

The high-minus-low decile on total accruals (Ta) earns an average return of -0.22% ($t=-1.63$) with NYSE-VW. In contrast, Richardson et al. (2005) report an average return of -13.3% per annum (-1.11% per month, $t=-10.25$) based on NYSE-Amex-NASDAQ breakpoints and equal-weighted size-adjusted returns. Although the individual size-adjusted returns are equal-weighted, the size portfolio returns in the adjustment are value-weighted, giving rise to internal inconsistency. Our All-EW estimate is -0.53% ($t=-3.29$), which is only one half of the original estimate.

The high-minus-low operating accruals (Oa) decile with NYSE-VW earns -0.27% per month ($t=-2.19$). This estimate is much lower in magnitude than -10.4% per annum (-0.87% per month, $t=-4.71$) in Sloan (1996). Sloan

uses NYSE-Amex breakpoints (no NASDAQ stocks) and equal-weighted size-adjusted returns, but the size portfolio returns in the adjustment are value-weighted. Our All-EW estimate is -0.44% ($t = -3.4$), which is only half of the original estimate.

3.2.4 Profitability. Panel D of Table 3 details the replication of the 79 anomalies in the profitability category. The return on equity (Roe) is significant mostly within short horizons with NYSE-VW. The high-minus-low deciles earn on average 0.68% , 0.42% , and 0.23% ($t = 3.12$, 1.98 , and 1.18) at the 1-, 6-, and 12-month horizons, respectively. The 1-month evidence reproduces that of Hou, Xue, and Zhang (2015).

More generally, many different profitability measures have recently been proposed to predict returns, but not all are equally effective. The high-minus-low gross profits-to-lagged assets (Gla) decile earns an average return of only 0.16% per month ($t = 1.04$). This estimate is lower than 0.38% ($t = 2.62$) for the high-minus-low gross profits-to-assets (Gpa) decile. The difference between Gla and Gpa is that Gla scales gross profits with 1-year-lagged assets, but Gpa scales with current assets. Because both profits and assets are measured at the end of a period in Compustat, profits should be scaled by lagged assets, which in turn produce current profits. In contrast, the current assets at the end of a period are accumulated through investment over the current period and generate profits only in future periods. More important, because Gpa equals Gla divided by asset growth, the Gpa premium is confounded with the investment premium. Purging the investment premium yields an economically small and statistically insignificant gross profitability premium.

Operating profits-to-book equity (Ope), which is the sorting variable underlying the Fama and French (2015) robust-minus-weak profitability factor (RMW), also fails to replicate. The high-minus-low Ope decile with NYSE-VW earns an average return of only 0.27% per month ($t = 1.34$). Ope scales operating profits with current book equity. Scaling with 1-year-lagged book equity in operating profits-to-lagged book equity (Ole) reduces the estimate further to 0.11% ($t = 0.58$).

Adding research and development expenses to operating profits, Ball et al. (2016) show that the high-minus-low operating profits-to-assets (Opa) decile earns on average 0.29% per month ($t = 1.95$). We obtain an average return of 0.41% ($t = 2.09$). However, scaling their operating profits with lagged assets (Ola) reduces the average return to 0.2% ($t = 1.11$).

The fundamental score (F) fails to replicate. The high-minus-low decile with NYSE-VW earns only 0.29% per month ($t = 1.11$), which is lower than 23.5% per annum (1.96% per month, $t = 5.59$) for the high-minus-low quintile in Piotroski (2000) based on equal-weighted returns in a subsample of value stocks. Our All-EW estimate is 0.46% ($t = 2.06$). Sampling variation plays a role. In Piotroski's 1976–1996 original sample, our estimates are 0.65% ($t = 2.19$) with NYSE-VW and 0.76% ($t = 3$) with All-EW.

Strikingly, the distress anomaly is virtually nonexistent. In annual sorts with NYSE-VW, the high-minus-low failure probability (Fp) decile earns an average return of -0.39% per month ($t = -1.35$) from July 1976 to December 2016. This estimate is lower in magnitude than -9.66% per annum (-0.81% per month) in Campbell, Hilscher, and Szilagyi (2008) in the 1981–2003 sample. We reproduce their estimate in their sample with an average return of -0.82% ($t = -2.09$). However, prior to their sample from July 1976 to December 1980, the average return is strongly positive, 0.69% (-0.02% from January 2004 onward). While Campbell, Hilscher, and Szilagyi (2008) use NYSE-Amex-NASDAQ breakpoints, we use NYSE breakpoints. For comparison, our All-VW estimate is -0.55% ($t = -1.5$).

Alternative measures of financial distress, such as Altman's (1968) Z-score (Z) and Ohlson's (1980) O-score (O), show even weaker forecasting power than failure probability. None of the high-minus-low deciles with NYSE-VW show any significant average returns. The high-minus-low O decile earns -0.09% per month ($t = -0.48$) in annual sorts, and the high-minus-low Z decile 0.01% ($t = 0.06$). These estimates contrast with those in Dichev (1998), who reports an average return of -1.17% ($t = -3.36$) for the highest-10%-minus-lowest-70% O portfolio based on NYSE-Amex-NASDAQ breakpoints and equal-weighted returns and a significantly positive slope for the Z-score in cross-sectional regressions. Sampling variation plays a role. The high-minus-low O decile earns -0.6% ($t = -2.05$) in the original sample, but the Z-score remains weak at -0.04% ($t = -0.13$).

3.2.5 Intangibles. Panel E of Table 3 details the replication for the 103 anomalies in the intangibles category. The high-minus-low hiring rate (Hn) decile earns an average return of -0.25% per month ($t = -1.63$) with NYSE-VW. Using NYSE-Amex-NASDAQ breakpoints and value-weighted returns (All-VW) yields -0.19% ($t = -1.08$). These estimates are lower in magnitude than -5.61% per annum (-0.47% per month, $t = -2.26$) in Belo, Lin, and Bazdresch (2014), who include only firms with December fiscal year end. This restrictive sample screen loses almost 40% of observations. Our All-EW estimate is -0.85% ($t = -6.34$). As such, the hiring rate premium is mostly driven by microcaps.

The high-minus-low corporate governance (Gind) decile earns a tiny average return of 0.02% per month ($t = 0.06$) from September 1990 to December 2006 (the last available date). In contrast, Gompers, Ishii, and Metrick (2003) report a significant high-minus-low alpha of -0.71% ($t = -2.73$) in the Carhart (1997) model from September 1990 to December 1999. We reproduce their evidence with a Carhart alpha of -0.59% ($t = -1.88$) and an average return of -0.73% ($t = -2.04$) in their original sample. However, outside their sample from January 2000 to December 2006, the high-minus-low decile earns a positive average return of 1.01% ($t = 2.09$) and a Carhart alpha of 0.2% ($t = 0.56$). Our evidence

accords with that of Core, Guay, and Rusticus (2006), who also show that the high-minus-low decile return exhibits a reversal from 2000 to 2003.

The high-minus-low accrual quality (Acq) decile earns an average return of -0.12% per month ($t = -0.6$) in annual sorts with NYSE-VW. The estimates from monthly sorts are even smaller in magnitude. Also, the average returns of the high-minus-low deciles on earnings persistence (Eper), earnings smoothness (Esm), value relevance of earnings (Evr), and earnings conservatism (Ecs) are all small and insignificant, ranging from -0.04% to 0.18% , with t -values from -0.31 to 1.31 .

Our estimates contrast with those of Francis et al. (2004, 2005), who show that these earnings attributes have significant relations with expected returns measured as accounting-based costs of capital. Although Francis et al. (2004, 2005) construct factors based on the earnings attributes, their average returns are not reported. Our evidence accords with that of Core, Guay, and Verdi (2008), who also report that Acq is not priced in asset pricing tests. However, we emphasize that two other attributes, earnings predictability (Eprd) and earnings timeliness (Etl), do produce significant average return spreads, -0.53% ($t = -2.96$) and 0.34% ($t = 2.79$), respectively.

The high-minus-low deciles on the dispersion of analysts' earnings forecasts (Dis) with NYSE-VW earn -0.19% , -0.18% , and -0.08% per month at the 1-, 6-, and 12-month horizons, all of which are within one standard error from zero. The evidence contrasts with Diether, Malloy, and Scherbina (2002), who report an average return of -0.79% ($t = -2.88$) for the low-minus-high quintile at the 1-month horizon with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns (and a \$5 price screen). Our All-EW estimate is -0.69% ($t = -3.14$). As such, the Dis effect is mostly driven by microcaps.

3.2.6 Trading Frictions. As noted, the biggest casualty of our replication is the trading frictions category, with 102 of 106 anomalies (96.2%) not replicated. Panel F of Table 3 details the replication for these individual anomalies.

Most strikingly, 15 of 16 volatility measures yield insignificant high-minus-low average returns with NYSE breakpoints and value-weighted returns. In particular, the high-minus-low deciles on idiosyncratic volatility from the Fama and French (1993) three-factor model (Ivff) earn on average -0.52% , -0.32% , and -0.18% per month ($t = -1.71$, -1.12 , and -0.67) at the 1-, 6-, and 12-month horizons, respectively. The high-minus-low deciles on total volatility (Tv) earn on average -0.39% , -0.24% , and -0.2% ($t = -1.18$, -0.77 , and -0.65) over the three horizons, respectively. Systematic volatility (Sv) is insignificant at the 6- and 12-month horizons, -0.18% and -0.14% ($t = -1.27$ and -1.22), respectively, but significant at 1-month with an average return of -0.49% ($t = -2.24$).

In terms of magnitude, our estimates more than halve Ang et al.'s (2006) estimates of -1.06% , -0.97% , and -1.04% per month ($t = -3.1$, -2.86 and

−3.9) for the high-minus-low Ivff, Tv, and Sv quintiles, respectively, all at the 1-month horizon with NYSE-Amex-NASDAQ breakpoints and value-weighted returns (All-VW). With this procedure, we obtain −1.22% ($t = -3.38$) and −1.18% ($t = -3.02$) for the high-minus-low Ivff and Tv deciles, respectively, in our sample. For the high-minus-low Sv decile, we obtain −1.1% ($t = -3.1$) in Ang et al.'s original 1986–2000 sample but only −0.42% ($t = -1.56$) in our sample. In the 2001–2016 period, its average return is only 0.04%.

Curiously, overweighting microcaps does not revive the low volatility anomaly. With All-EW, again 15 of 16 volatility measures produce economically small and statistically insignificant average return spreads. Five measures even produce positive average return spreads, albeit insignificant. Also, 14 measures have absolute t -values below one. The All-EW evidence is even weaker than the NYSE-VW evidence. With All-VW, nine of 16 measures yield significant average return spreads. As such, consistent with Bali and Cakici (2008), the low volatility anomaly is extremely fragile.

Traditional liquidity measures fare poorly. With NYSE-VW, the high-minus-low deciles on the Amihud (2002) absolute return-to-volume (Ami) earn on average 0.25%, 0.34%, and 0.39% per month ($t = 1.2, 1.64, \text{ and } 1.91$) at the 1-, 6-, and 12-month horizons, respectively. In contrast, Amihud reports a highly significant liquidity effect using cross-sectional regressions. We replicate this result with cross-sectional regressions with ordinary least squares (FM-OLS) in our sample. With weighted least squares, the effect is significant only at the 12-month horizon with a slope of 0.26 ($t = 1.99$). With All-EW, the high-minus-low deciles earn average returns above 1%, with t -values above 3. As such, this liquidity effect is mostly driven by microcaps.

Similarly, the high-minus-low short-term reversal (Srev) decile earns on average only −0.27% per month ($t = -1.4$) with NYSE-VW. This estimate is much lower in magnitude than −1.99% ($t = -12.55$) in Jegadeesh (1990) based on NYSE-Amex-NASDAQ breakpoints and equal-weighted returns. We replicate this results with an All-EW estimate of −2.57% ($t = -9.22$). As such, the short-term reversal is also mostly driven by microcaps.

The Acharya and Pedersen (2005) liquidity betas fare very poorly. With NYSE-VW, all five versions of the their liquidity betas, including return-return (β^{ret}), illiquidity-illiquidity (β^{lcc}), return-illiquidity (β^{lrc}), illiquidity-return (β^{lcr}), and net liquidity beta (β^{net}), earn insignificant average return spreads across all three monthly horizons. The average returns range from −0.04% to 0.31% per month, most of which are within one standard error from zero. In contrast, Acharya and Pedersen report significant results for β^{ret} and β^{net} based on cross-sectional regressions with 25 illiquidity portfolios as testing assets. This replication failure has nothing to do with microcaps. We do not find any significance from cross-sectional regressions with individual stocks, even with ordinary least squares. With All-EW, all 15 variables from interacting five liquidity betas with three monthly horizons earn economically small and statistically insignificant average return spreads.

The Pastor and Stambaugh (2003) liquidity beta also fails to replicate. With NYSE-VW, the high-minus-low deciles earn on average only 0.08%, 0.11%, and 0.17% per month ($t=0.47, 0.74$, and 1.24), respectively. This failure also has nothing to do with microcaps, as the insignificance is robust to breakpoints, return weights, cross-sectional regressions, and sample periods.

The high-minus-low deciles on maximum daily return (Mdr) with NYSE-VW earn -0.36% , -0.17% , and -0.07% per month ($t=-1.27, -0.65$, and -0.27) at the 1-, 6-, and 12-month horizons, respectively, which are much lower than the 1-month estimate of -1.03% ($t=-2.83$) in Bali, Cakici, and Whitelaw (2011) with NYSE-Amex-NASDAQ breakpoints. They report that the average return starts at 1.01% for decile one, remains flat through decile seven, drops to 0.52% for decile nine, and precipitously to -0.02% for decile ten. With NYSE-VW, the average return starts at 0.97% for decile one, remains flat at 1.03% for decile nine, and drops only to 0.6% for decile ten.

The high-minus-low decile on the Adrian, Etula, and Muir (2014) intermediary leverage beta with NYSE-VW earns on average 0.39% , 0.26% , and 0.25% ($t=1.9, 1.31$, and 1.3) at the 1-, 6-, and 12-month horizons, respectively. With All-EW, the average return is 0.4% ($t=2.29$) at the 1-month horizon, but the 6- and 12-month estimates remain insignificant. Adrian, Etula, and Muir (2014) claim that their “single-factor model prices size, book-to-market, momentum, and bond portfolios with an R^2 of 77% and an average annual pricing error of 1%—performing as well as standard multifactor benchmarks designed to price these assets (original emphasis, p. 2557).” However, this claim is based on a mimicking portfolio from regressing the broker-dealer leverage on the six size and book-to-market portfolio excess returns and the momentum factor. As such, direct evidence on the leverage effect is fairly weak and not comparable to standard multifactor models.

Finally, the Easley, Hvidkjaer, and O’Hara (2002, 2010) probability of information-based trading (Pin), which is a leading market microstructure variable, also fails to replicate. With NYSE-VW, the high-minus-low Pin decile earns an average return of -0.23% per month ($t=-0.91$). With All-EW, the average return becomes even positive, 0.4% , albeit insignificant ($t=1.35$). In cross-sectional regressions, the Pin slope is insignificant with both ordinary and weighted least squares. However, in the sample that consists of only microcaps, the high-minus-low Pin decile earns 0.68% ($t=2.6$) with value-weighted and 1.5% ($t=6.04$) with equal-weighted returns (the Internet Appendix).

3.3 Commonality among the replicated anomalies

In this subsection, we briefly explore the commonality among the 158 replicated anomalies in single tests under NYSE-VW. We calculate the pairwise cross-sectional correlations based on each anomaly variable’s NYSE percentile rankings as well as the pairwise time-series correlations based on each anomaly’s high-minus-low decile returns. All the anomaly variables

Table 4
Pairwise cross-sectional correlations and principle component analysis for the 158 significant anomalies under NYSE breakpoints and value-weighted returns

<i>A. Average rank correlations, January 1967–December 2016, 600 months</i>							<i>B. Average high-minus-low return correlations, January 1967–December 2016, 600 months</i>							
	Mom	VvG	Inv	Prof	Intan	Fric	Mom	VvG	Inv	Prof	Intan	Fric		
Mom	0.20	−0.02	0.02	0.12	0.00	−0.01	0.39	−0.17	0.03	0.21	−0.02	0.01		
VvG		0.44	0.10	−0.03	0.12	0.04		0.60	0.18	−0.15	0.10	0.09		
Inv			0.32	−0.01	0.06	0.03			0.26	0.03	0.05	0.04		
Prof				0.36	0.02	−0.04				0.40	0.00	−0.06		
Intan					0.11	0.03					0.07	0.05		
Fric						0.16						0.10		
<i>C. The proportion (in %) of variance explained by each principle component, July 1976–December 2016, 486 months</i>														
	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10				
All	25.93	15.53	9.21	5.66	4.18	2.97	2.25	2.18	1.70	1.57				
Mom	53.16	10.33	6.62	4.35	3.78	3.39	2.48	2.16	1.92	1.89				
VvG	59.81	11.85	6.17	4.40	3.01	2.30	2.27	1.95	1.57	0.99				
Inv	36.11	11.14	8.71	5.99	5.53	3.86	3.61	2.79	2.41	2.23				
Prof	55.15	15.92	6.73	5.73	2.95	2.57	1.83	1.53	1.11	0.91				
Intan	20.85	18.52	9.59	6.01	5.89	5.01	4.47	4.36	4.00	3.52				
Fric	59.49	30.00	10.51											

The six categories, momentum, value versus growth, investment, profitability, intangibles, and trading frictions, are denoted by “Mom,” “VvG,” “Inv,” “Prof,” “Intan,” and “Fric,” respectively. All anomalies are realigned to yield positive high-minus-low average returns. Panel A shows the average within- and cross-category rank correlations based on each anomaly’s NYSE percentile rankings. Panel B shows within- and cross-category time-series correlations based on each anomaly’s high-minus-low decile returns. The average within-category correlations are averaged across all the pairwise correlations within a category, and the average cross-category correlations are averaged across all possible pairwise correlations across a given pair of categories. Panel C reports the proportion of the sum of variances for all high-minus-low decile returns captured by a principle component, denoted by PC. We report the first ten principle components, PC1, PC2, ..., and PC10, respectively.

are realigned to yield positive high-minus-low average returns. The within-category correlations are averaged across all the pairwise correlations within a category, and the cross-category correlations are averaged across all the pairwise correlations across a given pair of categories. The upshot is that our categorization of anomalies based on ex ante economic concepts is largely consistent with ex post statistical clustering.

Panel A of Table 4 shows the rank correlations of NYSE percentiles, and panel B the time-series correlations of the high-minus-low returns. The average within-category correlations are generally large, whereas the average cross-category correlations are small. For example, the within-category correlations in the value versus growth category are 0.44 with NYSE rankings and 0.6 with high-minus-low returns. In contrast, the cross-category correlations between the value versus growth and investment categories are only 0.1 and 0.18, respectively. However, the within-category correlations in the intangibles category are low, only 0.11 and 0.07, respectively. This category is more diffused, consisting of different anomalies, such as the Heston and Sadka (2008) seasonality anomalies that have close to zero correlations with other anomalies.

Panel C conducts principle component analysis for the 158 high-minus-low decile returns. Unlike pairwise correlations, this analysis requires a common

sample for all the anomalies. As such, we start the sample in July 1976, which drops only 11 anomalies from our analysis. Starting from January 1967 would force us to drop 62. Consistent with the cluster analysis, the first principle component for each category explains a large amount of the total variance of the high-minus-low returns within the category, 53.2%, 59.8%, 36.1%, 55.2%, 20.9%, and 59.5% across the momentum, value versus growth, investment, profitability, intangibles, and trading frictions categories, respectively. For intangibles, the first four principle components combine to explain 55% of the total variance. Across all the 158 anomalies, the first principle component explains 25.9%, and the first 4, 6, and 8 components combine to explain 56.3%, 63.5%, and 67.9% of the total variance, respectively.

4. Conclusion

We have replicated the bulk of the anomalies literature by compiling an extensive data library of 452 anomalies. Most anomalies fail to replicate, falling short of the currently acceptable standards for empirical finance. First, after we control for microcaps via NYSE breakpoints and value-weighted returns, 65% of the anomalies cannot clear the single test hurdle of $|t| \geq 1.96$. In the category that contains liquidity, market microstructure, and other trading frictions variables, 102 of 106 variables (96%) fail to clear this low hurdle. Second, regardless of microcaps, most anomalies fail to replicate if we adjust for multiple testing with a higher $|t|$ -cutoff of 2.78 at the 5% significance level. The failure rate is 52% in sorts with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns that assign maximum weights to microcaps. The failure rate is again the highest, 73.6%, in the trading frictions category. Even for replicated anomalies, their economic magnitudes are much lower than originally reported. The results from the shorter samples in the original studies are quantitatively similar. In all, capital markets are more efficient than previously recognized.

Our work has important implications. First, anomalies are not created equal. The value and momentum anomalies replicate well, along with the investment and profitability anomalies. Most of these anomalies reside in value-weighted returns, which account for 97% of the aggregate market capitalization. In contrast, even with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns, most trading frictions variables (60.4%) still fail to replicate in single tests. As such, economic fundamentals are more important than trading frictions in driving the cross section of expected returns.

Second, our results show that microcaps account for many of the published anomalies. To mitigate their disproportionately large impact, we advocate NYSE breakpoints and value-weighted returns in portfolio sorts as well as cross-sectional regressions with weighted least squares. While alternative procedures are not technically wrong, their results can be very fragile, if not misleading. Empirical results from economically more important and statistically more reliable procedures are more credible.

Third, and finally, the credibility of the anomalies literature can improve via a closer connection with economic theory. Ioannidis (2005) emphasizes the importance of theoretical predictions, which raise the ratio of ex ante true relations to false relations tested in a given field. Harvey, Liu, and Zhu (2016) also argue that a theory-based factor should have a lower absolute t -cutoff than a purely empirical factor. Although theory is not immune to problems of its own, such as hypothesizing after the results are known (Kerr 1998), more applied studies grounded on first principles are likely to increase the credibility of the anomalies literature, which is still largely statistical in nature.

Appendix A. Variable Definitions and Portfolio Construction

A.1 Momentum

A.1.1 Sue1, Sue6, and Sue12, standardized unexpected earnings. Per Foster, Olsen, and Shevlin (1984), Sue denotes Standardized Unexpected Earnings, and is calculated as the change in split-adjusted quarterly earnings per share (Compustat quarterly item EPSPXQ divided by item AJEXQ) from its value four quarters ago divided by the standard deviation of this change in quarterly earnings over the prior eight quarters (six quarters minimum). At the beginning of each month t , we split all NYSE, Amex, and NASDAQ stocks into deciles based on their most recent past Sue. Before 1972, we use the most recent Sue computed with quarterly earnings from fiscal quarters ending at least four months prior to the portfolio formation. Starting from 1972, we use Sue computed with quarterly earnings from the most recent quarterly earnings announcement dates (Compustat quarterly item RDQ). For a firm to enter our portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Sue to be within six months prior to the portfolio formation. We do so to exclude stale information on earnings. To avoid potentially erroneous records, we also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly portfolio returns are calculated, separately, for the current month t (Sue1), from month t to $t+5$ (Sue6), and from month t to $t+11$ (Sue12). Holding periods longer than one month like in Sue6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior 6-month period. We average the subdecile returns as the monthly return of the Sue6 decile.

A.1.2 Abr1, Abr6, and Abr12, cumulative abnormal returns around earnings announcement dates. We calculate cumulative abnormal stock return (Abr) around the latest quarterly earnings announcement date (Compustat quarterly item RDQ) (Chan, Jegadeesh, and Lakonishok 1996):

$$\text{Abr}_i = \sum_{d=-2}^{+1} r_{id} - r_{md}, \quad (\text{A1})$$

in which r_{id} is stock i 's return on day d (with the earnings announced on day 0) and r_{md} is the value-weighted market index return. We cumulate returns until one (trading) day after the announcement date to account for the 1-day-delayed reaction to earnings news.

At the beginning of each month t , we split all stocks into deciles based on their most recent past Abr. For a firm to enter our portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Abr to be within six months prior to the portfolio formation. We do so to exclude stale information on earnings. To avoid potentially erroneous records, we also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for the current month t (Abr1), and, separately, from month t to $t+5$ (Abr6) and from month t to $t+11$ (Abr12). The deciles are rebalanced monthly. The 6-month holding period for Abr6 means that for a given decile in each month there exist six subdeciles, each

of which is initiated in a different month in the prior 6-month period. We average the subdecile returns as the monthly return of the Abr6 decile. Because quarterly earnings announcement dates are largely unavailable before 1972, the Abr portfolios start in January 1972.

A.1.3 Re1, Re6, and Re12, revisions in analyst earnings forecasts. Following Chan, Jegadeesh, and Lakonishok (1996), we measure earnings surprise as the revisions in analysts' forecasts of earnings obtained from the Institutional Brokers' Estimate System (IBES). Because analysts' forecasts are not necessarily revised each month, we construct a 6-month moving average of past changes in analysts' forecasts:

$$Re_{it} = \sum_{\tau=1}^6 \frac{f_{it-\tau} - f_{it-\tau-1}}{p_{it-\tau-1}}, \quad (A2)$$

in which $f_{it-\tau}$ is the consensus mean forecast (unadjusted IBES file, item MEANEST) issued in month $t-\tau$ for firm i 's current fiscal year earnings (fiscal period indicator = 1), and $p_{it-\tau-1}$ is the prior month's share price (unadjusted file, item PRICE). We require both earnings forecasts and share prices to be in U.S. dollars (currency code = USD). We also adjust for any stock splits and require a minimum of four monthly forecast changes when constructing Re. At the beginning of each month t , we split all stocks into deciles based on Re. Monthly returns are calculated for the current month t (Re1), and, separately, from month t to $t+5$ (Re6) and from month t to $t+11$ (Re12). The deciles are rebalanced monthly. The 6-month holding period for Re6 means that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Re6 decile. Because analyst forecast data start in January 1976, the Re portfolios start in July 1976.

A.1.4 R^6_1 , R^6_6 , and R^6_{12} , prior 6-month returns. At the beginning of each month t , we split all stocks into deciles based on their prior 6-month returns from month $t-7$ to $t-2$. Skipping month $t-1$, we calculate monthly decile returns, separately, for month t (R^6_1), from month t to $t+5$ (R^6_6), and from month t to $t+11$ (R^6_{12}). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in R^6_6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior 6-month period. We average the subdeciles returns as the monthly return of the R^6_6 decile.

A.1.5 R^{11}_1 , R^{11}_6 , and R^{11}_{12} , prior 11-month returns. We split stocks into deciles at the beginning of each month t based on prior 11-month returns from month $t-12$ to $t-2$. Skipping month $t-1$, we calculate monthly decile returns for month t (R^{11}_1), from month t to $t+5$ (R^{11}_6), and from month t to $t+11$ (R^{11}_{12}). All the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in R^{11}_6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior 6-month period. We average the subdecile returns as the monthly return of the R^{11}_6 decile.

A.1.6 Im1, Im6, and Im12, industry momentum. We start with the Fama-French (1997) 49-industry classifications. Excluding financial firms from the sample leaves 45 industries. At the beginning of each month t , we sort industries based on their prior 6-month value-weighted returns from $t-6$ to $t-1$. Following Moskowitz and Grinblatt (1999), we do not skip month $t-1$. We form nine portfolios ($9 \times 5 = 45$), each of which contains five different industries. We define the return of a given portfolio as the simple average of the five industry returns within the portfolio. We calculate portfolio returns for the nine portfolios for the current month t (Im1), from month t to $t+5$ (Im6), and from month t to $t+11$ (Im12). The portfolios are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in Im6 mean that for a given portfolio in each month there exist six subportfolios, each of which is initiated in a different month in the prior 6-month period. We average the subportfolio returns as the monthly return of the Im6 portfolio.

A.1.7 Rs1, Rs6, and Rs12, revenue surprises. Following Jegadeesh and Livnat (2006), we measure revenue surprises (Rs) as changes in revenue per share (Compustat quarterly item SALEQ/(item CSHPRQ times item AJEXQ)) from its value four quarters ago divided by the standard deviation of this change in quarterly revenue per share over the prior eight quarters (six minimum). At the beginning of each month t , we split stocks into deciles based on their most recent past Rs. Before 1972, we use the most recent Rs computed with quarterly revenue from fiscal quarters ending at least four months prior to the portfolio formation. Starting from 1972, we use Rs computed with quarterly revenue from the most recent quarterly earnings announcement dates (Compustat quarterly item RDQ). Jegadeesh and Livnat report that quarterly revenue data are generally available when earnings are announced. For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Rs to be within six months prior to the portfolio formation. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly deciles returns are calculated for the current month t (Rs1), from month t to $t+5$ (Rs6), and from month t to $t+11$ (Rs12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Rs6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdeciles returns as the return of the Rs6 decile.

A.1.8 Tes1, Tes6, and Tes12, tax expense surprises. Following Thomas and Zhang (2011), we measure tax expense surprises (Tes) as changes in tax expense, which is tax expense per share (Compustat quarterly item TXTQ/(item CSHPRQ times item AJEXQ)) in quarter q minus tax expense per share in quarter $q-4$, scaled by assets per share (item ATQ/(item CSHPRQ times item AJEXQ)) in quarter $q-4$. At the beginning of each month t , we sort stocks into deciles based on their Tes calculated with Compustat quarterly data items from at least four months ago. We exclude firms with zero Tes (most of these firms pay no taxes). We calculate decile returns the current month t (Tes1), from month t to $t+5$ (Tes6), and from month t to $t+11$ (Tes12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Tes6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior 6-month period. We average the subdeciles returns as the monthly return of the Tes6 decile. For sufficient data coverage, we start the sample in January 1976.

A.1.9 dEf1, dEf6, and dEf12, changes in analyst earnings forecasts. Following Hawkins, Chamberlin, and Daniel (1984), we define $dEf \equiv (f_{it-1} - f_{it-2}) / (0.5|f_{it-1}| + 0.5|f_{it-2}|)$, in which f_{it-1} is the consensus mean forecast (unadjusted IBES file, item MEANEST) issued in month $t-1$ for firm i 's current fiscal year earnings (fiscal period indicator = 1). We require earnings forecasts to be denominated in U.S. dollars (currency code = USD). We also adjust for any stock splits between months $t-2$ and $t-1$ when constructing dEf. At the beginning of each month t , we sort stocks into deciles on the prior month dEf, and calculate returns for the current month t (dEf1), from month t to $t+5$ (dEf6), and from month t to $t+11$ (dEf12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in dEf6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the dEf6 decile. Because analyst forecast data start in January 1976, the dEf portfolios start in March 1976.

A.1.10 Nei1, Nei6, and Nei12, The number of quarters with consecutive earnings increase. We follow Barth, Elliott, and Finn (1999) and Green, Hand, and Zhang (2013) in measuring Nei as the number of consecutive quarters (up to eight quarters) with an increase in earnings (Compustat quarterly item IBQ) over the same quarter in the prior year. At the beginning of each month t , we sort stocks into nine portfolios (with Nei = 0, 1, 2, ..., 7, and 8, respectively) based on their most recent past Nei. Before 1972, we use Nei computed with quarterly earnings from fiscal quarters ending at least four months prior to the portfolio formation. Starting from 1972, we use Nei computed with earnings from the most recent quarterly earnings announcement dates (Compustat quarterly

item RDQ). For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Nei to be within six months prior to the portfolio formation. This restriction is imposed to exclude stale earnings information. We also require the announcement date to be after the corresponding fiscal quarter end. We calculate monthly returns for the current month t (Nei1), from month t to $t+5$ (Nei6), and from month t to $t+11$ (Nei12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Nei6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdeciles returns as the return of the Nei6 decile. For sufficient data, the Nei portfolios start in January 1969.

A.1.11 52w1, 52w6, and 52w12, 52-week high. At the beginning of each month t , we split stocks into deciles based on 52w, which is the ratio of its split-adjusted price per share at the end of month $t-1$ to its highest (daily) split-adjusted price per share during the 12-month period ending on the last day of month $t-1$. Monthly decile returns are calculated for the current month t (52w1), from month t to $t+5$ (52w6), and from month t to $t+11$ (52w12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in 52w6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the 52w6 decile. Because a disproportionately large number of stocks can reach the 52-week high at the same time and have 52w equal to one, we use only 52w smaller than one to form the portfolio breakpoints. Doing so helps avoid missing portfolio observations.

A.1.12 ϵ^6 1, ϵ^6 6, and ϵ^6 12, 6-month residual momentum. We split stocks into deciles at the beginning of each month t based on prior 6-month average residual returns from month $t-7$ to $t-2$ scaled by their standard deviation over the same period. Skipping month $t-1$, we calculate monthly decile returns for month t (ϵ^6 1), from month t to $t+5$ (ϵ^6 6), and from month t to $t+11$ (ϵ^6 12). Residual returns are estimated each month over the prior 36 months from month $t-36$ to month $t-1$ from regressing stock excess returns on the Fama and French (1993) three factors. We require returns to be available for all prior 36 months. The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in ϵ^6 6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the ϵ^6 6 decile.

A.1.13 ϵ^{11} 1, ϵ^{11} 6, and ϵ^{11} 12, 11-month residual momentum. We split all stocks into deciles at the beginning of each month t based on their prior 11-month residual returns from month $t-12$ to $t-2$ scaled by their standard deviation over the same period. Skipping month $t-1$, we calculate monthly decile returns for month t (ϵ^{11} 1), from month t to $t+5$ (ϵ^{11} 6), and from month t to $t+11$ (ϵ^{11} 12). Residual returns are estimated each month for all stocks over the prior 36 months from month $t-36$ to month $t-1$ from regressing stock excess returns on the Fama and French (1993) three factors. We require returns to be available for all prior 36 months. All the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in ϵ^{11} 6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the ϵ^{11} 6 decile.

A.1.14 Sm1, Sm6, and Sm12, segment momentum. Following Cohen and Lou (2012), we extract firms' segment accounting and financial information from Compustat segment files. Industries are based on two-digit SIC codes. Stand-alone firms are those that operate in only one industry with segment sales in segment files accounting for more than 80% of total sales in Compustat annual files. Conglomerate firms are those that operating in more than one industry with aggregate sales from all reported segments accounting for more than 80% of total sales.

At the end of June of each year, we form a pseudoconglomerate for each conglomerate firm. The pseudoconglomerate is a portfolio of the conglomerate's industry segments constructed with solely the stand-alone firms in each industry. The segment portfolios (value-weighted across stand-alone firms) are then weighted by the percentage of sales contributed by each industry segment within the conglomerate. At the beginning of each month t (starting in July), using segment information from the previous fiscal year, we sort all conglomerate firms into deciles based on the returns of their pseudoconglomerate portfolios in month $t - 1$. Monthly deciles are calculated for month t (Sm1), from month t to $t + 5$ (Sm6), and from month t to $t + 11$ (Sm12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Sm6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior 6-month period. We average the subdecile returns as the monthly return of the Sm6 decile. Because the segment data start in 1976, the Sm portfolios start in July 1977.

A.1.15 Ilr1, Ilr6, Ilr12, Ile1, Ile6, Ile12, industry lead-lag effect in prior returns (earnings surprises). We start with the Fama-French (1997) 49-industry classifications. Excluding financial firms from the sample leaves 45 industries. At the beginning of each month t , we sort industries based on the month $t - 1$ value-weighted return of the portfolio consisting of the 30% biggest (market equity) firms within a given industry. We form nine portfolios ($9 \times 5 = 45$), each of which contains five different industries. We define the return of a given portfolio as the simple average of the five value-weighted industry returns within the portfolio. The nine portfolio returns are calculated for the current month t (Ilr1), from month t to $t + 5$ (Ilr6), and from month t to $t + 11$ (Ilr12), and the portfolios are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Ilr6 mean that for a given portfolio in each month there exist six subportfolios, each of which is initiated in a different month in the prior 6-month period. We average the subportfolio returns as the monthly return of the Ilr6 portfolio.

We calculate standardized unexpected earnings, Sue, as the change in split-adjusted quarterly earnings per share (Compustat quarterly item EPSPXQ divided by item AJEXQ) from its value four quarters ago divided by the standard deviation of this change in quarterly earnings over the prior eight quarters (six quarters minimum). At the beginning of each month t , we sort industries based on their most recent Sue averaged across the 30% biggest firms within a given industry.¹⁴ To mitigate outliers, we winsorize Sue at the 1st and 99th percentiles of its distribution each month. We form nine portfolios ($9 \times 5 = 45$), each of which contains five different industries. We define the return of a given portfolio as the simple average of the five value-weighted industry returns within the portfolio. The nine portfolio returns are calculated for the current month t (Ile1), from month t to $t + 5$ (Ile6), and from month t to $t + 11$ (Ile12), and the portfolios are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Ile6 mean that for a given portfolio in each month there exist six subportfolios, each of which is initiated in a different month in the prior six months. We average the subportfolio returns as the return of the Ile6 portfolio.

A.1.16 Cm1, Cm6, and Cm12, customer momentum. Following Cohen and Frazzini (2008), we extract firms' principal customers from Compustat segment files. For each firm we determine whether the customer is another company listed on the CRSP/Compustat tape, and we assign it the corresponding CRSP permno number. At the end of June of each year t , we form a customer portfolio for each firm with identifiable firm-customer relations for the fiscal year ending in calendar year $t - 1$. For firms with multiple customer firms, we form equal-weighted customer portfolios.

¹⁴ Before 1972, we use the most recent Sue with earnings from fiscal quarters ending at least four months prior to the portfolio month. Starting from 1972, we use Sue with earnings from the most recent quarterly earnings announcement dates (Compustat quarterly item RDQ). For a firm to enter our portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Sue to be within six months prior to the portfolio month. We also require the earnings announcement date to be after the corresponding fiscal quarter end.

The customer portfolio returns are calculated from July of year t to June of $t+1$, and the portfolios are rebalanced in June.

At the beginning of each month t , we sort all stocks into quintiles based on their customer portfolio returns, Cm , in month $t-1$. We do not form deciles because a disproportionate number of firms can have the same Cm , which leads to fewer than ten portfolios in some months. Monthly quintile returns are calculated for month t ($Cm1$), from month t to $t+5$ ($Cm6$), and from month t to $t+11$ ($Cm12$), and the quintiles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in $Cm6$ mean that for a given quintile in each month there exist six subquintiles, each of which is initiated in a different month in the prior 6-month period. We average the subquintile returns as the monthly return of the $Cm6$ quintile. For sufficient data coverage, we start the Cm portfolios in July 1979.

A.1.17 Sim1, Sim6, Sim12, Cim1, Cim6, and Cim12, supplier (customer) industries momentum. Following Menzly and Ozbas (2010), we use Benchmark Input-Output Accounts at the Bureau of Economic Analysis (BEA) to identify supplier and customer industries for a given industry. BEA Surveys are conducted roughly once every five years in 1958, 1963, 1967, 1972, 1977, 1982, 1987, 1992, 1997, 2002, and 2007. We delay the use of any data from a given survey until the end of the year in which the survey is publicly released during 1964, 1969, 1974, 1979, 1984, 1991, 1994, 1997, 2002, 2007, and 2013, respectively. The BEA industry classifications are based on SIC codes in the surveys from 1958 to 1992 and on NAICS codes afterward. From 1997 to 2007, we merge three separate industry accounts, 2301, 2302, and 2303 into a single account. We also merge “Housing” (HS) and “Other Real Estate” (ORE) in the 2007 Survey. From 1958 to 1992, we merge industry account pairs 1–2, 5–6, 9–10, 11–12, 20–21, and 33–34. We also merge industry account pairs 22–23 and 44–45 in the 1987 and 1992 surveys. We drop miscellaneous industry accounts related to government, import, and inventory adjustments.

At the end of June of each year t , we assign each stock to an BEA industry (at the summary level) based on its reported SIC or NAICS code in Compustat (fiscal year ending in $t-1$) or CRSP (June of t). Monthly value-weighted industry returns are calculated from July of year t to June of $t+1$, and the industry portfolios are rebalanced in June of $t+1$. For each industry, we further form two separate portfolios, the suppliers portfolio and the customers portfolios. The share of an industry’s total purchases from other industries is used to calculate the supplier industries portfolio returns, and the share of the industry’s total sales to other industries is used to calculate the customer industries portfolio returns. The cross-industry flows of goods and services are from the BEA Input-Output Accounts’ Use Table (based on producers’ prices).

At the beginning of each month t , we split industries into deciles based on the supplier portfolio returns, Sim , and separately, on the customer portfolio returns, Cim , in month $t-1$. We then assign the decile rankings of each industry to its member stocks. Monthly decile returns are calculated for month t ($Sim1$ and $Cim1$), from month t to $t+5$ ($Sim6$ and $Cim6$), and from month t to $t+11$ ($Sim12$ and $Cim12$), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in $Sim6$ mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the $Sim6$ decile.

A.2 Value versus growth

A.2.1 Bm, book-to-market equity. At the end of June of each year t , we split stocks into deciles based on Bm, which is the book equity for the fiscal year ending in calendar year $t-1$ divided by the market equity (from CRSP) at the end of December of $t-1$. For firms with more than one share class, we merge the market equity for all share classes before computing Bm. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. Following Davis, Fama, and French (2000), we measure book equity as stockholders’ book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. Stockholders’ equity is the value

reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

A.2.2 Bmj, book-to-June-end market equity. Following Asness and Frazzini (2013), at the end of June of each year t , we sort stocks into deciles based on Bmj, which is book equity per share for the fiscal year ending in calendar year $t-1$ divided by share price (from CRSP) at the end of June of t . We adjust for any stock splits between the fiscal year end and the end of June. Book equity per share is book equity divided by the number of shares outstanding (Compustat annual item CSHO). Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.2.3 Bm^q1, Bm^q6, and Bm^q12, quarterly book-to-market equity. At the beginning of each month t , we split stocks into deciles based on Bm^q, which is the book equity for the latest fiscal quarter ending at least four months ago divided by the market equity (from CRSP) at the end of month $t-1$. For firms with more than one share class, we merge the market equity for all share classes before computing Bm^q. We calculate decile returns for the current month t (Bm^q1), from month t to $t+5$ (Bm^q6), and from month t to $t+11$ (Bm^q12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Bm^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Bm^q6 decile. Book equity is shareholders' equity, plus balance sheet deferred taxes and investment tax credit (Compustat quarterly item TXDITCQ) if available, minus the book value of preferred stock (item PSTKQ). Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the book value of preferred stock, or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity.

Before 1972, the sample coverage is limited for quarterly book equity in Compustat quarterly files. We expand the coverage by using book equity from Compustat annual files as well as by imputing quarterly book equity with clean surplus accounting. Specifically, whenever available we first use quarterly book equity from Compustat quarterly files. We then supplement the coverage for fiscal quarter four with annual book equity from Compustat annual files. We measure annual book equity as stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if available. If not, stockholders' equity is the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

If both approaches are unavailable, we apply the clean surplus relation to impute the book equity. Specifically, we impute the book equity for quarter t forward based on book equity from prior quarters. Let BEQ_{t-j} , $1 \leq j \leq 4$ denote the latest available quarterly book equity as of quarter t , and $IBQ_{t-j+1,t}$ and $DVQ_{t-j+1,t}$ be the sum of quarterly earnings and quarterly dividends from quarter $t-j+1$ to t , respectively. BEQ_t can then be imputed as $BEQ_{t-j} + IBQ_{t-j+1,t} - DVQ_{t-j+1,t}$. We do not use prior book equity from more than four quarters ago (i.e., $1 \leq j \leq 4$) to reduce

imputation errors. Quarterly earnings are income before extraordinary items (Compustat quarterly item IBQ). Quarterly dividends are zero if dividends per share (item DVPSXQ) are zero. Otherwise, total dividends are dividends per share times beginning-of-quarter shares outstanding adjusted for stock splits during the quarter. Shares outstanding are from Compustat (quarterly item CSHOQ supplemented with annual item CSHO for fiscal quarter four) or CRSP (item SHROUT), and the share adjustment factor is from Compustat (quarterly item AJEXQ supplemented with annual item AJEX for fiscal quarter four) or CRSP (item CFACSHR). Because we use quarterly book equity at least four months after the fiscal quarter end, all the data used in the imputation are at least 4-month lagged prior to the portfolio formation. We do not impute quarterly book equity backward using future earnings and book equity information to avoid look-ahead bias.

A.2.4 Dm, debt-to-market. At the end of June of each year t , we split stocks into deciles based on debt-to-market, Dm, which is total debt (Compustat annual item DLC plus DLTT) for the fiscal year ending in calendar year $t - 1$ divided by the market equity (from CRSP) at the end of December of $t - 1$. For firms with more than one share class, we merge the market equity for all share classes. Firms with no debt are excluded. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.2.5 Dm^q1, Dm^q6, and Dm^q12, quarterly debt-to-market. At the beginning of each month t , we split stocks into deciles based on quarterly debt-to-market, Dm^q, which is total debt (Compustat quarterly item DLCQ plus item DLTTQ) for the latest fiscal quarter ending at least four months ago divided by the market equity (from CRSP) at the end of month $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Dm^q. Firms with no debt are excluded. We calculate decile returns for the current month t (Dm^q1), from month t to $t + 5$ (Dm^q6), and from month t to $t + 11$ (Dm^q12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Dm^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Dm^q6 decile. For sufficient data coverage, the Dm^q portfolios start in January 1972.

A.2.6 Am, assets-to-market. At the end of June of each year t , we split stocks into deciles based on asset-to-market, Am, which is total assets (Compustat annual item AT) for the fiscal year ending in calendar year $t - 1$ divided by the market equity (from CRSP) at the end of December of $t - 1$. For firms with more than 1 share class, we merge the market equity for all share classes before computing Am. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.2.7 Am^q1, Am^q6, and Am^q12, quarterly assets-to-market. At the beginning of each month t , we split stocks into deciles based on quarterly asset-to-market, Am^q, which is total assets (Compustat quarterly item ATQ) for the latest fiscal quarter ending at least four months ago divided by the market equity (from CRSP) at the end of month $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Am^q. We calculate decile returns for the current month t (Am^q1), from month t to $t + 5$ (Am^q6), and from month t to $t + 11$ (Am^q12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Am^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Am^q6 decile. For sufficient data coverage, the Am^q portfolios start in January 1972.

A.2.8 Rev1, Rev6, and Rev12, reversal. To capture the De Bondt and Thaler (1985) long-term reversal (Rev) effect, at the beginning of each month t , we split stocks into deciles based on the prior returns from month $t-60$ to $t-13$. Monthly decile returns are computed for the current month t (Rev1), from month t to $t+5$ (Rev6), and from month t to $t+11$ (Rev12), and the deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in Rev6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Rev6 decile. To be included in a portfolio for month t , a stock must have a valid price at the end of $t-61$ and a valid return for $t-13$. In addition, any missing returns from month $t-60$ to $t-14$ must be -99.0 , which is the CRSP code for a missing ending price.

A.2.9 Ep, earnings-to-price. At the end of June of each year t , we split stocks into deciles based on earnings-to-price, Ep, which is income before extraordinary items (Compustat annual item IB) for the fiscal year ending in calendar year $t-1$ divided by the market equity (from CRSP) at the end of December of $t-1$. For firms with more than one share class, we merge the market equity for all share classes. Firms with nonpositive earnings are excluded. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.2.10 Ep^q1, Ep^q6, and Ep^q12, quarterly earnings-to-price. At the beginning of each month t , we split stocks into deciles based on quarterly earnings-to-price, Ep^q, which is income before extraordinary items (Compustat quarterly item IBQ) divided by the market equity (from CRSP) at the end of month $t-1$. Before 1972, we use quarterly earnings from fiscal quarters ending at least four months prior to the portfolio formation. Starting from 1972, we use quarterly earnings from the most recent quarterly earnings announcement dates (item RDQ). For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent quarterly earnings to be within six months prior to the portfolio formation. This restriction is imposed to exclude stale earnings information. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Firms with nonpositive earnings are excluded. For firms with more than one share class, we merge the market equity for all share classes before computing Ep^q. We calculate decile returns for the current month t (Ep^q1), from month t to $t+5$ (Ep^q6), and from month t to $t+11$ (Ep^q12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ep^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ep^q6 decile.

A.2.11 Efp1, Efp6, and Efp12, earnings forecast-to-price. Following Elgers, Lo, and Pfeiffer (2001), we define analysts' earnings forecast-to-price, Efp, as the consensus median forecasts (unadjusted IBES file, item MEDEST) for the current fiscal year (fiscal period indicator = 1) divided by share price (unadjusted file, item PRICE). We require earnings forecasts to be denominated in U.S. dollars (currency code = USD). At the beginning of each month t , we sort stocks into deciles based on Efp estimated with forecasts in month $t-1$. Firms with nonpositive forecasts are excluded. Monthly decile returns are calculated for the current month t (Efp1), from month t to $t+5$ (Efp6), and from month t to $t+11$ (Efp12), and the deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in Efp6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Efp6 decile. Because the earnings forecast data start in January 1976, the Efp deciles start in February 1976.

A.2.12 Cp, cash flow-to-price. At the end of June of each year t , we split stocks into deciles based on cash flow-to-price, Cf, which is cash flows for the fiscal year ending in calendar year $t-1$ divided by the market equity (from CRSP) at the end of December of $t-1$. Cash flows are income

before extraordinary items (Compustat annual item IB) plus depreciation (item DP)). For firms with more than one share class, we merge the market equity for all share classes before computing Cp. Firms with nonpositive cash flows are excluded. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.2.13 Cp^{q1}, Cp^{q6}, and Cp^{q12}, quarterly cash flow-to-price. At the beginning of each month t , we split stocks into deciles based on quarterly cash flow-to-price, Cp^q, which is cash flows for the latest fiscal quarter ending at least four months ago divided by the market equity (from CRSP) at the end of month $t-1$. Quarterly cash flows are income before extraordinary items (Compustat quarterly item IBQ) plus depreciation (item DPQ). For firms with more than one share class, we merge the market equity for all share classes before computing Cp^q. Firms with nonpositive cash flows are excluded. We calculate decile returns for the current month t (Cp^{q1}), from month t to $t+5$ (Cp^{q6}), and from month t to $t+11$ (Cp^{q12}), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Cp^{q6} mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ep^{q6} decile.

A.2.14 Dp, dividend yield. At the end of June of each year t , we sort stocks into deciles based on dividend yield, Dp, which is the total dividends paid out from July of year $t-1$ to June of t divided by the market equity (from CRSP) at the end of June of t . We calculate monthly dividends as the begin-of-month market equity times the difference between returns with and without dividends. Monthly dividends are then accumulated from July of $t-1$ to June of t . We exclude firms that do not pay dividends. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.2.15 Dp^{q1}, Dp^{q6}, and Dp^{q12}, quarterly dividend yield. At the beginning of each month t , we split stocks into deciles on quarterly dividend yield, Dp^q, which is the total dividends paid out from months $t-3$ to $t-1$ divided by the market equity (from CRSP) at the end of month $t-1$. We calculate monthly dividends as the begin-of-month market equity times the difference between returns with and without dividends. Monthly dividends are then accumulated from month $t-3$ to $t-1$. We exclude firms that do not pay dividends. We calculate monthly decile returns for the current month t (Dp^{q1}), from month t to $t+5$ (Dp^{q6}), and from month t to $t+11$ (Dp^{q12}), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Dp^{q6} mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Dp^{q6} decile.

A.2.16 Op and Nop, (net) payout yield. Per Boudoukh et al. (2007), total payouts are dividends on common stock (Compustat annual item DVC) plus repurchases. Repurchases are the total expenditure on the purchase of common and preferred stocks (item PRSTKC) plus any reduction (negative change over the prior year) in the value of the net number of preferred stocks outstanding (item PSTKRV). Net payouts equal total payouts minus equity issuances, which are the sale of common and preferred stock (item SSTK) minus any increase (positive change over the prior year) in the value of the net number of preferred stocks outstanding (item PSTKRV). At the end of June of each year t , we sort stocks into deciles based on total payouts (net payouts) for the fiscal year ending in calendar year $t-1$ divided by the market equity (from CRSP) at the end of December of $t-1$ (Op and Nop, respectively). For firms with more than one share class, we merge the market equity for all share classes before computing Op and Nop. Firms with nonpositive total payouts (zero net payouts) are excluded. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. Because the data on total expenditure and the sale of common and preferred stocks start in 1971, the Op and Nop portfolios start in July 1972.

A.2.17 Op^q1, Op^q6, Op^q12, Nop^q1, Nop^q6, and Nop^q12, quarterly (net) payout yield.

Quarterly total payouts are dividends plus repurchases from the latest fiscal quarter. Quarterly dividends are zero if dividends per share (Compustat quarterly item DVPSXQ) are zero. Otherwise, quarterly dividends are dividends per share times beginning-of-quarter shares outstanding (item CSOQ) adjusted for stock splits during the quarter (item AJEXQ for the adjustment factor). Quarterly repurchases are the quarterly change in year-to-date expenditure on the purchase of common and preferred stocks (item PRSTKCY) plus any reduction (negative change in the prior quarter) in the book value of preferred stocks (item PSTKQ). Quarterly net payouts equal total payouts minus equity issuances, which are the quarterly change in year-to-date sale of common and preferred stock (item SSTKY) minus any increase (positive change over the prior quarter) in the book value of preferred stocks (item PSTKQ). At the beginning of month t , we split stocks into deciles based on quarterly payouts (net payouts) for the latest fiscal quarter ending at least 4 months ago, divided by the market equity at the end of month $t - 1$ (Op^q and Nop^q, respectively). For firms with more than one share class, we merge the market equity for all share classes before computing Op^q and Nop^q. Firms with nonpositive total payouts (zero net payouts) are excluded. We calculate monthly decile returns for the current month t (Op^q1 and Nop^q1), from month t to $t + 5$ (Op^q6 and Nop^q6), and from month t to $t + 11$ (Op^q12 and Nop^q12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Op^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Op^q6 decile. For sufficient data coverage, the Op^q and Nop^q portfolios start in January 1985.

A.2.18 Sr, 5-year sales growth rank. Following Lakonishok, Shleifer, and Vishny (1994), we measure 5-year sales growth rank, Sr, in June of year t as the weighted average of the annual sales growth ranks for the prior five years: $\sum_{j=1}^5 (6 - j) \times \text{Rank}(t - j)$. The sales growth for year $t - j$ is the growth rate in sales (Compustat annual item SALE) from the fiscal year ending in $t - j - 1$ to the fiscal year ending in $t - j$. Only firms with data for all five prior years are used to determine the annual sales growth ranks, and we exclude firms with nonpositive sales. For each year from $t - 5$ to $t - 1$, we rank stocks into deciles based on their annual sales growth, and then assign rank i ($i = 1, \dots, 10$) to a firm if its annual sales growth falls into the i^{th} decile. At the end of June of each year t , we assign stocks into deciles based on Sr. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced at the end of June in year $t + 1$.

A.2.19 Sg, sales growth. At the end of June of each year t , we assign stocks into deciles based on Sg, which is the growth in annual sales (Compustat annual item SALE) from the fiscal year ending in calendar year $t - 2$ to the fiscal year ending in $t - 1$. Firms with nonpositive sales are excluded. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced at the end of June in year $t + 1$.

A.2.20 Em, enterprise multiple. Em is enterprise value divided by operating income before depreciation (Compustat annual item OIBDP). Enterprise value is the market equity plus the total debt (item DLC plus item DLTT) plus the book value of preferred stocks (item PSTKRV) minus cash and short-term investments (item CHE). At the end of June of each year t , we split stocks into deciles based on Em for the fiscal year ending in calendar year $t - 1$. Market equity (from CRSP) is at the end of December of $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Em. Firms with nonpositive enterprise value or operating income before depreciation are excluded. Monthly returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.2.21 Em^q1, Em^q6, and Em^q12, quarterly enterprise multiple. Em^q is enterprise value scaled by operating income before depreciation (Compustat quarterly item OIBDPQ). Enterprise

value is the market equity plus total debt (item DLCQ plus item DLTTQ) plus the book value of preferred stocks (item PSTKQ) minus cash and short-term investments (item CHEQ). At the beginning of each month t , we split stocks into deciles on Em^q for the latest fiscal quarter ending at least four months ago. The market equity (from CRSP) is measured at the end of month $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Em^q . Firms with nonpositive enterprise value or operating income before depreciation are excluded. Monthly decile returns are calculated for the current month t (Em^q1), from month t to $t + 5$ (Em^q6), and from month t to $t + 11$ (Em^q12), and the deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Em^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Em^q6 decile. For sufficient data coverage, the Em^q portfolios start in January 1976.

A.2.22 Sp, sales-to-price. At the end of June of each year t , we sort stocks into deciles based on sales-to-price, Sp, which is sales (Compustat annual item SALE) for the fiscal year ending in calendar year $t - 1$ divided by the market equity (from CRSP) at the end of December of $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Sp. Firms with nonpositive sales are excluded. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.2.23 Sp^q1, Sp^q6, and Sp^q12, quarterly sales-to-price. At the beginning of each month t , we sort stocks into deciles based on quarterly sales-to-price, Sp^q, which is sales (Compustat quarterly item SALEQ) divided by the market equity at the end of month $t - 1$. Before 1972, we use quarterly sales from fiscal quarters ending at least four months prior to the portfolio formation. Starting from 1972, we use quarterly sales from the most recent quarterly earnings announcement dates (item RDQ). For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent quarterly sales to be within six months prior to the portfolio formation. This restriction is imposed to exclude stale earnings information. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Firms with nonpositive sales are excluded. For firms with more than one share class, we merge the market equity for all share classes. Monthly decile returns are calculated for the current month t (Sp^q1), from month t to $t + 5$ (Sp^q6), and from month t to $t + 11$ (Sp^q12), and the deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Sp^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Sp^q6 decile.

A.2.24 Ocp, operating cash flow-to-price. At the end of June of each year t , we sort stocks into deciles based on operating cash flows-to-price, Ocp, which is operating cash flows for the fiscal year ending in calendar year $t - 1$ divided by the market equity (from CRSP) at the end of December of $t - 1$. Operating cash flows are measured as funds from operation (Compustat annual item FOPT) minus change in working capital (item WCAP) prior to 1988, and then as net cash flows from operating activities (item OANCF) stating from 1988. For firms with more than one share class, we merge the market equity for all share classes before computing Ocp. Firms with nonpositive operating cash flows are excluded. Monthly decile returns are from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the data on funds from operation start in 1971, the Ocp portfolios start in July 1972.

A.2.25 Ocp^q1, Ocp^q6, and Ocp^q12, quarterly operating cash flow-to-price. At the beginning of each month t , we split stocks on quarterly operating cash flow-to-price, Ocp^q, which is operating cash flows for the latest fiscal quarter ending at least four months ago divided by the market equity at the end of month $t - 1$. Operating cash flows are measured as the quarterly change in year-to-date funds from operation (Compustat quarterly item FOPTY) minus change in quarterly working

capital (item WCAPQ) prior to 1988, and then as the quarterly change in year-to-date net cash flows from operating activities (item OANCFY) starting from 1988. For firms with more than one share class, we merge the market equity for all share classes before computing Ocp^q . Firms with nonpositive operating cash flows are excluded. Monthly decile returns are calculated for the current month t ($Ocp^q 1$), from month t to $t+5$ ($Ocp^q 6$), and from month t to $t+11$ ($Ocp^q 12$), and the deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in $Ocp^q 6$ mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the $Ocp^q 6$ decile. Because the data on year-to-date funds from operation start in 1984, the Ocp^q portfolios start in January 1985.

A.2.26 Ir, intangible return. At the end of June of each year t , we perform the cross-sectional regression of each firm's past 5-year log stock return on its 5-year-lagged log book-to-market and 5-year log book return:

$$r(t-5, t) = \gamma_0 + \gamma_1 bm_{t-5} + \gamma_2 r^B(t-5, t) + u_t \quad (A3)$$

in which $r(t-5, t)$ is the past 5-year log stock return from the end of year $t-6$ to the end of $t-1$, bm_{t-5} is the 5-year-lagged log book-to-market, and $r^B(t-5, t)$ is the 5-year log book return. The 5-year-lagged log book-to-market is computed as $bm_{t-5} = \log(B_{t-5}/M_{t-5})$, in which B_{t-5} is the book equity for the fiscal year ending in calendar year $t-6$ and M_{t-5} is the market equity (from CRSP) at the end of December of $t-6$. For firms with more than one share class, we merge the market equity for all share classes before computing bm_{t-5} . The 5-year log book return is computed as $r^B(t-5, t) = \log(B_t/B_{t-5}) + \sum_{s=t-5}^{t-1} (r_s - \log(P_s/P_{s-1}))$, in which B_t is the book equity for the fiscal year ending in calendar year $t-1$, r_s is the stock return from the end of year $s-1$ to the end of year s , and P_s is the stock price per share at the end of year s . Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

A firm's intangible return, Ir , is defined as its residual from the annual cross-sectional regression. At the end of June of each year t , we sort stocks based on Ir for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of year $t+1$.

A.2.27 Vhp and Vfp, (analyst-based) intrinsic value-to-market. Following Frankel and Lee (1998), at the end of June of each year t , we implement the residual income model to estimate the intrinsic value:

$$Vh_t = B_t + \frac{(E_t[Roe_{t+1}] - r)}{(1+r)} B_t + \frac{(E_t[Roe_{t+2}] - r)}{(1+r)r} B_{t+1} \quad (A4)$$

$$Vf_t = B_t + \frac{(E_t[Roe_{t+1}] - r)}{(1+r)} B_t + \frac{(E_t[Roe_{t+2}] - r)}{(1+r)^2} B_{t+1} + \frac{(E_t[Roe_{t+3}] - r)}{(1+r)^2 r} B_{t+2} \quad (A5)$$

in which Vh_t is the historical Roe-based intrinsic value and Vf_t is the analysts earnings forecast-based intrinsic value. B_t is the book equity (Compustat annual item CEQ) for the fiscal year ending in calendar year $t-1$. Future book equity is computed using the clean surplus accounting: $B_{t+1} = (1 + (1-k)E_t[Roe_{t+1}])B_t$, and $B_{t+2} = (1 + (1-k)E_t[Roe_{t+2}])B_{t+1}$. $E_t[Roe_{t+1}]$ and $E_t[Roe_{t+2}]$ are the return on equity expected for the current and next fiscal years. k is the dividend payout ratio, measured as common stock dividends (item DVC) divided by earnings (item IBCOM) for the fiscal

year ending in calendar year $t - 1$. For firms with negative earnings, we divide dividends by 6% of average total assets (item AT). r is a constant discount rate of 12%. When estimating Vh_t , we replace all Roe expectations with most recent Roe_t : $Roe_t = Ni_t / [(B_t + B_{t-1})/2]$, in which Ni_t is earnings for the fiscal year ending in $t - 1$, and B_t and B_{t-1} are the book equity from the fiscal years ending in $t - 1$ and $t - 2$.

When estimating Vf_t , we use analyst earnings forecasts from IBES to construct Roe expectations. Let $Fy1$ and $Fy2$ be the 1-year-ahead and 2-year-ahead consensus mean forecasts (unadjusted IBES file, item MEANEST; fiscal period indicator = 1 and 2) reported in June of year t . Let s be the number of shares outstanding from IBES (unadjusted file, item SHOUT). When IBES shares are not available, we use shares from CRSP (daily item SHROUT) on the IBES pricing date (item PRDAYS) that corresponds to the IBES report. Then $E_t[Roe_{t+1}] = sFy1 / [(B_{t+1} + B_t)/2]$, in which $B_{t+1} = (1 + s(1 - k)Fy1)B_t$. Analogously, $E_t[Roe_{t+2}] = sFy2 / [(B_{t+2} + B_{t+1})/2]$, in which $B_{t+2} = (1 + s(1 - k)Fy2)B_{t+1}$. Let Ltg denote the long-term earnings growth rate forecast from IBES (item MEANEST; fiscal period indicator = 0). Then $E_t[Roe_{t+3}] = sFy2(1 + Ltg) / [(B_{t+3} + B_{t+2})/2]$, in which $B_{t+3} = (1 + s(1 - k)Fy2(1 + Ltg))B_{t+2}$. If Ltg is missing, we set $E_t[Roe_{t+3}]$ to be $E_t[Roe_{t+2}]$. Firms are excluded if their expected Roe or dividend payout ratio is higher than 100%. We also exclude firms with negative book equity.

At the end of June of each year t , we sort stocks into deciles on the ratios of Vh and Vf scaled by the market equity (from CRSP) at the end of December of $t - 1$, denoted Vhp and Vfp , respectively. For firms with more than one share class, we merge the market equity for all share classes before computing intrinsic value-to-market. Firms with nonpositive intrinsic value are excluded. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because analyst forecast data start in 1976, the Vfp deciles start in July 1976.

A.2.28 Ebp, enterprise book-to-price, and Ndp, net debt-to-price. Following Penman, Richardson, and Tuna (2007), we measure enterprise book-to-price, Ebp, as the ratio of the book value of net operating assets (net debt plus book equity) to the market value of net operating assets (net debt plus market equity). Net Debt-to-price, Ndp, is the ratio of net debt to the market equity. Net debt is financial liabilities minus financial assets. We measure financial liabilities as the sum of long-term debt (Compustat annual item DLTT), debt in current liabilities (item DLC), carrying value of preferred stock (item PSTK), and preferred dividends in arrears (item DVPA, zero if missing), less preferred treasury stock (item TSTKP, zero if missing). We measure financial assets as cash and short-term investments (item CHE). Book equity is common equity (item CEQ) plus any preferred treasury stock (item TSTKP, zero if missing) less any preferred dividends in arrears (item DVPA, zero if missing). Market equity is the number of common shares outstanding times share price (from CRSP).

At the end of June of each year t , we sort stocks into deciles based on Ebp, and separately, on Ndp, for the fiscal year ending in calendar year $t - 1$. Market equity is measured at the end of December of $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Ebp and Ndp. When forming the Ebp portfolios, we exclude firms with nonpositive book or market value of net operating assets. For the Ndp portfolios, we exclude firms with nonpositive net debt. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.2.29 Ebp^q1, Ebp^q6, Ebp^q12, Ndp^q1, Ndp^q6, and Ndp^q12, quarterly enterprise book-to-price, quarterly net debt-to-price. We measure quarterly enterprise book-to-price, Ebp^q, as the ratio of the book value of net operating assets (net debt plus book equity) to the market value of net operating assets (net debt plus market equity). Quarterly net debt-to-price, Ndp^q, is the ratio of net debt to market equity. Net debt is financial liabilities minus financial assets. Financial liabilities are the sum of long-term debt (Compustat quarterly item DLTTQ), debt in current liabilities (item DLQ), and the carrying value of preferred stock (item PSTKQ). Financial assets are cash and

short-term investments (item CHEQ). Book equity is common equity (item CEQQ). Market equity is the number of common shares outstanding times share price (from CRSP).

At the beginning of each month t , we split stocks into deciles based on Ebp^q , and separately, on Ndp^q , for the latest fiscal quarter ending at least four months ago. Market equity is measured at the end of month $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Ebp^q and Ndp^q . When forming the Ebp^q portfolios, we exclude firms with nonpositive book or market value of net operating assets. For the Ndp^q portfolios, we exclude firms with nonpositive net debt. Monthly decile returns are calculated for the current month t (Ebp^q1 and Ndp^q1), from month t to $t + 5$ (Ebp^q6 and Ndp^q6), and from month t to $t + 11$ (Ebp^q12 and Ndp^q12), and the deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Ebp^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ebp^q6 decile. For sufficient data coverage, the Ebp^q and Ndp^q portfolios start in January 1976.

A.2.30 Dur, equity duration. Following Dechow, Sloan, and Soliman (2004), we calculate firm-level equity duration, Dur , as

$$Dur = \frac{\sum_{t=1}^T t \times CD_t / (1+r)^t}{Me} + \left(T + \frac{1+r}{r} \right) \frac{ME - \sum_{t=1}^T CD_t / (1+r)^t}{Me}, \quad (A6)$$

in which CD_t is the net cash distribution in year t , Me is market equity, T is the length of forecasting period, and r is the cost of equity. Market equity is price per share times shares outstanding (Compustat annual item PRCC_F times item CSHO). Net cash distribution, $CD_t = B_{t-1}(Roe_t - g_t)$, in which B_{t-1} is the book equity at the end of year $t - 1$, Roe_t is return on equity in year t , and g_t is the book equity growth in t . We use autoregressive processes to forecast Roe and book equity growth in future years. We model Roe as a first-order autoregressive process with an autocorrelation coefficient of 0.57 and a long-run mean of 0.12, and the growth in book equity as a first-order autoregressive process with an autocorrelation coefficient of 0.24 and a long-run mean of 0.06. For the starting year ($t=0$), we measure Roe as income before extraordinary items (item IB) divided by 1-year-lagged book equity (item CEQ), and the book equity growth rate as the annual change in sales (item SALE). Finally, we use a forecasting period of $T=10$ years and a cost of equity of $r=0.12$. Firms are excluded if book equity ever becomes negative during the forecasting period. We also exclude firms with nonpositive Dur . At the end of June of each year t , we sort stocks into deciles based on Dur constructed with data from the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.2.31 Ltg, Ltg^m1, Ltg^m6, and Ltg^m12, long-term growth forecasts. The long-term growth forecast, Ltg , is the consensus median forecast of the long-term earnings growth rate from IBES (item MEDEST, fiscal period indicator = 0). At the end of June of each year t , we assign stocks into deciles based on Ltg reported in December of $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced at the end of June in year $t + 1$. In addition, at the beginning of each month t , we sort stocks into deciles based on Ltg reported in $t - 1$. Monthly decile returns are calculated for the current month t (Ltg^m1), from month t to $t + 5$ (Ltg^m6), and from month t to $t + 11$ (Ltg^m12), and the deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Ltg^m6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ltg^m6 decile. Because the long-term growth forecasts data start in December 1981, the deciles start in January 1982.

A.3 Investment

A.3.1 Aci, abnormal corporate investment. At the end of June of year t , we measure Aci as $Ce_{t-1}/[(Ce_{t-2}+Ce_{t-3}+Ce_{t-4})/3]-1$, in which Ce_{t-j} is capital expenditure (Compustat annual item CAPX) scaled by sales (item SALE) for the fiscal year ending in calendar year $t-j$. The last 3-year average capital expenditure is designed to project the benchmark investment in the portfolio formation year. We exclude firms with sales less than 10 million dollars. At the end of June of each year t , we sort stocks into deciles based on Aci. Monthly decile returns are computed from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.2 I/A, investment-to-assets. At the end of June of each year t , we sort stocks into deciles based on investment-to-assets, I/A, which is measured as total assets (Compustat annual item AT) for the fiscal year ending in calendar year $t-1$ divided by total assets for the fiscal year ending in $t-2$ minus one. Monthly decile returns are computed from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.3 Ia^q1, Ia^q6, and Ia^q12, quarterly investment-to-assets. Quarterly investment-to-assets, Ia^q, is defined as quarterly total assets (Compustat quarterly item ATQ) divided by 4-quarter-lagged total assets minus one. At the beginning of each month t , we sort stocks into deciles based on Ia^q for the latest fiscal quarter ending at least four months ago. Monthly decile returns are calculated for the current month t (Ia^q1), from month t to $t+5$ (Ia^q6), and from month t to $t+11$ (Ia^q12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ia^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ia^q6 decile. For sufficient coverage of quarterly assets data, the Ia^q portfolios start in January 1973.

A.3.4 dPia, changes in PPE and inventory-to-assets. Changes in PPE and Inventory-to-assets, dPia, is defined as the annual change in gross property, plant, and equipment (Compustat annual item PPEGT) plus the annual change in inventory (item INVT) scaled by 1-year-lagged total assets (item AT). At the end of June of each year t , we sort stocks into deciles based on dPia for the fiscal year ending in calendar year $t-1$. Monthly decile returns are computed from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.5 Noa and dNoa, (changes in) net operating assets. We measure net operating assets as operating assets minus operating liabilities. Operating assets are total assets (Compustat annual item AT) minus cash and short-term investment (item CHE). Operating liabilities are total assets minus debt included in current liabilities (item DLC, zero if missing), minus long-term debt (item DLT, zero if missing), minus minority interests (item MIB, zero if missing), minus preferred stocks (item PSTK, zero if missing), and minus common equity (item CEQ). Noa is net operating assets scaled by 1-year-lagged total assets. Changes in net operating assets, dNoa, is the annual change in net operating assets scaled by 1-year-lagged total assets. At the end of June of each year t , we sort stocks into deciles based on Noa, and separately, on dNoa, for the fiscal year ending in calendar year $t-1$. Monthly decile returns are computed from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.6 dLno, changes in long-term net operating assets. We measure dLno as the annual change in net property, plant, and equipment (Compustat item PPENT) plus the change in intangibles (item INTAN) plus the change in other long-term assets (item AO) minus the change in other long-term liabilities (item LO) and plus depreciation and amortization expense (item DP). dLno is the change in long-term net operating assets scaled by the average of total assets (item AT) from the current and prior years. At the end of June of each year t , we sort stocks into deciles based on dLno for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.7 Ig, investment growth. At the end of June of each year t , we sort stocks into deciles based on investment growth, Ig, which is the growth rate in capital expenditure (Compustat annual item CAPX) from the fiscal year ending in calendar year $t - 2$ to the fiscal year ending in $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.8 2Ig, 2-year investment growth. At the end of June of each year t , we sort stocks into deciles based on 2-year investment growth, 2Ig, which is the growth rate in capital expenditure (Compustat annual item CAPX) from the fiscal year ending in calendar year $t - 3$ to the fiscal year ending in $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.9 3Ig, 3-year investment growth. At the end of June of each year t , we sort stocks into deciles based on 3-year investment growth, 3Ig, which is the growth rate in capital expenditure (Compustat annual item CAPX) from the fiscal year ending in calendar year $t - 4$ to the fiscal year ending in $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.10 Nsi, net stock issues. At the end of June of year t , we measure net stock issues, Nsi, as the natural log of the ratio of the split-adjusted shares outstanding at the fiscal year ending in calendar year $t - 1$ to the split-adjusted shares outstanding at the fiscal year ending in $t - 2$. The split-adjusted shares outstanding is shares outstanding (Compustat annual item CSHO) times the adjustment factor (item AJEX). At the end of June of each year t , we sort stocks with negative Nsi into two portfolios (1 and 2), stocks with zero Nsi into 1 portfolio (3), and stocks with positive Nsi into seven portfolios (4 to 10). Monthly decile returns are from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.11 dIi, percentage change in investment relative to industry. Following Abarbanell and Bushee (1998), we define the $\%d(\cdot)$ operator as the percentage change in the variable in the parentheses from its average over the prior two years. For example, $\%d(\text{Investment}) = [\text{Investment}(t) - E[\text{Investment}(t)]]/E[\text{Investment}(t)]$, in which $E[\text{Investment}(t)] = [\text{Investment}(t - 1) + \text{Investment}(t - 2)]/2$. dIi is defined as $\%d(\text{Investment}) - \%d(\text{Industry investment})$, in which investment is capital expenditure in property, plant, and equipment (Compustat annual item CAPXV). Industry investment is the aggregate investment across all firms with the same 2-digit SIC code. Firms with nonpositive $E[\text{Investment}(t)]$ are excluded and we require at least two firms in each industry. At the end of June of each year t , we sort stocks into deciles based on dIi for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.12 Cei, composite equity issuance. At the end of June of each year t , we sort stocks into deciles based on composite equity issuance, Cei, which is the log growth rate in the market equity not attributable to stock return, $\log(\text{Me}_t/\text{Me}_{t-5}) - r(t - 5, t)$. $r(t - 5, t)$ is the cumulative log stock return from the last trading day of June in year $t - 5$ to the last trading day of June in year t , and Me_t is the market equity (from CRSP) on the last trading day of June in year t . Monthly decile returns are from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.13 Cdi, composite debt issuance. At the end of June of each year t , we sort stocks into deciles based on composite debt issuance, Cdi, which is the log growth rate of the book value of debt (Compustat annual item DLC plus item DLTT) from the fiscal year ending in calendar year $t - 6$ to the fiscal year ending in year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of year $t + 1$.

A.3.14 Ivg, inventory growth. At the end of June of each year t , we sort stocks into deciles based on inventory growth, Ivg, which is the annual growth rate in inventory (Compustat annual item INVT) from the fiscal year ending in calendar year $t-2$ to the fiscal year ending in $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.15 Ivc, inventory changes. At the end of June of each year t , we sort stocks into deciles based on inventory changes, Ivc, which is the annual change in inventory (Compustat annual item INVT) scaled by the average of total assets (item AT) for the fiscal years ending in $t-2$ and $t-1$. We exclude firms that carry no inventory for the past two fiscal years. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.16 Oa, operating accruals. Prior to 1988, we use the balance sheet approach in Sloan (1996) to measure operating accruals, Oa, as changes in noncash working capital minus depreciation, in which the noncash working capital is changes in noncash current assets minus changes in current liabilities less short-term debt and taxes payable. In particular, $Oa = (dCA - dCASH) - (dCL - dSTD - dTP) - DP$, in which dCA is the change in current assets (Compustat annual item ACT), dCASH is the change in cash or cash equivalents (item CHE), dCL is the change in current liabilities (item LCT), dSTD is the change in debt included in current liabilities (item DLC), dTP is the change in income taxes payable (item TXP), and DP is depreciation and amortization (item DP). Missing changes in income taxes payable are set to zero. Starting from 1988, we follow Hribar and Collins (2002) to measure Oa using the statement of cash flows as net income (item NI) minus net cash flow from operations (item OANCF). Doing so helps mitigate measurement errors that can arise from nonoperating activities such as acquisitions and divestitures. Data from the statement of cash flows are only available since 1988. At the end of June of each year t , we sort stocks into deciles on Oa for the fiscal year ending in calendar year $t-1$ scaled by total assets (item AT) for the fiscal year ending in $t-2$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.17 Ta, total accruals. Prior to 1988, we use the balance sheet approach in Richardson et al. (2005) to measure total accruals, $Ta = dWc + dNco + dFin$. dWc is the change in net noncash working capital. Net noncash working capital is current operating asset (Coa) minus current operating liabilities (Col), with $Coa = \text{current assets (Compustat annual item ACT)} - \text{cash and short-term investments (item CHE)}$ and $Col = \text{current liabilities (item LCT)} - \text{debt in current liabilities (item DLC)}$. dNco is the change in net noncurrent operating assets. Net noncurrent operating assets are noncurrent operating assets (Nca) minus noncurrent operating liabilities (Ncl), with $Nca = \text{total assets (item AT)} - \text{current assets} - \text{long-term investments (item IVAO)}$, and $Ncl = \text{total liabilities (item LT)} - \text{current liabilities} - \text{long-term debt (item DLTT)}$. dFin is the change in net financial assets. Net financial assets are financial assets (Fna) minus financial liabilities (Fnl), with $Fna = \text{short-term investments (item IVST)} + \text{long-term investments}$, and $Fnl = \text{long-term debt} + \text{debt in current liabilities} + \text{preferred stocks (item PSTK)}$. Missing changes in debt in current liabilities, long-term investments, long-term debt, short-term investments, and preferred stocks are set to zero.

Starting from 1988, we use the cash flow approach to measure Ta as net income (item NI) minus total operating, investing, and financing cash flows (items OANCF, IVNCF, and FINCF) plus sales of stocks (item SSTK, zero if missing) minus stock repurchases and dividends (items PRSTKC and DV, zero if missing). Data from the statement of cash flows are only available since 1988. At the end of June of each year t , we sort stocks into deciles based on Ta for the fiscal year ending in calendar year $t-1$ scaled by total assets for the fiscal year ending in $t-2$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.3.18 dWc, dCoa, and dCol, changes in net noncash working capital, in current operating assets, and in current operating liabilities. dWc is the change in net noncash working capital.

Net noncash working capital is current operating assets (Coa) minus current operating liabilities (Col), with Coa = current assets (Compustat annual item ACT) – cash and short term investments (item CHE) and Col = current liabilities (item LCT) – debt in current liabilities (item DLC). dCoa is the change in current operating assets, and dCol is the change in current operating liabilities. Missing changes in debt in current liabilities are set to zero. At the end of June of each year t , we sort stocks into deciles based on dWc, dCoa, and dCol for the fiscal year ending in calendar year $t - 1$, all scaled by total assets (item AT) for the fiscal year ending in calendar year $t - 2$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.19 dNco, dNca, and dNcl, changes in net noncurrent operating assets, in noncurrent operating assets, and in noncurrent operating liabilities. dNco is the change in net noncurrent operating assets. Net noncurrent operating assets are noncurrent operating assets (Nca) minus noncurrent operating liabilities (Ncl), with Nca = total assets (Compustat annual item AT) – current assets (item ACT) – long-term investments (item IVAO), and Ncl = total liabilities (item LT) – current liabilities (item LCT) – long-term debt (item DLTT). dNca is the change in noncurrent operating assets, and dNcl is the change in noncurrent operating liabilities. Missing changes in long-term investments and long-term debt are set to zero. At the end of June of each year t , we sort stocks into deciles based, on dNco, dNca, and dNcl for the fiscal year ending in calendar year $t - 1$, all scaled by total assets for the fiscal year ending in calendar year $t - 2$. Monthly decile returns are from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.20 dFin, dSti, dLti, dFnl, and dBe, changes in net financial assets, in short-term investments, in long-term investments, in financial liabilities, and in book equity. dFin is the change in net financial assets. Net financial assets are financial assets (Fna) minus financial liabilities (Fnl), with Fna = short-term investments (Compustat annual item IVST) + long-term investments (item IVAO), and Fnl = long-term debt (item DLTT) + debt in current liabilities (item DLC) + preferred stock (item PSTK). dSti is the change in short-term investments, dLti is the change in long-term investments, and dFnl is the change in financial liabilities. dBe is the change in book equity (item CEQ). Missing changes in debt in current liabilities, long-term investments, long-term debt, short-term investments, and preferred stocks are set to zero (at least 1 change must be non-missing). When constructing dSti (dLti), we exclude firms that do not have short-term (long-term) investments in the past two fiscal years. At the end of June of each year t , we sort stocks into deciles based, separately, on dFin, dSti, dLti, dFnl, and dBe for the fiscal year ending in calendar year $t - 1$, all scaled by total assets (item AT) for the fiscal year ending in calendar year $t - 2$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. For sufficient data coverage, the dSti portfolios start in July 1971.

A.3.21 Dac, discretionary accruals. We measure discretionary accruals, Dac, from:

$$\frac{Oa_{it}}{A_{it-1}} = \alpha_1 \frac{1}{A_{it-1}} + \alpha_2 \frac{dSALE_{it} - dREC_{it}}{A_{it-1}} + \alpha_3 \frac{PPE_{it}}{A_{it-1}} + e_{it}, \quad (A7)$$

in which Oa_{it} is operating accruals for firm i (see Appendix A.3.16), A_{it-1} is total assets (Compustat annual item AT) at the end of year $t - 1$, $dSALE_{it}$ is the annual change in sales (item SALE) from year $t - 1$ to t , $dREC_{it}$ is the annual change in net receivables (item RECT) from year $t - 1$ to t , and PPE_{it} is gross property, plant, and equipment (item PPEGT) at the end of year t . We winsorize the variables at the right hand side of equation (A7) at the 1st and 99th percentiles of their distributions each year. We estimate the cross-sectional regression (A7) for each 2-digit SIC industry and year combination, formed separately for NYSE/AMEX firms and for NASDAQ firms. We require at least six firms for each regression. The discretionary accrual for stock i is defined as the residual

from the regression, e_{it} . At the end of June of each year t , we sort stocks into deciles based on Dac for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.22 Poa, percent operating accruals. Accruals are traditionally scaled by total assets. Hafzalla, Lundholm, and Van Winkle (2011) show that scaling accruals by the absolute value of earnings (percent accruals) is more effective in selecting firms for which the differences between sophisticated and naive forecasts of earnings are the most extreme. To construct the percent operating accruals (Poa) deciles, at the end of June of each year t , we sort stocks into deciles based on operating accruals scaled by the absolute value of net income (Compustat annual item NI) for the fiscal year ending in calendar year $t - 1$. See Appendix A.3.16 for the measurement of operating accruals. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.23 Pta, percent total accruals. At the end of June of each year t , we sort stocks into deciles on percent total accruals, Pta, calculated as total accruals scaled by the absolute value of net income (Compustat annual item NI) for the fiscal year ending in calendar year $t - 1$. See Appendix A.3.17 for the measurement of total accruals. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of year $t + 1$.

A.3.24 Pda, percent discretionary accruals. At the end of June of each year t , we split stocks into deciles based on percent discretionary accruals, Pda, calculated as the discretionary accruals, Dac , for the fiscal year ending in calendar year $t - 1$ multiplied with total assets (Compustat annual item AT) for the fiscal year ending in $t - 2$ scaled by the absolute value of net income (item NI) for the fiscal year ending in $t - 1$. See Appendix A.3.21 for the measurement of discretionary accruals. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.3.25 Nxf, Nef, and Ndf, net external, equity, and debt financing. Net external financing, Nxf, is the sum of net equity financing, Nef, and net debt financing, Ndf. Nef is the proceeds from the sale of common and preferred stocks (Compustat annual item SSTK) less cash payments for the repurchases of common and preferred stocks (item PRSTKC) less cash payments for dividends (item DV). Ndf is the cash proceeds from the issuance of long-term debt (item DLTIS) less cash payments for long-term debt reductions (item DLTR) plus the net changes in current debt (item DLCCH, zero if missing). At the end of June of each year t , we sort stocks into deciles based on Nxf, and, separately, on Nef and Ndf, for the fiscal year ending in calendar year $t - 1$ scaled by the average of total assets for fiscal years ending in $t - 2$ and $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the data on financing activities start in 1971, the portfolios start in July 1972.

A.4 Profitability

A.4.1 Roe1, Roe6, and Roe12, return on equity. Return on equity, Roe, is income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged book equity (Hou, Xue, and Zhang 2015). Book equity is shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock (item PSTKQ). Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the book value of preferred stock, or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity.

Before 1972, the sample coverage is limited for quarterly book equity in Compustat quarterly files. We expand the coverage by using book equity from Compustat annual files as well as by imputing quarterly book equity with clean surplus accounting. Specifically, whenever available we

first use quarterly book equity from Compustat quarterly files. We then supplement the coverage for fiscal quarter four with annual book equity from Compustat annual files. Annual book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if available. If not, stockholders' equity is the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

If both approaches are unavailable, we apply the clean surplus relation to impute the book equity. First, if available, we backward impute the beginning-of-quarter book equity as the end-of-quarter book equity minus quarterly earnings plus quarterly dividends. Quarterly earnings are income before extraordinary items (Compustat quarterly item IBQ). Quarterly dividends are zero if dividends per share (item DVPSXQ) are zero. Otherwise, total dividends are dividends per share times beginning-of-quarter shares outstanding adjusted for stock splits during the quarter. Shares outstanding are from Compustat (quarterly item CSHOQ supplemented with annual item CSHO for fiscal quarter four) or CRSP (item SHROUT), and the share adjustment factor is from Compustat (quarterly item AJEXQ supplemented with annual item AJEX for fiscal quarter four) or CRSP (item CFACSHR). Because we impose a 4-month lag between earnings and the holding period month (and the book equity in the denominator of Roe is 1-quarter-lagged relative to earnings), all the Compustat data in the backward imputation are at least 4-month lagged prior to the portfolio formation. If data are unavailable for the backward imputation, we impute the book equity for quarter t forward based on book equity from prior quarters. Let BEQ_{t-j} , $1 \leq j \leq 4$ denote the latest available quarterly book equity as of quarter t , and $IBQ_{t-j+1,t}$ and $DVQ_{t-j+1,t}$ be the sum of quarterly earnings and quarterly dividends from quarter $t-j+1$ to t , respectively. BEQ_t can then be imputed as $BEQ_{t-j} + IBQ_{t-j+1,t} - DVQ_{t-j+1,t}$. We do not use prior book equity from more than 4 quarters ago (i.e., $1 \leq j \leq 4$) to reduce imputation errors.

At the beginning of each month t , we sort all stocks into deciles based on their most recent past Roe. Before 1972, we use the most recent Roe computed with quarterly earnings from fiscal quarters ending at least four months prior to the portfolio formation. Starting from 1972, we use Roe computed with quarterly earnings from the most recent quarterly earnings announcements (Compustat quarterly item RDQ). For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Roe to be within six months prior to the portfolio formation. This restriction is imposed to exclude stale earnings information. To avoid potentially erroneous records, we also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for the current month t (Roe1), from month t to $t+5$ (Roe6), and from month t to $t+11$ (Roe12). The deciles are rebalanced monthly. Holding periods longer than one month like in Roe6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Roe6 decile.

A.4.2 dRoe1, dRoe6, and dRoe12, 4-quarter change in return on equity. Change in return on equity, dRoe, is return on equity minus its value from four quarters ago. At the beginning of each month t , we sort all stocks into deciles on their most recent past dRoe. Before 1972, we use the most recent dRoe with quarterly earnings from fiscal quarters ending at least four months ago. Starting from 1972, we use dRoe computed with quarterly earnings from the most recent quarterly earnings announcement dates (Compustat quarterly item RDQ). We require a firm's end of the fiscal quarter that corresponds to its most recent dRoe to be within six months prior to the portfolio formation. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for the current month t (dRoe1), from month t to $t+5$ (dRoe6), and from month t to $t+11$ (dRoe12). The deciles are rebalanced monthly. Holding periods longer than one month like in dRoe6 mean that for a given decile in each month there exist

six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the dRoe6 decile.

A.4.3 Roa1, Roa6, and Roa12, return on assets. Return on assets, Roa, is income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged total assets (item ATQ). At the beginning of each month t , we sort stocks into deciles on Roa computed with quarterly earnings from the most recent earnings announcement dates (item RDQ). For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Roa to be within six months prior to the portfolio formation. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for month t (Roa1), from month t to $t+5$ (Roe6), and from month t to $t+11$ (Roe12). The deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in Roa6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Roa6 decile. For sufficient data, the Roa deciles start in January 1972.

A.4.4 dRoa1, dRoa6, and dRoa12, 4-quarter change in return on assets. Change in return on assets, dRoa, is return on assets minus its value from four quarters ago. At the beginning of each month t , we sort all stocks into deciles based on dRoa computed with quarterly earnings from the most recent earnings announcement dates (Compustat quarterly item RDQ). We require a firm's end of the fiscal quarter that corresponds to its most recent dRoa to be within six months prior to the portfolio formation. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for month t (dRoa1), from month t to $t+5$ (dRoa6), and from month t to $t+11$ (dRoa12). The deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in dRoa6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the dRoa6 decile. For sufficient data coverage, the dRoa portfolios start in January 1973.

A.4.5 Rna, Pm, and Ato, return on net operating assets, profit margin, assets turnover. Soliman (2008) decomposes $Roe = Rna + FLEV \times SPREAD$, in which Roe is return on equity, Rna is return on net operating assets, FLEV is financial leverage, and SPREAD is the difference between return on net operating assets and borrowing costs. We further decompose Rna as $Pm \times Ato$, in which Pm is profit margin and Ato is asset turnover.

We use annual sorts to form Rna, Pm, and Ato deciles. At the end of June of year t , we measure Rna as operating income after depreciation (Compustat annual item OIADP) for the fiscal year ending in calendar year $t-1$ divided by net operating assets (Noa) for the fiscal year ending in $t-2$. Noa is operating assets minus operating liabilities. Operating assets are total assets (item AT) minus cash and short-term investment (item CHE), and minus other investment and advances (item IVAO, zero if missing). Operating liabilities are total assets minus debt in current liabilities (item DLC, zero if missing), minus long-term debt (item DLTT, zero if missing), minus minority interests (item MIB, zero if missing), minus preferred stocks (item PSTK, zero if missing), and minus common equity (item CEQ). Pm is operating income after depreciation divided by sales (item SALE) for the fiscal year ending in calendar year $t-1$. Ato is sales for the fiscal year ending in calendar year $t-1$ divided by Noa for the fiscal year ending in $t-2$.

At the end of June of each year t , we sort stocks into three sets of deciles based on Rna, Pm, and Ato. We exclude firms with nonpositive Noa for the fiscal year ending in calendar year $t-2$ when forming the Rna and the Ato portfolios. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.4.6 Cto, capital turnover. At the end of June of each year t , we split stocks into deciles based on capital turnover, Cto, measured as sales (Compustat annual item SALE) for the fiscal

year ending in calendar year $t - 1$ divided by total assets (item AT) for the fiscal year ending in $t - 2$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.4.7 Rna^q1, Rna^q6, Rna^q12, Pm^q1, Pm^q6, Pm^q12, Ato^q1, Ato^q6, and Ato^q12, quarterly return on net operating assets, quarterly profit margin, quarterly assets turnover. Quarterly return on net operating assets, Rna^q, is quarterly operating income after depreciation (Compustat quarterly item OIADPQ) divided by 1-quarter-lagged net operating assets (Noa). Noa is operating assets minus operating liabilities. Operating assets are total assets (item ATQ) minus cash and short-term investments (item CHEQ), and minus other investment and advances (item IVAOQ, zero if missing). Operating liabilities are total assets minus debt in current liabilities (item DLCQ, zero if missing), minus long-term debt (item DLTTQ, zero if missing), minus minority interests (item MIBQ, zero if missing), minus preferred stocks (item PSTKQ, zero if missing), and minus common equity (item CEQQ). Quarterly profit margin, Pm^q, is quarterly operating income after depreciation divided by quarterly sales (item SALEQ). Quarterly asset turnover, Ato^q, is quarterly sales divided by 1-quarter-lagged Noa.

At the beginning of each month t , we sort stocks into deciles based on Rna^q or Pm^q for the latest fiscal quarter ending at least four months ago. Separately, we sort stocks into deciles based on Ato^q computed with quarterly sales from the most recent earnings announcement dates (item RDQ). Sales are generally announced with earnings during quarterly earnings announcements (Jegadeesh and Livnat 2006). For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Ato^q to be within six months prior to the portfolio formation. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for month t (Rna^q1, Pm^q1, and Ato^q1), from month t to $t + 5$ (Rna^q6, Pm^q6, and Ato^q6), and from month t to $t + 11$ (Rna^q12, Pm^q12, and Ato^q12). The deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Ato^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ato^q6 decile. For sufficient data coverage, the Rna^q portfolios start in January 1976 and the Ato^q portfolios start in January 1972.

A.4.8 Cto^q1, Cto^q6, and Cto^q12, quarterly capital turnover. Quarterly capital turnover, Cto^q, is quarterly sales (Compustat quarterly item SALEQ) scaled by 1-quarter-lagged total assets (item ATQ). At the beginning of each month t , we sort stocks into deciles based on Cto^q computed with quarterly sales from the most recent earnings announcement dates (item RDQ). For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Ato^q to be within six months prior to the portfolio formation. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for month t (Cto^q1), from month t to $t + 5$ (Cto^q6), and from month t to $t + 11$ (Cto^q12). The deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Cto^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Cto^q6 decile. For sufficient data, the Cto^q portfolios start in January 1972.

A.4.9 Gpa, gross profits-to-assets. We measure gross profits-to-assets, Gpa, as total revenue (Compustat annual item REVT) minus cost of goods sold (item COGS) divided by total assets (item AT, current, not lagged, total assets). At the end of June of each year t , we sort stocks into deciles on Gpa for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.4.10 Gla, gross profits-to-lagged assets. We measure gross profits-to-lagged assets, Gla, as total revenue (Compustat annual item REVT) minus cost of goods sold (item COGS) divided by

1-year-lagged total assets (item AT). At the end of June of each year t , we sort stocks into deciles based on Gla for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.4.11 Gla^q1 , Gla^q6 , and Gla^q12 , quarterly gross profits-to-lagged assets. Gla^q is quarterly total revenue (Compustat quarterly item REV_{TQ}) minus cost of goods sold (item COGS_Q) divided by 1-quarter-lagged total assets (item AT_Q). At the beginning of each month t , we sort stocks into deciles on Gla^q for the fiscal quarter ending at least four months ago. Monthly decile returns are calculated for month t (Gla^q1), from month t to $t + 5$ (Gla^q6), and from month t to $t + 11$ (Gla^q12). The deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Gla^q6 mean that for a given decile in each month there exist six subdeciles, each of which is initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Gla^q6 decile. For sufficient data, the Gla^q deciles start in January 1976.

A.4.12 Ope, operating profits to equity. Following Fama and French (2015), we measure operating profitability to equity, Ope, as total revenue (Compustat annual item REV_T) minus cost of goods sold (item COGS, zero if missing), minus selling, general, and administrative expenses (item XSGA, zero if missing), and minus interest expense (item XINT, zero if missing), scaled by book equity (the denominator is current, not lagged, book equity). We require at least one of the three expense items (COGS, XSGA, and XINT) to be nonmissing. Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. At the end of June of each year t , we sort stocks into deciles based on Ope for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.4.13 Ole, operating profits-to-lagged equity. Ole is total revenue (Compustat annual item REV_T) minus cost of goods sold (item COGS, zero if missing), minus selling, general, and administrative expenses (item XSGA, zero if missing), and minus interest expense (item XINT, zero if missing), scaled by 1-year-lagged book equity. We require at least one of the three expense items (COGS, XSGA, and XINT) to be nonmissing. Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. At the end of June of each year t , we sort stocks into deciles on Ole for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.4.14 Ole^q1 , Ole^q6 , and Ole^q12 , quarterly operating profits-to-lagged equity. Quarterly operating profits-to-lagged equity, Ole^q , is quarterly total revenue (Compustat quarterly item REV_{TQ}) minus cost of goods sold (item COGS_Q, zero if missing), minus selling, general, and administrative expenses (item XSGA_Q, zero if missing), and minus interest expense (item XINT_Q, zero if missing), scaled by 1-quarter-lagged book equity. We require at least one of the three expense items (COGS_Q, XSGA_Q, and XINT_Q) to be nonmissing. Book equity is shareholders' equity, plus

balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock (item PSTKQ). Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the book value of preferred stock, or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity.

At the beginning of each month t , we split stocks on Ole^q for the fiscal quarter ending at least four months ago. Monthly decile returns are calculated for month t (Ole^q1), from month t to $t+5$ (Ole^q6), and from month t to $t+11$ (Ole^q12). The deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in Ole^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ole^q6 decile. For sufficient data coverage, the Ole^q portfolios start in January 1972.

A.4.15 Opa, operating profits-to-assets. Following Ball et al. (2016), we measure operating profits-to-assets, Opa, as total revenue (Compustat annual item REVT) minus cost of goods sold (item COGS), minus selling, general, and administrative expenses (item XSGA), and plus research and development expenditures (item XRD, zero if missing), scaled by book assets (item AT, the denominator is current, not lagged, total assets). At the end of June of each year t , we sort stocks into deciles based on Opa for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.4.16 Ola, operating profits-to-lagged assets. We measure operating profits-to-lagged assets, Ola, as total revenue (Compustat annual item REVT) minus cost of goods sold (item COGS), minus selling, general, and administrative expenses (item XSGA), and plus research and development expenditures (item XRD, zero if missing), scaled by 1-year-lagged book assets (item AT). At the end of June of each year t , we sort stocks into deciles based on Ola for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.4.17 Ola^q1, Ola^q6, and Ola^q12, quarterly operating profits-to-lagged assets. We measure quarterly operating profits-to-lagged assets, Ola^q, as quarterly total revenue (Compustat quarterly item REVTQ) minus cost of goods sold (item COGSQ), minus selling, general, and administrative expenses (item XSGAQ), plus research and development expenditures (item XRDQ, zero if missing), scaled by 1-quarter-lagged book assets (item ATQ). At the beginning of each month t , we sort stocks into deciles based on Ola^q for the fiscal quarter ending at least four months ago. Monthly decile returns are calculated for month t (Ola^q1), from month t to $t+5$ (Ola^q6), and from month t to $t+11$ (Ola^q12). The deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in Ola^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ola^q6 decile. For sufficient data coverage, the Ola^q portfolios start in January 1976.

A.4.18 Cop, cash-based operating profitability. Following Ball et al. (2016), we measure cash-based operating profitability, Cop, as total revenue (Compustat annual item REVT) minus cost of goods sold (item COGS), minus selling, general, and administrative expenses (item XSGA), plus research and development expenditures (item XRD, zero if missing), minus change in accounts receivable (item RECT), minus change in inventory (item INVT), minus change in prepaid expenses (item XPP), plus change in deferred revenue (item DRC plus item DRLT), plus change in trade accounts payable (item AP), and plus change in accrued expenses (item XACC), all scaled by book assets (item AT, current, not lagged, total assets). All changes are annual changes in balance sheet items and we set missing changes to zero. At the end of June of each year t , we sort stocks into deciles based on Cop for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.4.19 Cla, cash-based operating profits-to-lagged assets. We measure cash-based operating profits-to-lagged assets, Cla, as total revenue (Compustat annual item REVT) minus cost of goods sold (item COGS), minus selling, general, and administrative expenses (item XSGA), plus research and development expenditures (item XRD, zero if missing), minus change in accounts receivable (item RECT), minus change in inventory (item INVT), minus change in prepaid expenses (item XPP), plus change in deferred revenue (item DRC plus item DRLT), plus change in trade accounts payable (item AP), and plus change in accrued expenses (item XACC), all scaled by 1-year-lagged book assets (item AT). All changes are annual changes in balance sheet items and we set missing changes to zero. At the end of June of each year t , we sort stocks into deciles based on Cla for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.4.20 Cla^{q1}, Cla^{q6}, and Cla^{q12}, quarterly cash-based operating profits-to-lagged assets.

We measure quarterly cash-based operating profits-to-lagged assets, Cla, as quarterly total revenue (Compustat quarterly item REVTQ) minus cost of goods sold (item COGSQ), minus selling, general, and administrative expenses (item XSGAQ), plus research and development expenditures (item XRDQ, zero if missing), minus change in accounts receivable (item RECTQ), minus change in inventory (item INVTQ), plus change in deferred revenue (item DRCQ plus item DRLTQ), and plus change in trade accounts payable (item APQ), all scaled by 1-quarter-lagged book assets (item ATQ). All changes are quarterly changes in balance sheet items and we set missing changes to zero. At the beginning of each month t , we split stocks on Cla^q for the fiscal quarter ending at least 4 months ago. Monthly decile returns are calculated for month t (Cla^{q1}), from month t to $t + 5$ (Cla^{q6}), and from month t to $t + 11$ (Cla^{q12}). The deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Cla^{q6} mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Cla^{q6} decile. For sufficient data coverage, the Cla^q portfolios start in January 1976.

A.4.21 F, fundamental score. Piotroski (2000) classifies each fundamental signal as either good or bad depending on the signal's implication for future stock prices and profitability. An indicator variable for a particular signal is one if its realization is good and zero if it is bad. The aggregate signal, denoted F, is the sum of the nine binary signals. F is designed to measure the overall quality, or strength, of the firm's financial position. The nine fundamental signals are chosen to measure three areas of a firm's financial condition, profitability, liquidity, and operating efficiency.

We use four variables to measure profitability: (1) Roa is income before extraordinary items (Compustat annual item IB) scaled by 1-year-lagged total assets (item AT). If the firm's Roa is positive, the indicator variable F_{Roa} equals one and zero otherwise. (2) Cf/A is cash flow from operation scaled by 1-year-lagged total assets. Cash flow from operation is net cash flow from operating activities (item OANCF) if available, or funds from operation (item FOPT) minus the annual change in working capital (item WCAP). If the firm's Cf/A is positive, the indicator variable $F_{Cf/A}$ equals one and zero otherwise. (3) dRoa is the current year's Roa less the prior year's Roa. If dRoa is positive, the indicator variable F_{dRoa} is one and zero otherwise. Finally, (4) the indicator F_{Acc} equals one if $Cf/A > Roa$ and zero otherwise.

We use three variables to measure changes in capital structure and a firm's ability to meet debt obligations. Piotroski (2000) assumes that an increase in leverage, a deterioration of liquidity, or the use of external financing is a bad signal about financial risk. (1) dLever is the change in the ratio of total long-term debt (Compustat annual item DLTT) to the average of current and 1-year-lagged total assets. F_{dLever} is one if the firm's leverage ratio falls, that is, $dLever < 0$, and zero otherwise. (2) dLiquid measures the change in a firm's current ratio from the prior year, in which the current ratio is the ratio of current assets (item ACT) to current liabilities (item LCT). An improvement in liquidity ($\Delta dLiquid > 0$) is a good signal about the firm's ability to service debt obligations. The indicator $F_{dLiquid}$ equals one if the firm's liquidity improves and zero otherwise. (3) The indicator,

Eq, equals one if the firm does not issue common equity during the current year and zero otherwise. The issuance of common equity is sales of common and preferred stocks (item SSTK) minus any increase in preferred stocks (item PSTK). Issuing equity is interpreted as a bad signal (inability to generate sufficient internal funds).

The remaining two signals are designed to measure changes in the efficiency of the firm's operations that reflect two key constructs underlying the decomposition of return on assets. (1) dMargin is the firm's current gross margin ratio, measured as gross margin (Compustat annual item SALE minus item COGS) scaled by sales (item SALE), less the prior year's gross margin ratio. An improvement in margins signifies a potential improvement in factor costs, a reduction in inventory costs, or a rise in the price of the firm's product. The indicator $F_{dMargin}$ equals one if $dMargin > 0$ and zero otherwise. (2) dTurn is the firm's current year asset turnover ratio, measured as total sales scaled by 1-year-lagged total assets (item AT), minus the prior year's asset turnover ratio. An improvement in asset turnover ratio signifies greater productivity from the asset base. The indicator, F_{dTurn} , equals one if $dTurn > 0$ and zero otherwise.

Piotroski (2000) forms a composite score, F , as the sum of the individual binary signals:

$$F \equiv F_{Roa} + F_{dRoa} + F_{Cf/A} + F_{Acc} + F_{dMargin} + F_{dTurn} + F_{dLever} + F_{dLiquid} + Eq. \quad (A8)$$

At the end of June of each year t , we sort stocks based on F for the fiscal year ending in calendar year $t-1$ to form seven portfolios: low ($F = 0, 1, 2$), 3, 4, 5, 6, 7, and high ($F = 8, 9$). Because extreme F scores are rare, we combine scores 0, 1, and 2 into the low portfolio and scores 8 and 9 into the high portfolio. Monthly portfolio returns are calculated from July of year t to June of $t+1$, and the portfolios are rebalanced in June of $t+1$. For sufficient data coverage, the F portfolio returns start in July 1972.

A.4.22 F^q1 , F^q6 , and F^q12 , quarterly fundamental score. To construct quarterly F -score, F^q , we use quarterly accounting data and the same nine binary signals from Piotroski (2000). Among the four signals related to profitability, (1) Roa is quarterly income before extraordinary items (Compustat quarterly item IBQ) scaled by 1-quarter-lagged total assets (item ATQ). If the firm's Roa is positive, the indicator variable F_{Roa} equals one and zero otherwise. (2) Cf/A is quarterly cash flow from operation scaled by 1-quarter-lagged total assets. Cash flow from operation is the quarterly change in year-to-date net cash flow from operating activities (item OANCFY) if available, or the quarterly change in year-to-date funds from operation (item FOPTY) minus the quarterly change in working capital (item WCAPQ). If the firm's Cf/A is positive, the indicator variable $F_{Cf/A}$ equals one and zero otherwise. (3) $dRoa$ is the current quarter's Roa less the Roa from four quarters ago. If $dRoa$ is positive, the indicator variable F_{dROA} is one and zero otherwise. Finally, (iv) the indicator F_{Acc} equals one if $Cf/A > Roa$ and zero otherwise.

Among the three signals related changes in capital structure and a firm's ability to meet debt obligations: (1) $dLever$ is the change in the ratio of total long-term debt (Compustat quarterly item DLTQT) to the average of current and 1-quarter-lagged total assets. F_{dLever} is one if the firm's leverage ratio falls, that is, $dLever < 0$, relative to its value four quarters ago, and zero otherwise. (2) $dLiquid$ measures the change in a firm's current ratio between the current quarter and four quarters ago, in which the current ratio is the ratio of current assets (item ACTQ) to current liabilities (item LCTQ). An improvement in liquidity ($dLiquid > 0$) is a good signal about the firm's ability to service current debt obligations. The indicator $F_{dLiquid}$ equals one if the firm's liquidity improves and zero otherwise. (3) The indicator, Eq , equals one if the firm does not issue common equity during the past four quarters and zero otherwise. The issuance of common equity is sales of common and preferred stocks minus any increase in preferred stocks (item PSTKQ). To measure sales of common and preferred stocks, we first compute the quarterly change in year-to-date sales of common and preferred stocks (item SSTKY) and then take the total change for the past four quarters. Issuing equity is interpreted as a bad signal (inability to generate sufficient internal funds).

For the remaining two signals, (1) $dMargin$ is the firm's current gross margin ratio, measured as gross margin (item SALEQ minus item COGSQ) scaled by sales (item SALEQ), less the gross

margin ratio from four quarters ago. The indicator $F_{dMargin}$ equals one if $dMargin > 0$ and zero otherwise. (2) $dTurn$ is the firm's current asset turnover ratio, measured as (item SALEQ) scaled by 1-quarter-lagged total assets (item ATQ), minus the asset turnover ratio from four quarters ago. The indicator, F_{dTurn} , equals one if $dTurn > 0$ and zero otherwise.

The composite score, F^q , is the sum of the individual binary signals:

$$F^q \equiv F_{Roa} + F_{dRoa} + F_{Cf/A} + F_{Acc} + F_{dMargin} + F_{dTurn} + F_{dLever} + F_{dLiquid} + Eq. \quad (A9)$$

At the beginning of each month t , we sort stocks based on F^q for the fiscal quarter ending at least 4 quarters ago to form seven portfolios: low ($F^q = 0, 1, 2$), 3, 4, 5, 6, 7, and high ($F^q = 8, 9$). Monthly portfolio returns are calculated for month t ($F^q 1$), from month t to $t+5$ ($F^q 6$), and from month t to $t+11$ ($F^q 12$), and the portfolios are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in $F^q 6$ mean that for a given portfolio in each month there exist six subportfolios, each initiated in a different month in prior six months. We average the subportfolio returns as the monthly return of the $F^q 6$ portfolio. For sufficient data coverage, the F^q portfolios start in January 1985.

A.4.23 F_p , $F_p^m 1$, $F_p^m 6$, and $F_p^m 12$, failure probability, annual and monthly sorts. Failure probability (F_p) is from Campbell, Hilscher, and Szilagyi (2008, table IV, column 3):

$$F_p \equiv -9.164 - 20.264NIMTAAVG_t + 1.416TLMTA_t - 7.129EXRETAVG_t \\ + 1.411SIGMA_t - 0.045RSIZE_t - 2.132CASHMTA_t + 0.075MB_t - 0.058PRICE_t \quad (A10)$$

in which

$$NIMTAAVG_{t-1,t-12} \equiv \frac{1-\phi^3}{1-\phi^{12}} \left(NIMTA_{t-1,t-3} + \dots + \phi^9 NIMTA_{t-10,t-12} \right) \quad (A11)$$

$$EXRETAVG_{t-1,t-12} \equiv \frac{1-\phi}{1-\phi^{12}} \left(EXRET_{t-1} + \dots + \phi^{11} EXRET_{t-12} \right), \quad (A12)$$

and $\phi = 2^{-1/3}$. $NIMTA$ is net income (Compustat quarterly item NIQ) divided by the sum of market equity (share price times the number of shares outstanding from CRSP) and total liabilities (item LTQ). The moving average $NIMTAAVG$ captures that a long history of losses is a better predictor of bankruptcy than one large quarterly loss in a single month. $EXRET \equiv \log(1 + R_{it}) - \log(1 + R_{S\&P500,t})$ is the monthly log excess return on each firm's equity relative to the S&P 500 index. The moving average $EXRETAVG$ captures that a sustained decline in stock market value is a better predictor of bankruptcy than a sudden decline in a single month.

$TLMTA$ is total liabilities divided by the sum of market equity and total liabilities. $SIGMA$ is the annualized 3-month rolling sample standard deviation: $\sqrt{\frac{252}{N-1} \sum_{k \in \{t-1, t-2, t-3\}} r_k^2}$, in which k is the index of trading days in months $t-1$, $t-2$, and $t-3$, r_k is the firm-level daily return, and N is the total number of trading days in the 3-month period. $SIGMA$ is treated as missing if there are fewer than five nonzero observations over the three months in the rolling window. $RSIZE$ is the relative size of each firm measured as the log ratio of its market equity to that of the S&P 500 index. $CASHMTA$ is cash and short-term investments (Compustat quarterly item CHEQ) divided by the sum of market equity and total liabilities (item LTQ). MB is the market-to-book equity, in which we add 10% of the difference between the market equity and the book equity to the book equity to alleviate measurement issues for extremely small book equity values (Campbell, Hilscher, and Szilagyi 2008). For firm-month observations that still have negative book equity after this adjustment, we replace these negative values with \$1 to ensure that the market-to-book ratios for these firms are in the right tail of the distribution. $PRICE$ is each firm's log price per share, truncated above at \$15. We further eliminate stocks with prices less than \$1 at the portfolio formation date. We winsorize the variables on the right-hand side of equation (A10) at the 1st and 99th percentiles of their distributions each month.

To form the Fp deciles, we sort stocks at the end of June of year t based on Fp calculated with accounting data from the fiscal quarter ending at least four months ago. Because unlike earnings, other quarterly data items in the definition of Fp might not be available upon earnings announcement, we impose a 4-month gap between the fiscal quarter end and portfolio formation to guard against look-ahead bias. We calculate decile returns from July of year t to June of year $t+1$, and the deciles are rebalanced in June. For sufficient data coverage, the Fp deciles start in July 1976.

At the beginning of each month t , we split stocks into deciles based on Fp calculated with accounting data from the fiscal quarter ending at least four months ago. We calculate decile returns for the current month t (Fp^m1), from month t to $t+5$ (Fp^m6), and from month t to $t+11$ (Fp^m12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Fp^m6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Fp^m6 decile. For sufficient data coverage, the Fp^m deciles start in January 1976.

A.4.24 O, Ohlson's O-score. We follow Ohlson (1980, model 1 in Table 4) to construct O-score (Dichev 1998):

$$O \equiv -1.32 - 0.407 \log(TA) + 6.03 TLTA - 1.43 WCTA + 0.076 CLCA \\ - 1.72 OENEG - 2.37 NITA - 1.83 FUTL + 0.285 IN2 - 0.521 CHIN, \quad (A13)$$

in which TA is total assets (Compustat annual item AT). TLTA is the leverage ratio defined as total debt (item DLC plus item DLTT) divided by total assets. WCTA is working capital (item ACT minus item LCT) divided by total assets. CLCA is current liability (item LCT) divided by current assets (item ACT). OENEG is one if total liabilities (item LT) exceeds total assets and zero otherwise. NITA is net income (item NI) divided by total assets. FUTL is the fund provided by operations (item PI plus item DP) divided by total liabilities. IN2 is equal to one if net income is negative for the last two years and zero otherwise. CHIN is $(NI_s - NI_{s-1}) / (|NI_s| + |NI_{s-1}|)$, in which NI_s and NI_{s-1} are the net income for the current and prior years. We winsorize all nondummy variables on the right-hand side of equation (A13) at the 1st and 99th percentiles of their distributions each year. At the end of June of each year t , we sort stocks into deciles based on O-score for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.4.25 O^q1, O^q6, and O^q12, Quarterly O-score. We use quarterly accounting data to construct the quarterly O-score as:

$$O^q \equiv -1.32 - 0.407 \log(TA^q) + 6.03 TLTA^q - 1.43 WCTA^q + 0.076 CLCA^q \\ - 1.72 OENEG^q - 2.37 NITA^q - 1.83 FUTL^q + 0.285 IN2^q - 0.521 CHIN^q, \quad (A14)$$

in which TA^q is total assets (Compustat quarterly item ATQ). $TLTA^q$ is the leverage ratio defined as total debt (item DLCQ plus item DLTTQ) divided by total assets. $WCTA^q$ is working capital (item ACTQ minus item LCT) divided by total assets. $CLCA^q$ is current liability (item LCTQ) divided by current assets (item ACTQ). $OENEG^q$ is one if total liabilities (item LTQ) exceeds total assets and zero otherwise. $NITA^q$ is the sum of net income (item NIQ) for the trailing four quarters divided by total assets at the end of the current quarter. $FUTL^q$ is the sum of funds provided by operations (item PIQ plus item DPQ) for the trailing four quarters divided by total liabilities at the end of the current quarter. $IN2^q$ is equal to one if net income is negative for the current quarter and four quarters ago, and zero otherwise. $CHIN^q$ is $(NIQ_s - NIQ_{s-4}) / (|NIQ_s| + |NIQ_{s-4}|)$, in which NIQ_s and NIQ_{s-4} are the net income for the current quarter and four quarters ago. We winsorize all nondummy variables on the right-hand side of equation (A14) at the 1st and 99th percentiles of their distributions each month.

At the beginning of each month t , we sort stocks into deciles based on O^q calculated with accounting data from the fiscal quarter ending at least four months ago. We calculate decile returns for the current month t (O^q1), from month t to $t+5$ (O^q6), and from month t to $t+11$ (O^q12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in O^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the O^q6 decile. For sufficient data coverage, the O^q portfolios start in January 1976.

A.4.26 Z, Altman's Z-score. We follow Altman (1968) to construct the Z-score (Dichev 1998):

$$Z \equiv 1.2WCTA + 1.4RETA + 3.3EBITTA + 0.6METL + SALETA, \quad (A15)$$

in which WCTA is working capital (Compustat annual item ACT minus item LCT) divided by total assets (item AT), RETA is retained earnings (item RE) divided by total assets, EBITTA is earnings before interest and taxes (item OIADP) divided by total assets, METL is the market equity (from CRSP, at fiscal year end) divided by total liabilities (item LT), and SALETA is sales (item SALE) divided by total assets. For firms with more than one share class, we merge the market equity for all share classes before computing Z. We winsorize all nondummy variables on the right-hand side of equation (A15) at the 1st and 99th percentiles of their distributions each year. At the end of June of each year t , we split stocks into deciles based on Z-score for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.4.27 Z^q1 , Z^q6 , and Z^q12 , quarterly Z-score. We use quarterly accounting data to construct the quarterly Z-score as

$$Z^q \equiv 1.2WCTA^q + 1.4RETA^q + 3.3EBITTA^q + 0.6METL^q + SALETA^q, \quad (A16)$$

in which $WCTA^q$ is working capital (Compustat quarterly item ACTQ minus item LCTQ) divided by total assets (item ATQ), $RETA^q$ is retained earnings (item REQ) divided by total assets, $EBITTA^q$ is the sum of earnings before interest and taxes (item OIADPQ) for the trailing four quarters divided by total assets at the end of the current quarter, $METL^q$ is the market equity (from CRSP, at fiscal quarter end) divided by total liabilities (item LTQ), and $SALETA^q$ is the sum of sales (item SALEQ) for the trailing four quarters divided by total assets at the end of the current quarter. For firms with more than one share class, we merge the market equity for all share classes before computing Z^q . We winsorize all nondummy variables on the right-hand side of equation (A16) at the 1st and 99th percentiles of their distributions each month.

At the beginning of each month t , we split stocks into deciles based on Z^q calculated with accounting data from the fiscal quarter ending at least four months ago. We calculate decile returns for the current month t (Z^q1), from month t to $t+5$ (Z^q6), and from month t to $t+11$ (Z^q12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Z^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Z^q6 decile. For sufficient data coverage, the Z^q portfolios start in January 1976.

A.4.28 Cr1, Cr6, and Cr12, credit ratings. Following Avramov et al. (2009), we measure credit ratings, Cr, by transforming S&P ratings into numerical scores as follows: AAA=1, AA+=2, AA=3, AA-=4, A+=5, A=6, A-=7, BBB+=8, BBB=9, BBB-=10, BB+=11, BB=12, BB-=13, B+=14, B=15, B-=16, CCC+=17, CCC=18, CCC-=19, CC=20, C=21, and D=22. At the beginning of each month t , we sort stocks into quintiles based on Cr at the end of $t-1$. We do not form deciles because a disproportional number of firms can have the same rating, which leads to fewer than ten portfolios. We calculate quintile returns for the current month t (Cr1), from month t to $t+5$ (Cr6), and from month t to $t+11$ (Cr12). The quintiles are rebalanced at the beginning of month $t+1$.

Holding periods longer than one month like in Cr6 mean that for a given quintile in each month there exist six subquintiles, each initiated in a different month in the prior six months. We average the subquintile returns as the monthly return of the Cr6 quintile. For sufficient data coverage, the Cr portfolios start in January 1986.

A.4.29 Tbi, taxable income-to-book income. Following Green, Hand, and Zhang (2013), we measure taxable income-to-book income, Tbi, as pretax income (Compustat annual item PI) divided by net income (item NI). At the end of June of each year t , we sort stocks into deciles based on Tbi for the fiscal year ending in calendar year $t - 1$. We exclude firms with nonpositive pretax income or net income. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.4.30 Tbi^q1, Tbi^q6, and Tbi^q12, quarterly taxable income-to-book income. Tbi^q is quarterly pretax income (Compustat quarterly item PIQ) divided by net income (NIQ). At the beginning of each month t , we split stocks into deciles based on Tbi^q calculated with accounting data from the fiscal quarter ending at least four months ago. We exclude firms with nonpositive pretax income or net income. We calculate monthly decile returns for the current month t (Tbi^q1), from month t to $t + 5$ (Tbi^q6), and from month t to $t + 11$ (Tbi^q12). The deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Tbi^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Tbi^q6 decile.

A.4.31 G, growth score. Following Mohanram (2005), we construct the G-score as the sum of eight binary signals: $G \equiv G_1 + \dots + G_8$. G_1 equals one if a firm's return on assets (Roa) is greater than the median Roa in the same industry (2-digit SIC code), and zero otherwise. Roa is net income before extraordinary items (Compustat annual item IB) scaled by the average of total assets (item AT) from the current and prior years. We also calculate an alternative measure of Roa using cash flow from operations instead of net income. Cash flow from operation is net cash flow from operating activities (item OANCF) if available, or funds from operation (item FOPT) minus the annual change in working capital (item WCAP). G_2 equals one if a firm's cash flow Roa exceeds the industry median, and zero otherwise. G_3 equals one if a firm's cash flow from operations exceeds net income, and zero otherwise.

G_4 equals one if a firm's earnings variability is less than the industry median. Earnings variability is the variance of a firm's quarterly Roa during the past 16 quarters (six quarters minimum). Quarterly Roa is quarterly net income before extraordinary items (Compustat quarterly item IBQ) scaled by 1-quarter-lagged total assets (item ATQ). G_5 equals one if a firm's sales growth variability is less the industry median, and zero otherwise. Sales growth variability is the variance of a firm's quarterly sales growth during the past 16 quarters (six quarters minimum). Quarterly sales growth is the growth in quarterly sales (item SALEQ) from its value four quarters ago.

G_6 equals one if a firm's R&D (Compustat annual item XRD) deflated by 1-year-lagged total assets is greater than the industry median, and zero otherwise. G_7 equals one if a firm's capital expenditure (item CAPX) deflated by 1-year-lagged total assets is greater than the industry median, and zero otherwise. G_8 equals one if a firm's advertising expenses (item XAD) deflated by 1-year-lagged total assets is greater than the industry median, and zero otherwise.

At the end of June of each year t , we sort stocks on G for the fiscal year ending in calendar year $t - 1$ to form seven portfolios: low ($F = 0, 1$), 2, 3, 4, 5, 6, and high ($F = 7, 8$). Because extreme G scores are rare, we combine scores 0, and 1 into the low portfolio and scores 7 and 8 into the high portfolio. Monthly returns are from July of year t to June of $t + 1$, and the portfolios are rebalanced in June of $t + 1$. For sufficient data coverage, the G portfolio returns start in July 1976.

A.4.32 Bl, book leverage. Bl is total assets (Compustat annual item AT) divided by book equity. Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment

tax credit (item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value in Compustat (item SEQ), if available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. At the end of June of each year t , we sort stocks into deciles on BI for the fiscal year ending in calendar year $t-1$. Monthly returns are from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.4.33 BI^q1, BI^q6, and BI^q12, quarterly book leverage. Quarterly book leverage, BI^q, is total assets (Compustat quarterly item ATQ) divided by book equity. Book equity is shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock (item PSTKQ). Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the book value of preferred stock, or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity. At the beginning of each month t , we split stocks into deciles on BI^q for the fiscal quarter ending at least 4 months ago. We calculate monthly decile returns for the current month t (BI^q1), from month t to $t+5$ (BI^q6), and from month t to $t+11$ (BI^q12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in BI^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the BI^q6 decile. For sufficient data coverage, the BI^q portfolios start in January 1972.

A.4.34 Sg^q1, Sg^q6, and Sg^q12, quarterly sales growth. Quarterly sales growth, Sg^q, is quarterly sales (Compustat quarterly item SALEQ) divided by its value four quarters ago. At the beginning of each month t , we sort stocks into deciles based on the latest Sg^q. Before 1972, we use the most recent Sg^q from fiscal quarters ending at least four months ago. Starting from 1972, we use Sg^q from the most recent quarterly earnings announcement dates (item RDQ). We require a firm's fiscal quarter end that corresponds to its most recent Sg^q to be within six months prior to the portfolio formation. We also require the earnings announcement date to be after the corresponding fiscal quarter end. We calculate monthly decile returns for the current month t (Sg^q1), from month t to $t+5$ (Sg^q6), and from month t to $t+11$ (Sg^q12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Sg^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Sg^q6 decile.

A.5 Intangibles

A.5.1 Oca and Ioca, (industry-adjusted) organizational capital-to-assets. Following Eisfeldt and Papanikolaou (2013), we construct the stock of organization capital, Oc, using the perpetual inventory method:

$$Oc_{it} = (1 - \delta)Oc_{it-1} + SG\&A_{it}/CPI_t, \quad (A17)$$

in which Oc_{it} is the organization capital of firm i at the end of year t , $SG\&A_{it}$ is selling, general, and administrative (SG&A) expenses (Compustat annual item XSGA) in t , CPI_t is the average consumer price index during year t , and δ is the annual depreciation rate of Oc. The initial stock of Oc is $Oc_{i0} = SG\&A_{i0}/(g + \delta)$, in which $SG\&A_{i0}$ is the first valid SG&A observation (zero or positive) for firm i and g is the long-term growth rate of SG&A. We assume a depreciation rate of 15% for Oc and a long-term growth rate of 10% for SG&A. Missing SG&A values after the starting date are treated as zero. For portfolio formation at the end of June of year t , we require SG&A to be nonmissing for the fiscal year ending in calendar year $t-1$ because this SG&A value

receives the highest weight in Oc . In addition, we exclude firms with zero Oc . Organizational capital-to-assets, Oca , is Oc scaled by total assets (item AT). We industry-standardize Oca with the Fama-French (1997) 17-industry classification. To calculate the industry-adjusted Oca , $Ioca$, we demean a firm's Oca by its industry mean and then divide the demeaned Oca by the standard deviation of Oca within its industry. To alleviate the impact of outliers, we winsorize Oca at the 1st and 99th percentiles of all firms each year before the industry standardization. At the end of June of each year t , we sort stocks into deciles based on Oca , and separately, on $Ioca$, for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.2 Adm, advertising expense-to-market. At the end of June of each year t , we sort stocks into deciles based on advertising expenses-to-market, Adm , which is advertising expenses (Compustat annual item XAD) for the fiscal year ending in calendar year $t - 1$ divided by the market equity (from CRSP) at the end of December of $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Adm . We keep only firms with positive advertising expenses. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because sufficient XAD data start in 1972, the Adm portfolios start in July 1973.

A.5.3 gAd, growth in advertising expense. At the end of June of each year t , we sort stocks into deciles based on growth in advertising expenses, gAd , which is the growth rate of advertising expenses (Compustat annual item XAD) from the fiscal year ending in calendar year $t - 2$ to the fiscal year ending in calendar year $t - 1$. Following Lou (2014), we keep only firms with advertising expenses of at least 0.1 million dollars. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because sufficient XAD data start in 1972, the gAd portfolios start in July 1974.

A.5.4 Rdm, R&D expense-to-market. At the end of June of each year t , we sort stocks into deciles based on R&D-to-market, Rdm , which is R&D expenses (Compustat annual item XRD) for the fiscal year ending in calendar year $t - 1$ divided by the market equity (from CRSP) at the end of December of $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Rdm . We keep only firms with positive R&D expenses. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the accounting treatment of R&D expenses was standardized in 1975, the Rdm portfolios start in July 1976.

A.5.5 Rdm^q1, Rdm^q6, and Rdm^q12, quarterly R&D expense-to-market. At the beginning of each month t , we split stocks into deciles on Rdm^q , which is quarterly R&D expense (Compustat quarterly item $XRDQ$) for the fiscal quarter ending at least four months ago scaled by the market equity (from CRSP) at the end of $t - 1$. For firms with more than one share class, we merge the market equity for all share classes before computing Rdm^q . We keep only firms with positive R&D expenses. We calculate decile returns for the current month t (Rdm^q1), from month t to $t + 5$ (Rdm^q6), and from month t to $t + 11$ (Rdm^q12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Rdm^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Rdm^q6 decile. Because the quarterly R&D data start in late 1989, the Rdm^q deciles start in January 1990.

A.5.6 Rds, R&D expenses-to-sales. At the end of June of each year t , we sort stocks into deciles based on R&D-to-sales, Rds , which is R&D expenses (Compustat annual item XRD) divided by sales (item $SALE$) for the fiscal year ending in calendar year $t - 1$. We keep only firms with positive

R&D expenses. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. Because the accounting treatment of R&D expenses was standardized in 1975, the Rds portfolios start in July 1976.

A.5.7 Rds^{q1}, Rds^{q6}, and Rds^{q12}, quarterly R&D expense-to-sales. At the beginning of each month t , we split stocks into deciles based on quarterly R&D-to-sales, Rds^q, which is quarterly R&D expense (Compustat quarterly item XRDQ) scaled by sales (item SALEQ) for the fiscal quarter ending at least four months ago. We keep only firms with positive R&D expenses. We calculate decile returns for the current month t (Rds^{q1}), from month t to $t+5$ (Rds^{q6}), and from month t to $t+11$ (Rds^{q12}), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Rds^{q6} mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Rds^{q6} decile. Because the quarterly R&D data start in late 1989, the Rds^q portfolios start in January 1990.

A.5.8 Ol, operating leverage. Operating leverage, Ol, is operating costs scaled by total assets (Compustat annual item AT, the denominator is current, not lagged, total assets). Operating costs are cost of goods sold (item COGS) plus selling, general, and administrative expenses (item XSGA). At the end of June of year t , we sort stocks into deciles based on Ol for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.5.9 Ol^{q1}, Ol^{q6}, and Ol^{q12}, quarterly operating leverage. At the beginning of each month t , we split stocks into deciles based on quarterly operating leverage, Ol^q, which is quarterly operating costs divided by assets (Compustat quarterly item ATQ) for the fiscal quarter ending at least four months ago. Operating costs are the cost of goods sold (item COGSQ) plus selling, general, and administrative expenses (item XSGAQ). We calculate decile returns for the current month t (Ol^{q1}), from month t to $t+5$ (Ol^{q6}), and from month t to $t+11$ (Ol^{q12}), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ol^{q6} mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ol^{q6} decile. For sufficient data coverage, the Ol^q portfolios start in January 1973.

A.5.10 Hn, hiring rate. Following Belo, Lin, and Bazzdresch (2014), at the end of June of year t , we measure the hiring rate (Hn) as $(N_{t-1} - N_{t-2}) / (0.5N_{t-1} + 0.5N_{t-2})$, in which N_{t-j} is the number of employees (Compustat annual item EMP) from the fiscal year ending in calendar year $t-j$. At the end of June of year t , we sort stocks into deciles based on Hn. We exclude firms with zero Hn (often due to stale information on employment). Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.5.11 Rca, R&D capital-to-assets. Following Li (2011), we measure R&D capital, Rc, by accumulating annual R&D expenses over the past five years with a depreciation rate of 20%:

$$Rc_{it} = XRD_{it} + 0.8XRD_{it-1} + 0.6XRD_{it-2} + 0.4XRD_{it-3} + 0.2XRD_{it-4}, \quad (A18)$$

in which XRD_{it-j} is firm i 's R&D expenses (Compustat annual item XRD) in year $t-j$. R&D capital-to-assets, Rca, is Rc scaled by total assets (item AT). At the end of June of each year t , we sort stocks into deciles based on Rca for the fiscal year ending in calendar year $t-1$. We keep only firms with positive Rc. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. For the portfolio formation at the end of June of year t , we require R&D expenses to be nonmissing for the fiscal year ending in calendar year $t-1$, because this value of R&D expenses receives the highest weight in Rc. Because Rc requires past five years of R&D expenses data and the accounting treatment of R&D expenses was standardized in 1975, the Rca portfolios start in July 1980.

A.5.12 Bca, brand capital-to-assets. Following Belo, Lin, and Vitorino (2014), we construct brand capital, Bc, by accumulating advertising expenses with the perpetual inventory method:

$$Bc_{it} = (1 - \delta)Bc_{it-1} + XAD_{it}. \quad (A19)$$

in which Bc_{it} is the brand capital for firm i at the end of year t , XAD_{it} is the advertising expenses (Compustat annual item XAD) in t , and δ is the annual depreciation rate of Bc. The initial stock of Bc is $Bc_{i0} = XAD_{i0}/(g + \delta)$, in which XAD_{i0} is first valid XAD (zero or positive) for firm i and g is the long-term growth rate of XAD. We assume a depreciation rate of 50% for Bc and a long-term growth rate of 10% for XAD. Missing values of XAD after the starting date are treated as zero. For the portfolio formation at the end of June of year t , we exclude firms with zero Bc and require XAD to be nonmissing for the fiscal year ending in calendar year $t - 1$. Brand capital-to-assets, Bca, is Bc scaled by total assets (item AT). At the end of June of each year t , we sort stocks into deciles based on Bca for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because sufficient XAD data start in 1972, the Bc portfolios start in July 1973.

A.5.13 Aop, analysts optimism. Following Frankel and Lee (1998), we measure analysts optimism, Aop, as $(V_f - V_h)/V_h$, in which V_f is the analysts forecast-based intrinsic value, and V_h is the historical Roe-based intrinsic value. See Appendix A.2.27 for the construction of intrinsic values. At the end of June of each year t , we sort stocks into deciles based on Aop. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because analyst forecast data start in 1976, the Aop deciles start in July 1976.

A.5.14 Pafe, predicted analysts forecast error. Following Frankel and Lee (1998), we define analysts forecast errors for year t as the actual realized Roe in year $t + 3$ minus the predicted Roe for $t + 3$ based on analyst forecasts. See Appendix A.2.27 for the measurement of realized and predicted Roe. To calculate predicted analysts forecast errors, Pafe, for the portfolio formation at the end of June of year t , we estimate the intercept and slopes of the annual cross-sectional regressions of $Roe_{t-1} - E_{t-4}[Roe_{t-1}]$ on four firm characteristics for the fiscal year ending in calendar year $t - 4$, including prior 5-year sales growth, book-to-market, long-term earnings growth forecast, and analysts optimism. Prior 5-year sale growth is the growth rate in sales (Compustat annual item SALE) from the fiscal year ending in calendar year $t - 9$ to the fiscal year ending in $t - 4$. Book-to-market is book equity (item CEQ) for the fiscal year ending in calendar year $t - 4$ divided by the market equity (form CRSP) at the end of June in $t - 3$. Long-term earnings growth forecast is from IBES (unadjusted file, item MEANEST; fiscal period indicator = 0), reported in June of $t - 3$. See Appendix A.5.13 for the construction of analyst optimism. We winsorize the regressors at the 1st and 99th percentiles of their respective pooled distributions each year, and standardize all the regressors (by subtracting mean and dividing by standard deviation). Pafe for the portfolio formation year t is then obtained by applying the estimated intercept and slopes on the winsorized and standardized regressors for the fiscal year ending in calendar year $t - 1$. At the end of June of each year t , we sort stocks into deciles based on Pafe. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the long-term earnings growth forecast data start in 1981, the Pafe portfolios start in July 1985.

A.5.15 Parc, patent-to-R&D capital. Following Hirshleifer, Hsu, and Li (2013), we measure patent-to-R&D capital, Parc, as the ratio of firm i 's patents granted in year t , $Patents_{it}$, scaled by its R&D capital for the fiscal year ending in calendar year $t - 2$, $Patents_{it}/(XRD_{it-2} + 0.8XRD_{it-3} + 0.6XRD_{it-4} + 0.4XRD_{it-5} + 0.2XRD_{it-6})$, in which XRD_{it-j} is R&D expenses (Compustat annual item XRD) for the fiscal year ending in calendar year $t - j$. We require nonmissing R&D expenses for the fiscal year ending in $t - 2$ but set missing values to zero for other years ($t - 6$ to $t - 3$). The patent data are from the NBER patent database and are available

from 1976 to 2006. At the end of June of each year t , we use Parc for $t-1$ to form deciles. Stocks with zero Parc are grouped into 1 portfolio (1) and stocks with positive Parc are sorted into 9 portfolios (2 to 10). Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. Because the accounting treatment of R&D expenses was standardized in 1975 and the NBER patent data stop in 2006, the Parc portfolios are available from July 1982 to June 2008.

A.5.16 Crd, citations-to-R&D expenses. Following Hirshleifer, Hsu, and Li (2013), we measure citations-to-R&D expenses, Crd, in year t as the adjusted number of citations in year t to firm i 's patents granted over the previous five years scaled by the sum of R&D expenses:

$$\text{Crd}_t = \frac{\sum_{s=1}^5 \sum_{k=1}^{N_{t-s}} C_{ik}^{t-s}}{\sum_{s=1}^5 \text{XRD}_{it-2-s}}, \quad (\text{A20})$$

in which C_{ik}^{t-s} is the number of citations received in year t by patent k , granted in year $t-s$ scaled by the average number of citations received in year t by all patents of the same subcategory granted in year $t-s$. N_{t-s} is the total number of patents granted in year $t-s$ to firm i . XRD_{it-2-s} is R&D expenses (Compustat annual item XRD) for the fiscal year ending in calendar year $t-2-s$. The patent citation data are from the NBER patent database. At the end of June of each year t , we use Crd for $t-1$ to form deciles. Stocks with zero Crd are grouped into one portfolio (1) and stocks with positive Crd are sorted into nine portfolios (2 to 10). Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. Because the NBER data are from 1977 to 2006, the Crd portfolios are available from July 1983 to June 2008.

A.5.17 Hs, Ha, and He, industry concentration (sales, assets, book equity). Following Hou and Robinson (2006), we measure a firm's industry concentration with the Herfindahl index, $\sum_{i=1}^{N_j} s_{ij}^2$, in which s_{ij} is the market share of firm i in industry j , and N_j is the total number of firms in the industry. We calculate the market share of a firm using sales (Compustat annual item SALE), total assets (item AT), or book equity. We measure book equity as stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. Industries are defined by 3-digit SIC codes. We exclude financial firms (SIC between 6000 and 6999) and firms in regulated industries. The regulated industries include: railroads (SIC=4011) through 1980, trucking (4210 and 4213) through 1980, airlines (4512) through 1978, telecommunication (4812 and 4813) through 1982, and gas and electric utilities (4900 to 4939). To improve the accuracy of the concentration measure, we exclude an industry if the market share data are available for fewer than five firms or 80% of all firms in the industry. We measure industry concentration as the average Herfindahl index during the past three years. Industry concentrations calculated with sales, assets, and book equity are denoted, Hs, Ha, and He, respectively. At the end of June of each year t , we sort stocks into deciles based on Hs, Ha, and He for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.5.18 Age1, Age6, and Age12, firm age. Following Jiang, Lee, and Zhang (2005), we measure firm age, Age, as the number of months between the portfolio formation date and the first month that a firm appears in Compustat or CRSP (item permco). At the beginning of each month t , we sort stocks into quintiles based on Age at the end of $t-1$. We do not form deciles because a

disproportional number of firms can have the same Age (e.g., caused by the inception of NASDAQ coverage in 1973). Monthly quintile returns are calculated for the current month t (Age1), from month t to $t+5$ (Age6), and from month t to $t+11$ (Age12), and the quintiles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Age6 mean that for a given quintile in each month there exist six subquintiles, each initiated in a different month in the prior six months. We average the subquintiles returns as the monthly return of the Age6 quintile.

A.5.19 D1, D2, and D3, price delay. At the end of June of each year, we regress each stock's weekly returns over the prior year on the contemporaneous and four weeks of lagged market returns:

$$r_{it} = \alpha_i + \beta_i R_{mt} + \sum_{n=1}^4 \delta_i^{(-n)} R_{mt-n} + \epsilon_{it}, \quad (\text{A21})$$

in which r_{it} is the return on stock j in week t , and R_{mt} is the return on the CRSP value-weighted market index. Weekly returns are measured from Wednesday market close to the next Wednesday market close. Following Hou and Moskowitz (2005), we calculate three price delay measures:

$$D1_i \equiv 1 - \frac{R_{\delta_i^{(-4)}=\delta_i^{(-3)}=\delta_i^{(-2)}=\delta_i^{(-1)}=0}^2}{R^2}, \quad (\text{A22})$$

in which $R_{\delta_i^{(-4)}=\delta_i^{(-3)}=\delta_i^{(-2)}=\delta_i^{(-1)}=0}^2$ is the R^2 from regression equation (A21) with the restriction $\delta_i^{(-4)}=\delta_i^{(-3)}=\delta_i^{(-2)}=\delta_i^{(-1)}=0$, and R^2 is without this restriction. In addition,

$$D2_i \equiv \frac{\sum_{n=1}^4 n \delta_i^{(-n)}}{\beta_i + \sum_{n=1}^4 \delta_i^{(-n)}} \quad (\text{A23})$$

$$D3_i \equiv \frac{\sum_{n=1}^4 \frac{n \delta_i^{(-n)}}{\text{se}(\delta_i^{(-n)})}}{\frac{\beta_i}{\text{se}(\beta_i)} + \sum_{n=1}^4 \frac{\delta_i^{(-n)}}{\text{se}(\delta_i^{(-n)})}}, \quad (\text{A24})$$

in which $\text{se}(\cdot)$ is the standard error of the point estimate in parentheses.

To improve precision of the price delay estimate, we sort firms into portfolios based on market equity and individual delay measure, compute the delay measure for the portfolio, and assign the portfolio delay measure to each firm in the portfolio. At the end of June of each year t , we sort stocks into size deciles based on the market equity (from CRSP) at the end of June in $t-j$ ($j=1, 2, \dots$). Within each size decile, we then sort stocks into deciles based on their first-stage individual delay measure, estimated using weekly return data from July of year $t-j-1$ to June of year $t-j$. The equal-weighted weekly returns of the 100 size-delay portfolios are computed over the following year from July of year $t-j$ to June of $t-j+1$. We then reestimate the delay measure for each of the 100 portfolios using the entire past sample of weekly returns up to June of year t . The second-stage portfolio delay measure is then assigned to individual stocks within the 100 portfolios formed at end of June in year t . At the end of June of year t , we sort stocks into deciles based on D1, D2, and D3. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.5.20 dSi, % change in sales minus % change in inventory. Following Abarbanell and Bushee (1998), we define the $\%d(\cdot)$ operator as the percentage change in the variable in the parentheses from its average over the prior two years, for example, $\%d(\text{Sales}) = [\text{Sales}(t) - E[\text{Sales}(t)]]/E[\text{Sales}(t)]$, in which $E[\text{Sales}(t)] = [\text{Sales}(t-1) + \text{Sales}(t-2)]/2$. dSi is calculated as $\%d(\text{Sales}) - \%d(\text{Inventory})$, in which sales is net sales (Compustat annual item SALE), and

inventory is finished goods inventories (item INVFG) if available, or total inventories (item INVT). Firms with nonpositive average sales or inventory during the past two years are excluded. At the end of June of each year t , we sort stocks into deciles based on dSi for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.21 dSa, % change in sales minus % change in accounts receivable. Following Abarbanell and Bushee (1998), we define the $\%d(\cdot)$ operator as the percentage change in the variable in the parentheses from its average over the prior two years, for example, $\%d(\text{Sales}) = [\text{Sales}(t) - E[\text{Sales}(t)]]/E[\text{Sales}(t)]$, in which $E[\text{Sales}(t)] = [\text{Sales}(t - 1) + \text{Sales}(t - 2)]/2$. dSa is calculated as $\%d(\text{Sales}) - \%d(\text{Accounts receivable})$, in which sales is net sales (Compustat annual item SALE) and accounts receivable is total receivables (item RECT). Firms with nonpositive average sales or receivables during the past two years are excluded. At the end of June of each year t , we sort stocks into deciles based on dSa for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.22 dGs, % change in gross margin minus % change in sales. Following Abarbanell and Bushee (1998), we define the $\%d(\cdot)$ operator as the percentage change in the variable in the parentheses from its average over the prior two years, for example, $\%d(\text{Sales}) = [\text{Sales}(t) - E[\text{Sales}(t)]]/E[\text{Sales}(t)]$, in which $E[\text{Sales}(t)] = [\text{Sales}(t - 1) + \text{Sales}(t - 2)]/2$. dGs is calculated as $\%d(\text{Gross margin}) - \%d(\text{Sales})$, in which sales is net sales (Compustat annual item SALE) and gross margin is sales minus cost of goods sold (item COGS). Firms with nonpositive average gross margin or sales during the past two years are excluded. At the end of June of each year t , we sort stocks into deciles based on dGs for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.23 dSs, % change in sales minus % change in SG&A. Following Abarbanell and Bushee (1998), we define the $\%d(\cdot)$ operator as the percentage change in the variable in the parentheses from its average over the prior two years, e.g., $\%d(\text{Sales}) = [\text{Sales}(t) - E[\text{Sales}(t)]]/E[\text{Sales}(t)]$, in which $E[\text{Sales}(t)] = [\text{Sales}(t - 1) + \text{Sales}(t - 2)]/2$. dSs is calculated as $\%d(\text{Sales}) - \%d(\text{SG\&A})$, in which sales is net sales (Compustat annual item SALE) and SG&A is selling, general, and administrative expenses (item XSGA). Firms with nonpositive average sales or SG&A during the past two years are excluded. At the end of June of each year t , we sort stocks into deciles based on dSs for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.24 Etr, effective tax rate. Following Abarbanell and Bushee (1998), we measure effective tax rate, Etr, as

$$\text{Etr}(t) = \left[\frac{\text{TaxExpense}(t)}{\text{EBT}(t)} - \frac{1}{3} \sum_{\tau=1}^3 \frac{\text{TaxExpense}(t-\tau)}{\text{EBT}(t-\tau)} \right] \times \text{dEPS}(t), \quad (\text{A25})$$

in which $\text{TaxExpense}(t)$ is total income taxes (Compustat annual item TXT) paid in year t , $\text{EBT}(t)$ is pretax income (item PI) plus amortization of intangibles (item AM), and dEPS is the change in split-adjusted earnings per share (item EPSPX divided by item AJEX) between years $t - 1$ and t , deflated by stock price (item PRCC_F) at the end of $t - 1$. At the end of June of each year t , we sort stocks into deciles based on Etr for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.25 Lfe, labor force efficiency. Following Abarbanell and Bushee (1998), we measure labor force efficiency, Lfe, as

$$\text{Lfe}(t) = \left[\frac{\text{Sales}(t)}{\text{Employees}(t)} - \frac{\text{Sales}(t-1)}{\text{Employees}(t-1)} \right] / \frac{\text{Sales}(t-1)}{\text{Employees}(t-1)}, \quad (\text{A26})$$

in which $\text{Sales}(t)$ is net sales (Compustat annual item SALE) in year t , and $\text{Employees}(t)$ is the number of employees (item EMP). At the end of June of each year t , we sort stocks into deciles based on Lfe for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.5.26 Ana1, Ana6, and Ana12, analysts coverage. Following Elgers, Lo, and Pfeiffer (2001), we measure analysts coverage, Ana, as the number of analysts' earnings forecasts from IBES (item NUMEST) for the current fiscal year (fiscal period indicator = 1). We require earnings forecasts to be denominated in U.S. dollars (currency code = USD). At the beginning of each month t , we sort stocks into quintiles on Ana from the IBES report in $t-1$. We do not form deciles because a disproportional number of firms can have the same Ana before 1980. Monthly quintile returns are calculated for the current month t (Ana1), from month t to $t+5$ (Ana6), and from month t to $t+11$ (Ana12). The quintiles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ana6 mean that for a given quintile in each month there exist six subquintiles, each initiated in a different month in the prior six months. We average the subquintile returns as the monthly return of the Ana6 quintile. Because the earnings forecast data start in January 1976, the Ana portfolios start in February 1976.

A.5.27 Tan, tangibility. We measure tangibility, Tan, as cash holdings (Compustat annual item CHE) + $0.715 \times$ accounts receivable (item RECT) + $0.547 \times$ inventory (item INVT) + $0.535 \times$ gross property, plant, and equipment (item PPEGT), all scaled by total assets (item AT). At the end of June of each year t , we sort stocks into deciles on Tan for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.5.28 Tan^q1, Tan^q6, and Tan^q12, quarterly tangibility. Tan^q is cash holdings (Compustat quarterly item CHEQ) + $0.715 \times$ accounts receivable (item RECTQ) + $0.547 \times$ inventory (item INVTQ) + $0.535 \times$ gross property, plant, and equipment (item PPEGTQ), all scaled by total assets (item ATQ). At the beginning of each month t , we sort stocks into deciles based on Tan^q for the fiscal quarter ending at least four months ago. Monthly decile returns are calculated for the current month t (Tan^q1), from month t to $t+5$ (Tan^q6), and from month t to $t+11$ (Tan^q12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Tan^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Tan^q6 decile. For sufficient data coverage, the Tan^q portfolios start in January 1972.

A.5.29 Rer, industry-adjusted real estate ratio. Following Tuzel (2010), we measure the real estate ratio as the sum of buildings (Compustat annual item PPENB) and capital leases (item PPENLS) divided by net property, plant, and equipment (item PPENT) prior to 1983. From 1984 onward, the real estate ratio is the sum of buildings at cost (item FATB) and leases at cost (item FATL) divided by gross property, plant, and equipment (item PPEGT). Industry-adjusted real estate ratio, Rer, is the real estate ratio minus its industry average. Industries are defined by 2-digit SIC codes. To alleviate the impact of outliers, we winsorize the real estate ratio at the 1st and 99th percentiles of its distribution each year before computing Rer. Following Tuzel (2010), we exclude industries with fewer than five firms. At the end of June of each year t , we sort stocks into deciles based on Rer for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. Because the real estate data start in 1969, the Rer portfolios start in July 1970.

A.5.30 Kz, financial constraints (the Kaplan-Zingales index). Following Lamont, Polk, and Saa-Requejo (2001), we construct the Kaplan-Zingales index, Kz_{it} , as

$$\begin{aligned} & -1.002 \times \frac{CF_{it}}{K_{it-1}} + 0.283 \times Q_{it} + 3.139 \times \frac{Debt_{it}}{\text{Total Capital}_{it}} - 39.368 \times \frac{\text{Dividends}_{it}}{K_{it-1}} \\ & - 1.315 \times \frac{\text{Cash}_{it}}{K_{it-1}}, \end{aligned} \quad (A27)$$

in which CF_{it} is firm i 's cash flows in year t , measured as income before extraordinary items (Compustat annual item IB) plus depreciation and amortization (item DP). K_{it-1} is net property, plant, and equipment (item PPENT) at the end of year $t-1$. Q_{it} is Tobin's Q , measured as total assets (item AT) plus the December-end market equity (from CRSP), minus book equity (item CEQ), and minus deferred taxes (item TXDB), scaled by total assets. $Debt_{it}$ is the sum of short-term debt (item DLC) and long-term debt (item DLTT). $\text{Total Capital}_{it}$ is the sum of total debt and stockholders' equity (item SEQ). Dividends_{it} is total dividends (item DVC plus item DVP). Cash_{it} is cash holdings (item CHE). At the end of June of each year t , we sort stocks into deciles based on Kz for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.5.31 Kz^q1 , Kz^q6 , and Kz^q12 , quarterly Kaplan-Zingales index. We construct the quarterly Kaplan-Zingales index, Kz^q , as

$$\begin{aligned} Kz^q_{it} & \equiv -1.002 \frac{CF^q_{it}}{K^q_{it-1}} + 0.283 Q^q_{it} + 3.139 \frac{Debt^q_{it}}{\text{Total Capital}^q_{it}} - 39.368 \frac{\text{Dividends}^q_{it}}{K^q_{it-1}} \\ & - 1.315 \frac{\text{Cash}^q_{it}}{K^q_{it-1}}, \end{aligned} \quad (A28)$$

in which CF^q_{it} is firm i 's trailing 4-quarter total cash flows from quarter $t-3$ to t . Quarterly cash flows are measured as income before extraordinary items (Compustat quarterly item IBQ) plus depreciation and amortization (item DPQ). K^q_{it-1} is net property, plant, and equipment (item PPENTQ) at the end of quarter $t-1$. Q^q_{it} is Tobin's Q , measured as total assets (item ATQ) plus the fiscal-quarter-end market equity (from CRSP), minus book equity (item CEQQ), and minus deferred taxes (item TXDBQ, zero if missing), scaled by total assets. $Debt^q_{it}$ is the sum of short-term debt (item DLCQ) and long-term debt (item DLTTQ). $\text{Total Capital}^q_{it}$ is the sum of total debt and stockholders' equity (item SEQQ). Dividends^q_{it} is the total dividends (item DVPSXQ times item CSOQ), accumulated over the past four quarters from $t-3$ to t .

At the beginning of each month t , we sort stocks into deciles based on Kz^q for the fiscal quarter ending at least four months ago. Monthly decile returns are computed for the current month t (Kz^q1), from month t to $t+5$ (Kz^q6), and from month t to $t+11$ (Kz^q12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Kz^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Kz^q6 decile. For sufficient data coverage, the Kz^q portfolios start in January 1977.

A.5.32 Ww, financial constraints (the Whited-Wu index). Following Whited and Wu (2006, equation 13), we construct the Whited-Wu index, Ww_{it} , as

$$-0.091CF_{it} - 0.062DIVPOS_{it} + 0.021TLTD_{it} - 0.044LNTA_{it} + 0.102ISG_{it} - 0.035SG_{it}, \quad (A29)$$

in which CF_{it} is the ratio of firm i 's cash flows in year t scaled by total assets (Compustat annual item AT) at the end of t . Cash flows are measured as income before extraordinary items (item

IB) plus depreciation and amortization (item DP). $DIVPOS_{it}$ is an indicator that takes the value of one if the firm pays cash dividends (item DVPSX), and zero otherwise. $TLTD_{it}$ is the ratio of the long-term debt (item DLTT) to total assets. $LNTA_{it}$ is the natural log of total assets. ISG_{it} is the firm's industry sales growth, computed as the sum of current sales (item SALE) across all firms in the industry divided by the sum of 1-year-lagged sales minus one. Industries are defined by 3-digit SIC codes. We exclude industries with fewer than two firms. SG_{it} is the firm's annual growth in sales. Because the coefficients in equation (A29) were estimated with quarterly data in Whited and Wu (2006), we convert annual cash flow and sales growth rates into quarterly terms. We divide CF_{it} by four and use the compounded quarterly growth for sales $((1+ISG_{it})^{1/4} - 1)$ and $(1+SG_{it})^{1/4} - 1$. At the end of June of each year t , we split stocks into deciles based on Ww for the fiscal year ending in calendar year $t-1$. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$.

A.5.33 Ww^q1 , Ww^q6 , and Ww^q12 , the quarterly Whited-Wu index. We construct the quarterly Whited-Wu index, Ww^q_{it} , as

$$-0.091CF_{it}^q - 0.062DIVPOS_{it}^q + 0.021TLTD_{it}^q - 0.044LNTA_{it}^q + 0.102ISG_{it}^q - 0.035SG_{it}^q, \quad (A30)$$

in which CF_{it}^q is the ratio of firm i 's cash flows in quarter t scaled by total assets (Compustat quarterly item ATQ) at the end of t . Cash flows are measured as income before extraordinary items (item IBQ) plus depreciation and amortization (item DPQ). $DIVPOS_{it}^q$ is an indicator that takes the value of 1 if the firm pays cash dividends (item DVPSXQ), and zero otherwise. $TLTD_{it}^q$ is the ratio of the long-term debt (item DLTTQ) to total assets. $LNTA_{it}^q$ is the natural log of total assets. ISG_{it}^q is the firm's industry sales growth, the sum of current sales (item SALEQ) across all firms in the industry divided by the sum of 1-quarter-lagged sales minus one. Industries are defined by 3-digit SIC codes. We exclude industries with fewer than two firms. SG_{it}^q is the firm's quarterly growth in sales. At the beginning of each month t , we sort stocks into deciles based on Ww^q for the fiscal quarter ending at least four months ago. Monthly decile returns are calculated for the current month t (Ww^q1), from month t to $t+5$ (Ww^q6), and from month t to $t+11$ (Ww^q12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ww^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ww^q6 decile. For sufficient data the Ww^q deciles start in January 1972.

A.5.34 Sdd, secured debt-to-total debt. Following Valta (2016), we measure secured debt-to-total debt, Sdd, as mortgages and other secured debt (Compustat annual item DM) divided by total debt. Total debt is debt in current liabilities (item DLC) plus long-term debt (item DLTT). At the end of June of each year t , we sort stocks into deciles based on Sdd for the fiscal year ending in calendar year $t-1$. Firms with no secured debt are excluded. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. Because the data on secured debt start in 1981, the Sdd portfolios start in July 1982.

A.5.35 Cdd, convertible debt-to-total debt. Following Valta (2016), we measure convertible debt-to-total debt, Cdd, as convertible debt (Compustat annual item DCVT) divided by total debt. Total debt is debt in current liabilities (item DLC) plus long-term debt (item DLTT). At the end of June of each year t , we sort stocks into deciles based on Cdd for the fiscal year ending in calendar year $t-1$. Firms with no convertible debt are excluded. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. Because a disproportionately large number of stocks can have Cdd equal to one, we use only Cdd smaller than one to form portfolio breakpoints. Because the data on convertible debt start in 1969, the Cdd portfolios start in July 1970.

A.5.36 Vcf1, Vcf6, and Vcf12, cash flow volatility. Cash flow volatility, Vcf, is the standard deviation of the ratio of operating cash flows to sales (Compustat quarterly item SALEQ) during the past 16 quarters (eight nonmissing quarters minimum). Operating cash flows are income before extraordinary items (item IBQ) plus depreciation and amortization (item DPQ), and plus the change in working capital (item WCAPQ) from the last quarter. At the beginning of each month t , we sort stocks into deciles based on Vcf for the fiscal quarter ending at least four months ago. Monthly decile returns are calculated for the current month t (Vcf1), from month t to $t+5$ (Vcf6), and from month t to $t+11$ (Vcf12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Vcf6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Vcf6 decile. For sufficient data coverage, the Vcf portfolios start in January 1978.

A.5.37 Cta1, Cta6, and Cta12, cash-to-assets. Following Palazzo (2012), we measure cash-to-assets, Cta, as cash holdings (Compustat quarterly item CHEQ) scaled by total assets (item ATQ). At the beginning of each month t , we sort stocks into deciles based on Cta from the fiscal quarter ending at least four months ago. Monthly decile returns are calculated for the current month t (Cta1), from month t to $t+5$ (Cta6), and from month t to $t+11$ (Cta12), and the deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in Cta6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Cta6 decile. For sufficient data coverage, the Cta portfolios start in January 1972.

A.5.38 Gind, corporate governance. The data for the firm-level corporate governance index (Gind, from September 1990 to December 2006) are from Andrew Metrick's Web site. Following Gompers, Ishii, and Metrick (2003, table VI), we use the following breakpoints to form the Gind portfolios: $Gind \leq 5, 6, 7, 8, 9, 10, 11, 12, 13$, and ≥ 14 . Firms with dual share classes are excluded. We rebalance the portfolios in the months immediately following each publication of Gind, and calculate monthly portfolio returns between two adjacent publication dates. The first months following the publication dates are September 1990, July 1993, July 1995, February 1998, November 1999, January 2002, January 2004, and January 2006.

A.5.39 Acq, Acq^m1, Acq^m6, Acq^m12, accrual quality. Following Francis et al. (2005), we estimate accrual quality (Acq) with the following cross-sectional regression:

$$TCA_{it} = \phi_{0,i} + \phi_{1,i} CFO_{it-1} + \phi_{2,i} CFO_{it} + \phi_{3,i} CFO_{it+1} + \phi_{4,i} dREV_{it} + \phi_{5,i} PPE_{it} + v_{it}, \quad (A31)$$

in which TCA_{it} is firm i 's total current accruals in year t , CFO_{it} is cash flow from operations, $dREV_{it}$ is change in revenues (Compustat annual item SALE) from $t-1$ to t , and PPE_{it} is gross property, plant, and equipment (item PPEGT). $TCA_{it} = dCA_{it} - dCL_{it} - dCASH_{it} + dSTDEBT_{it}$, in which dCA_{it} is the change in current assets (item ACT) from year $t-1$ to t , dCL_{it} is the change in current liabilities (item LCT), $dCASH_{it}$ is the change in cash (item CHE), and $dSTDEBT_{it}$ is the change in debt in current liabilities (item DLC). $CFO_{it} = NIBE_{it} - (dCA_{it} - dCL_{it} - dCASH_{it} + dSTDEBT_{it} - DEPN_{it})$, in which $NIBE_{it}$ is income before extraordinary items (item IB), and $DEPN_{it}$ is depreciation and amortization expense (item DP). All variables are scaled by the average of total assets in t and $t-1$. We estimate annual cross-sectional regressions in equation (A31) for each of Fama-French (1997) 48 industries (excluding financial industries) with at least 20 firms in year t . We winsorize both dependent and independent variables at the 1st and 99th percentiles of their distributions each year. The annual cross-sectional regressions yield firm- and year-specific residuals, v_{it} . We measure accrual quality of firm i , $Acq_i = \sigma(v_i)$, as the standard deviation of firm i 's residuals during the past five years from $t-4$ to t . For a firm to be included in our portfolio, its residual has to be available for all five years.

At the end of June of each year t , we sort stocks into deciles based on Acq for the fiscal year ending in calendar year $t - 2$. To avoid look-ahead bias, we do not sort on Acq for the fiscal year ending in $t - 1$, because the regression in equation (A31) requires the next year's CFO. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. In addition, at the beginning of each month t , we sort stocks into deciles based on Acq calculated with data up to the fiscal year ending at least four months ago. Monthly decile returns are calculated for the current month t (Acq^m1), from month t to $t + 5$ (Acq^m6), and from month t to $t + 11$ (Acq^m12), and the deciles are rebalanced at the beginning of $t + 1$. Holding periods longer than one month like in Acq^m6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Acq^m6 decile.

A.5.40 Ob, order backlog. At the end of June of each year t , we sort stocks into deciles based on order backlog, Ob (Compustat annual item OB) for the fiscal year ending in calendar year $t - 1$, scaled by the average of total assets (item AT) from the fiscal years ending in $t - 2$ and $t - 1$. Firms with no order backlog are excluded (most of them never have any order backlog). Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the order backlog data start in 1970, the Ob portfolios start in July 1971.

A.5.41 Eper and Eprd, earnings persistence, earnings predictability. Following Francis et al. (2004), we estimate earnings persistence, Eper, and earnings predictability, Eprd, from a first-order autoregressive model for annual split-adjusted earnings per share (Compustat annual item EPSPX divided by item AJEX). At the end of June of each year t , we estimate the autoregressive model in the 10-year rolling window up to the fiscal year ending in calendar year $t - 1$. Only firms with a complete 10-year history are included. Eper is measured as the slope coefficient and Eprd is measured as the residual volatility. We sort stocks into deciles based on Eper, and separately, on Eprd. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.42 Esm, earnings smoothness. Following Francis et al. (2004), we measure earnings smoothness, Esm, as the ratio of the standard deviation of earnings (Compustat annual item IB) scaled by 1-year-lagged total assets (item AT) to the standard deviation of cash flow from operations scaled by 1-year-lagged total assets. Cash flow from operations is income before extraordinary items minus operating accruals. We measure operating accruals as the 1-year change in current assets (item ACT) minus the change in current liabilities (item LCT), minus the change in cash (item CHE), plus the change in debt in current liabilities (item DLC), and minus depreciation and amortization (item DP). At the end of June of each year t , we sort stocks into deciles based on Esm, calculated over the 10-year rolling window up to the fiscal year ending in calendar year $t - 1$. Only firms with a complete 10-year history are included. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.43 Evr, value relevance of earnings. Following Francis et al. (2004), we measure value relevance of earnings, Evr, as the R^2 from the following rolling-window regression:

$$R_{it} = \delta_{i0} + \delta_{i1} \text{EARN}_{it} + \delta_{i2} \text{dEARN}_{it} + \epsilon_{it}, \quad (\text{A32})$$

in which R_{it} is firm i 's 15-month stock return ending three months after the end of fiscal year ending in calendar year t . EARN_{it} is earnings (Compustat annual item IB) for the fiscal year ending in t , scaled by the fiscal year-end market equity (from CRSP). dEARN_{it} is the 1-year change in earnings scaled by the market equity. For firms with more than one share class, we merge the market equity for all share classes. At the end of June of each year t , we split stocks into deciles on Evr, calculated over the 10-year rolling window up to the fiscal year ending in calendar year $t - 1$. Only firms with a complete 10-year history are included. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.44 Etl and Ecs, earnings timeliness, earnings conservatism. Following Francis et al. (2004), we measure earnings timeliness, Etl, and earnings conservatism, Ecs, from the following rolling-window regression:

$$\text{EARN}_{it} = \alpha_{i0} + \alpha_{i1} \text{NEG}_{it} + \beta_{i1} R_{it} + \beta_{i2} \text{NEG}_{it} R_{it} + e_{it}, \quad (\text{A33})$$

in which EARN_{it} is earnings (Compustat annual item IB) for the fiscal year ending in calendar year t , scaled by the fiscal year-end market equity. R_{it} is firm i 's 15-month stock return ending three months after the end of fiscal year ending in calendar year t . NEG_{it} equals one if $R_{it} < 0$, and zero otherwise. For firms with more than one share class, we merge the market equity for all share classes. We measure Etl as the R^2 and Ecs as $(\beta_{i1} + \beta_{i2})/\beta_{i1}$ from the regression in (A33). At the end of June of each year t , we sort stocks into deciles based on Etl, and separately, on Ecs, both of which are calculated over the 10-year rolling window up to the fiscal year ending in calendar year $t - 1$. Only firms with a complete 10-year history are included. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.45 Frm and Fra, pension plan funding rate. Following Franzoni and Marin (2006), we define market pension plan funding rates as $(\text{PA} - \text{PO})/\text{Me}$ (denoted Frm) and $(\text{PA} - \text{PO})/\text{AT}$ (denoted Fra), in which PA is the fair value of pension plan assets, PO is the projected benefit obligation, Me is the market equity, and AT is total assets (Compustat annual item AT). Between 1980 and 1997, PA is measured as the sum of overfunded pension plan assets (item PPLAO) and underfunded pension plan assets (item PPLAU), and PO is the sum of overfunded pension obligation (item PBPRO) and underfunded pension obligation (item PBPRU). When the above data are not available, we also measure PA as pension benefits (item PBNAA) and PO as the present value of vested benefits (item PBNVV) from 1980 to 1986. Starting from 1998, firms are not required to report separate items for overfunded and underfunded plans, and Compustat collapses PA and PO into corresponding items reserved previously for overfunded plans (item PPLAO and item PBPRO). Me is from CRSP measured at the end of December. For firms with more than one share class, we merge the market equity for all share classes.

At the end of June of each year t , we split stocks into deciles on Frm, and separately, on Fra, both of which are for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the pension data start in 1980, the Frm and Fra portfolios start in July 1981.

A.5.46 Ala and Alm, asset liquidity. We measure asset liquidity as $\text{cash} + 0.75 \times \text{noncash current assets} + 0.50 \times \text{tangible fixed assets}$, cash as cash and short-term investments (Compustat annual item CHE), noncash current assets as current assets (item ACT) minus cash, and tangible fixed assets as total assets (item AT) minus current assets (item ACT), minus goodwill (item GDWL, zero if missing), and minus intangibles (item INTAN, zero if missing). Ala is asset liquidity scaled by 1-year-lagged total assets. Alm is asset liquidity scaled by 1-year-lagged market value of assets. The market value of assets is total assets plus market equity (item PRCC_F times item CSHO) minus book equity (item CEQ). At the end of June of each year t , we sort stocks into deciles based on Ala, and separately, on Alm, both of which are for the fiscal year ending in calendar year $t - 1$. Monthly decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.5.47 Ala^{q1}, Ala^{q6}, Ala^{q12}, Alm^{q1}, Alm^{q6}, and Alm^{q12}, quarterly asset liquidity. We measure quarterly asset liquidity as $\text{cash} + 0.75 \times \text{noncash current assets} + 0.50 \times \text{tangible fixed assets}$, cash as cash and short-term investments (Compustat quarterly item CHEQ), noncash current assets as current assets (item ACTQ) minus cash, and tangible fixed assets as total assets (item ATQ) minus current assets (item ACTQ), minus goodwill (item GDWLQ, zero if missing), and minus intangibles (item INTANQ, zero if missing). Ala^q is quarterly asset liquidity scaled by

1-quarter-lagged total assets. Alm^q is quarterly asset liquidity scaled by 1-quarter-lagged market value of assets. The market value of assets is total assets plus market equity (item PRCCQ times item CSHOQ) minus book equity (item CEQQ).

At the beginning of each month t , we sort stocks into deciles based on Ala^q , and separately, on Alm^q for the fiscal quarter ending at least four months ago. Monthly decile returns are calculated for the current month t (Ala^q1 and Alm^q1), from month t to $t+5$ (Ala^q6 and Alm^q6), and from month t to $t+11$ (Ala^q12 and Alm^q12). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ala^q6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ala^q6 decile. For sufficient data coverage, the quarterly asset liquidity portfolios start in January 1976.

A.5.48 Dls1, Dls6, and Dls12, disparity between long- and short-term earnings growth forecasts. Following Da and Warachka (2011), we measure the implied short-term earnings growth forecast as $100 \times (A1_t - A0_t) / |A0_t|$, in which $A1_t$ is analysts' consensus median forecast (unadjusted IBES file, item MEDEST) for the current fiscal year (fiscal period indicator = 1), and $A0_t$ is the actual earnings per share for the latest reported fiscal year (item FY0A, measure indicator = EPS). We require both earnings forecasts and actual earnings to be denominated in U.S. dollars (currency code = USD). The disparity between long- and short-term earnings growth forecasts, Dls, is analysts' consensus median forecast of the long-term earnings growth (item MEDEST, fiscal period indicator = 0) minus the implied short-term earnings growth forecast. At the beginning of each month t , we sort stocks into deciles based on Dls computed with analyst forecasts reported in $t-1$. Monthly decile returns are calculated for the current month t (Dls1), from month t to $t+5$ (Dls6), and from month t to $t+11$ (Dls12), and the deciles are rebalanced at the beginning of $t+1$. Holding periods longer than one month like in Dls6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Dls6 decile. Because the long-term growth forecast data start in December 1981, the deciles start in January 1982.

A.5.49 Dis1, Dis6, and Dis12, dispersion in analyst forecasts. We measure dispersion in analyst earnings forecasts, Dis, as the ratio of the standard deviation of earnings forecasts (unadjusted IBES file, item STDEV) to the absolute value of the consensus mean forecast (unadjusted file, item MEANEST). We use the earnings forecasts for the current fiscal year (fiscal period indicator = 1) and we require them to be denominated in U.S. dollars (currency code = USD). Stocks with a mean forecast of zero are assigned to the highest dispersion group. Firms with fewer than two forecasts are excluded. At the beginning of each month t , we sort stocks into deciles based on Dis computed with analyst forecasts reported in month $t-1$. Monthly decile returns are calculated for the current month t (Dis1), from month t to $t+5$ (Dis6), and from month t to $t+11$ (Dis12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Dis6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Dis6 decile. Because the analyst forecasts data start in January 1976, the Dis portfolios start in February 1976.

A.5.50 Dlg1, Dlg6, and Dlg12, dispersion in analyst long-term growth forecasts. Following Anderson, Ghysels, and Juergens (2005), we measure dispersion in analyst long-term growth forecasts, Dlg, as the standard deviation of the long-term earnings growth rate forecasts from IBES (item STDEV, fiscal period indicator = 0). Firms with fewer than two forecasts are excluded. At the beginning of each month t , we sort stocks into deciles based on Dlg reported in month $t-1$. Monthly decile returns are calculated for the current month t (Dlg1), from month t to $t+5$ (Dlg6), and from month t to $t+11$ (Dlg12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Dlg6 mean that for a given decile in each month

there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Dlg6 decile. Because the long-term growth forecast data start in December 1981, the Dlg portfolios start in January 1982.

A.5.51 $R_a^1, R_n^1, R_a^{[2,5]}, R_n^{[2,5]}, R_a^{[6,10]}, R_n^{[6,10]}, R_a^{[11,15]}, R_n^{[11,15]}, R_a^{[16,20]},$ and $R_n^{[16,20]}$, seasonality. Following Heston and Sadka (2008), at the beginning of each month t , we sort stocks into deciles based on various measures of past performance, including returns in month $t-12$ (R_a^1), average returns from month $t-11$ to $t-1$ (R_n^1), average returns across months $t-24, t-36, t-48$, and $t-60$ ($R_a^{[2,5]}$), average returns from month $t-60$ to $t-13$ except for lags 24, 36, 48, and 60 ($R_n^{[2,5]}$), average returns across months $t-72, t-84, t-96, t-108$, and $t-120$ ($R_a^{[6,10]}$), average returns from month $t-120$ to $t-61$ except for lags 72, 84, 96, 108, and 120 ($R_n^{[6,10]}$), average returns across months $t-132, t-144, t-156, t-168$, and $t-180$ ($R_a^{[11,15]}$), average returns from month $t-180$ to $t-121$ except for lags 132, 144, 156, 168, and 180 ($R_n^{[11,15]}$), average returns across months $t-192, t-204, t-216, t-228$, and $t-240$ ($R_a^{[16,20]}$), average returns from month $t-240$ to $t-181$ except for lags 192, 204, 216, 228, and 240 ($R_n^{[16,20]}$). Monthly decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t+1$.

A.6 Trading Frictions

A.6.1 Me, market equity. Market equity, Me, is price times shares outstanding from CRSP. At the end of June of each year t , we sort stocks into deciles based on the June-end Me. Monthly decile returns are calculated from July of year t to June of $t+1$, and the deciles are rebalanced in June of $t+1$. In the cross-sectional regressions, we use the logarithm of Me as the regressor.

A.6.2 Ivff1, Ivff6, and Ivff12, idiosyncratic volatility per the Fama and French (1993) 3-factor model. We calculate idiosyncratic volatility relative to the Fama-French 3-factor model, Ivff, as the residual volatility from regressing a stock's excess returns on the Fama-French three factors. At the beginning of each month t , we sort stocks into deciles based on the Ivff estimated with daily returns from month $t-1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Ivff1), from month t to $t+5$ (Ivff6), and from month t to $t+11$ (Ivff12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ivff6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ivff6 decile.

A.6.3 Iv, idiosyncratic volatility. At the end of June of each year t , we sort stocks into deciles on idiosyncratic volatility, Iv, which is the residual volatility from regressing a stock's daily excess returns on the market excess return over the prior one year from July of year $t-1$ to June of t . We require a minimum of 100 daily returns. Monthly decile returns are from July of year t to June of $t+1$, and the deciles are rebalanced at the end of June of year $t+1$.

A.6.4 Ivc1, Ivc6, and Ivc12, idiosyncratic volatility per the CAPM. We calculate idiosyncratic volatility per the CAPM, Ivc, as the residual volatility from regressing a stock's excess returns on the value-weighted market excess return. At the beginning of each month t , we sort stocks into deciles based on the Ivc estimated with daily returns from month $t-1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Ivc1), from month t to $t+5$ (Ivc6), and from month t to $t+11$ (Ivc12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ivc6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ivc6 decile.

A.6.5 Ivq1, Ivq6, and Ivq12, idiosyncratic volatility per the q -factor model. We calculate idiosyncratic volatility per the Hou, Xue, and Zhang (2015) q -factor model, Ivq, as the residual volatility from regressing a stock's excess returns on the q -factors. At the beginning of each month t , we sort stocks into deciles based on the Ivq estimated with daily returns from month $t - 1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Ivq1), from month t to $t + 5$ (Ivq6), and from month t to $t + 11$ (Ivq12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Ivq6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Ivq6 decile. Because the q -factors start in January 1967, the Ivq portfolios start in February 1967.

A.6.6 Tv1, Tv6, and Tv12, total volatility. At the beginning of each month t , we sort stocks into deciles based on total volatility, Tv, estimated as the volatility of a stock's daily returns from month $t - 1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Tv1), from month t to $t + 5$ (Tv6), and from month t to $t + 11$ (Tv12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Tv6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Tv6 decile.

A.6.7 Sv1, Sv6, and Sv12, systematic volatility risk. We measure systematic volatility risk, Sv, as β_{dVXO}^i from the bivariate regression:

$$r_d^i = \beta_0^i + \beta_{MKT}^i MKT_d + \beta_{dVXO}^i dVXO_d + \epsilon_d^i, \quad (A34)$$

in which r_d^i is stock i 's excess return on day d , MKT_d is the market factor return, and $dVXO_d$ is the aggregate volatility shock measured as the daily change in the Chicago Board Options Exchange S&P 100 volatility index (VXO). At the beginning of each month t , we sort stocks into deciles based on β_{dVXO}^i estimated with the daily returns from month $t - 1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Sv1), from month t to $t + 5$ (Sv6), and from month t to $t + 11$ (Sv12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Sv6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Sv6 decile. Because the VXO data start in January 1986, the Sv portfolios start in February 1986.

A.6.8 $\beta 1$, $\beta 6$, and $\beta 12$, market beta. At the beginning of each month t , we sort stocks into deciles on their market beta, β , which is estimated from month $t - 60$ to $t - 1$. We require a minimum of 24 monthly returns. Monthly decile returns are calculated for the current month t ($\beta 1$), from month t to $t + 5$ ($\beta 6$), and from month t to $t + 11$ ($\beta 12$), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in $\beta 6$ means that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the $\beta 6$ decile.

A.6.9 $\beta^{FP} 1$, $\beta^{FP} 6$, and $\beta^{FP} 12$, the Frazzini-Pedersen beta. We estimate the market beta for stock i , β^{FP} , as $\hat{\rho} \hat{\sigma}_i / \hat{\sigma}_m$, in which $\hat{\sigma}_i$ and $\hat{\sigma}_m$ are the estimated return volatilities for the stock and the market, and $\hat{\rho}$ is their return correlation. To estimate return volatilities, we compute the standard deviations of daily log returns over a 1-year rolling window (with at least 120 daily returns). To estimate return correlations, we use overlapping 3-day log returns, $r_{it}^{3d} = \sum_{k=0}^2 \log(1 + r_{t+k}^i)$, over a 5-year rolling window (with at least 750 daily returns). At the beginning of each month t , we sort stocks into deciles based on β^{FP} estimated at the end of month $t - 1$. Monthly decile returns are

calculated for the current month t (β^{FP1}), from month t to $t+5$ (β^{FP6}), and from month t to $t+11$ (β^{FP12}), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in β^{FP6} means that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the β^{FP6} decile.

A.6.10 β^D1 , β^D6 , and β^D12 , the Dimson beta. We use the current as well as the lead and lag of the market return when estimating the market beta:

$$r_{id} - r_{fd} = \alpha_i + \beta_{i1}(r_{md-1} - r_{fd-1}) + \beta_{i2}(r_{md} - r_{fd}) + \beta_{i3}(r_{md+1} - r_{fd+1}) + \epsilon_{id}, \quad (A35)$$

in which r_{id} is stock i 's return on day d , r_{md} is the market return, and r_{fd} is the risk-free rate. The Dimson beta of stock i , β^D , is calculated as $\hat{\beta}_{i1} + \hat{\beta}_{i2} + \hat{\beta}_{i3}$. At the beginning of each month t , we sort stocks into deciles based on β^D estimated with the daily returns from month $t-1$ (the lead market return is within month $t-1$ to avoid look-ahead bias). We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (β^D1), from month t to $t+5$ (β^D6), and from month t to $t+11$ (β^D12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in β^D6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the β^D6 decile.

A.6.11 Tur1, Tur6, and Tur12, share turnover. A stock's share turnover, Tur, is its average daily share turnover over the prior six months. We require a minimum of 50 days. Daily turnover is the number of shares traded on a given day divided by the number of shares outstanding on that day.¹⁵ At the beginning of each month t , we sort stocks into deciles on Tur over the prior six months from $t-6$ to $t-1$. Monthly decile returns are calculated for the current month t (Tur1), from month t to $t+5$ (Tur6), and from month t to $t+11$ (Tur12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Tur6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdecile returns as the monthly return of the Tur6 decile.

A.6.12 Cvt1, Cvt6, and Cvt12, coefficient of variation of share turnover. We calculate a stock's coefficient of variation (the ratio of the standard deviation to the mean) for its daily share turnover, Cvt, over the prior six months. We require a minimum of 50 daily observations. Daily turnover is the number of shares traded on a given day divided by the number of shares outstanding on that day. We adjust the trading volume of NASDAQ stocks per Gao and Ritter (2010) (see footnote 15). At the beginning of each month t , we sort stocks into deciles based on Cvt over the prior six months from $t-6$ to $t-1$. Monthly decile returns are calculated for the current month t (Cvt1), from month t to $t+5$ (Cvt6), and from month t to $t+11$ (Cvt12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Cvt6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the return of the Cvt6 decile.

¹⁵ We adjust the NASDAQ trading volume to account for the institutional differences between NASDAQ and NYSE-Amex volumes (Gao and Ritter 2010). Prior to February 1, 2001, we divide NASDAQ volume by two. On February 1, 2001, a "riskless principal" rule goes into effect and results in a reduction of approximately 10% in reported volume. From February 1, 2001 to December 31, 2001, we divide NASDAQ volume by 1.8. During 2002, securities firms began to charge institutional investors commissions on NASDAQ trades, rather than the prior practice of marking up or down the net price. This practice reduces reported volume by roughly 10%. For 2002 and 2003, we divide NASDAQ volume by 1.6. For 2004 and later years, we use a divisor of one.

A.6.13 Dtv1, Dtv6, and Dtv12, dollar trading volume. At the beginning of each month t , we sort stocks into deciles based on their average daily dollar trading volume, Dtv, over the prior six months from $t-6$ to $t-1$. We require a minimum of 50 days. Dollar trading volume is share price times the number of shares traded. We adjust the trading volume of NASDAQ stocks per Gao and Ritter (2010) (see footnote 15). Monthly decile returns are calculated for the current month t (Dtv1), from month t to $t+5$ (Dtv6), and from month t to $t+11$ (Dtv12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Dtv6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the return of the Dtv6 decile.

A.6.14 Cvd1, Cvd6, and Cvd12, coefficient of variation of dollar trading volume. We calculate a stock's coefficient of variation (the ratio of the standard deviation to the mean) for its daily dollar trading volume, Cvd, over the prior six months. We require a minimum of 50 daily observations. Dollar trading volume is share price times the number of shares. We adjust the trading volume of NASDAQ stocks per Gao and Ritter (2010) (see footnote 15). At the beginning of each month t , we sort stocks into deciles based on Cvd over the prior six months from $t-6$ to $t-1$. Monthly decile returns are calculated for the current month t (Cvd1), from month t to $t+5$ (Cvd6), and from month t to $t+11$ (Cvd12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Cvd6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Cvd6 decile.

A.6.15 Pps1, Pps6, and Pps12, share price. At the beginning of each month t , we sort stocks into deciles based on share price, Pps, at the end of month $t-1$. Monthly decile returns are calculated for the current month t (Pps1), from month t to $t+5$ (Pps6), and from month t to $t+11$ (Pps12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Pps6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Pps6 decile.

A.6.16 Ami1, Ami6, and Ami12, absolute return-to-volume. We calculate the Amihud (2002) illiquidity measure, Ami, as the ratio of absolute daily stock return to daily dollar trading volume, averaged over the prior six months. We require a minimum of 50 daily observations. Dollar trading volume is share price times the number of shares traded. We adjust the trading volume of NASDAQ stocks per Gao and Ritter (2010) (see footnote 15). At the beginning of each month t , we sort stocks into deciles based on Ami over the prior six months from $t-6$ to $t-1$. Monthly decile returns are calculated for the current month t (Ami1), from month t to $t+5$ (Ami6), and from month t to $t+11$ (Ami12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ami6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Ami6 decile.

A.6.17 Lm¹1, Lm¹6, Lm¹12, Lm⁶1, Lm⁶6, Lm⁶12, Lm¹²1, Lm¹²6, Lm¹²12, turnover-adjusted number of zero daily volume. Following Liu (2006), we calculate the standardized turnover-adjusted number of zero daily trading volume over the prior x month, Lm^x , as follows:

$$Lm^x \equiv \left[\text{Number of zero daily volume in prior } x \text{ months} + \frac{1/(x - \text{month TO})}{\text{Deflator}} \right] \frac{21x}{\text{NoTD}}, \quad (\text{A36})$$

in which x -month TO is the sum of daily turnover over the prior x months ($x=1, 6$, and 12). Daily turnover is the number of shares traded on a given day divided by the number of shares outstanding on that day. We adjust the trading volume of NASDAQ stocks per Gao and Ritter (2010) (see footnote 15). NoTD is the total number of trading days over the prior x months. We set the deflator

to $\max\{1/(x - \text{month TO})\} + 1$, in which the maximization is taken across all sample stocks each month. Our choice of the deflator ensures that $(1/(x - \text{month TO}))/\text{Deflator}$ is between zero and 1 for all stocks. We require a minimum of 15 daily turnover observations when estimating Lm^1 , 50 for Lm^6 , and 100 for Lm^{12} .

At the beginning of each month t , we sort stocks into deciles based on Lm^x , with $x = 1, 6$, and 12. We calculate decile returns for the current month t ($\text{Lm}^x 1$), from month t to $t+5$ ($\text{Lm}^x 6$), and from month t to $t+11$ ($\text{Lm}^x 12$). The deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in $\text{Lm}^x 6$ mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the $\text{Lm}^x 6$ decile.

A.6.18 Mdr1, Mdr6, and Mdr12, maximum daily return. At the beginning of each month t , we sort stocks into deciles based on maximal daily return, Mdr, in month $t-1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Mdr1), from month t to $t+5$ (Mdr6), and from month t to $t+11$ (Mdr12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Mdr6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Mdr6 decile.

A.6.19 Ts1, Ts6, and Ts12, total skewness. At the beginning of each month t , we sort stocks into deciles based on total skewness, Ts, calculated with daily returns from month $t-1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Ts1), from month t to $t+5$ (Ts6), and from month t to $t+11$ (Ts12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Ts6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Ts6 decile.

A.6.20 Isc1, Isc6, and Isc12, idiosyncratic skewness per the CAPM. At the beginning of each month t , we sort stocks into deciles based on idiosyncratic skewness, Isc, calculated as the skewness of the residuals from regressing a stock's excess return on the market excess return using daily observations from month $t-1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Isc1), from month t to $t+5$ (Isc6), and from month t to $t+11$ (Isc12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Isc6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Isc6 decile.

A.6.21 Isff1, Isff6, and Isff12, idiosyncratic skewness per the Fama and French (1993) 3-factor model. At the beginning of each month t , we sort stocks into deciles based on idiosyncratic skewness, Isff, calculated as the skewness of the residuals from regressing a stock's excess return on the Fama and French (1993) three factors using daily observations from month $t-1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Isff1), from month t to $t+5$ (Isff6), and from month t to $t+11$ (Isff12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Isff6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Isff6 decile.

A.6.22 Isq1, Isq6, and Isq12, idiosyncratic skewness per the q -factor model. At the beginning of each month t , we sort stocks into deciles based on idiosyncratic skewness, Isq, calculated as the skewness of the residuals from regressing a stock's excess return on the Hou, Xue, and Zhang (2015) q -factors using daily observations from month $t-1$. We require a minimum of 15 daily

returns. Monthly decile returns are calculated for the current month t (Isq1), from month t to $t+5$ (Isq6), and from month t to $t+11$ (Isq12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Isq6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Isq6 decile. Because the q -factors start in January 1967, the Ivq portfolios start in February 1967.

A.6.23 Cs1, Cs6, and Cs12, coskewness. Following Harvey and Siddique (2000), we measure coskewness, Cs, as

$$Cs = \frac{E[\epsilon_i \epsilon_m^2]}{\sqrt{E[\epsilon_i^2]E[\epsilon_m^2]}}, \quad (\text{A37})$$

in which ϵ_i is the residual from regressing stock i 's excess return on the market excess return, and ϵ_m is the demeaned market excess return. At the beginning of each month t , we sort stocks into deciles based on Cs calculated with daily returns from month $t-1$. We require a minimum of 15 daily returns. Monthly decile returns are calculated for the current month t (Cs1), from month t to $t+5$ (Cs6), and from month t to $t+11$ (Cs12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Cs6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Cs6 decile.

A.6.24 Tail1, Tail6, and Tail12, tail risk. Following Kelly and Jiang (2014), we estimate common tail risk, λ_t , by pooling daily returns for all stocks in month t , as follows:

$$\lambda_t = \frac{1}{K_t} \sum_{k=1}^{K_t} \log \frac{R_{kt}}{\mu_t}, \quad (\text{A38})$$

in which μ_t is the fifth percentile of all daily returns in month t , R_{kt} is the k th daily return that is below μ_t , and K_t is the total number of daily returns that are below μ_t . At the beginning of each month t , we split stocks on tail risk, Tail, estimated as the slope from regressing a stock's excess returns on 1-month-lagged common tail risk over the most recent 120 months from $t-120$ to $t-1$. We require a minimum of least 36 monthly observations. Monthly decile returns are calculated for the current month t (Tail1), from month t to $t+5$ (Tail6), and from month t to $t+11$ (Tail12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Tail6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Tail6 decile.

A.6.25 $\beta^{\text{ret}}1$, $\beta^{\text{ret}}6$, $\beta^{\text{ret}}12$, $\beta^{\text{lcc}}1$, $\beta^{\text{lcc}}6$, $\beta^{\text{lcc}}12$, $\beta^{\text{lrc}}1$, $\beta^{\text{lrc}}6$, $\beta^{\text{lrc}}12$, $\beta^{\text{lcr}}1$, $\beta^{\text{lcr}}6$, $\beta^{\text{lcr}}12$, $\beta^{\text{net}}1$, $\beta^{\text{net}}6$, and $\beta^{\text{net}}12$, liquidity betas (return-return, illiquidity-illiquidity, return-illiquidity, illiquidity-return, and net). Following Acharya and Pedersen (2005), we measure illiquidity using the Amihud (2002) measure, Ami_t^i , is the average ratio of absolute daily return to daily dollar trading volume. We require a minimum of 15 daily observations. Dollar trading volume is share price times the number of shares traded. We adjust the trading volume of NASDAQ stocks per Gao and Ritter (2010) (see footnote 15). The Market illiquidity, Ami_t^M , is the value-weighted average of $\min(\text{Ami}_t^i, (30-0.25)/(0.30P_{t-1}^M))$, in which P_{t-1}^M is the ratio of the total market capitalization of S&P 500 at the end of month $t-1$ to its value at the end of July 1962. We measure market illiquidity innovations, ϵ_{Mt}^i , as the residual from the regression below:

$$\begin{aligned} (0.25+0.30\text{Ami}_t^M P_{t-1}^M) = & a_0 + a_1(0.25+0.30\text{Ami}_{t-1}^M P_{t-1}^M) \\ & + a_2(0.25+0.30\text{Ami}_{t-2}^M P_{t-1}^M) + \epsilon_{Mt}^i \end{aligned} \quad (\text{A39})$$

Innovations to individual stocks' illiquidity, ϵ_{it}^i , are measured analogously by replacing Ami_t^M with $\min(\text{Ami}_t^i, (30-0.25)/(0.30P_{t-1}^M))$ in equation (A39). Finally, innovations to the market return are

measured as the residual, ϵ_{Mt}^r , from the second-order autoregression of the market return. Following Acharya and Pedersen, we define five measures of liquidity betas:

$$\text{Return—return: } \beta_i^{\text{ret}} \equiv \frac{\text{Cov}(r_{it}, \epsilon_{Mt}^r)}{\text{var}(\epsilon_{Mt}^r - \epsilon_{Mt}^c)} \quad (\text{A40})$$

$$\text{Illiquidity—illiquidity: } \beta_i^{\text{lcc}} \equiv \frac{\text{Cov}(\epsilon_{it}^c, \epsilon_{Mt}^c)}{\text{var}(\epsilon_{Mt}^r - \epsilon_{Mt}^c)} \quad (\text{A41})$$

$$\text{Return—illiquidity: } \beta_i^{\text{lrc}} \equiv \frac{\text{Cov}(r_{it}, \epsilon_{Mt}^c)}{\text{var}(\epsilon_{Mt}^r - \epsilon_{Mt}^c)} \quad (\text{A42})$$

$$\text{Illiquidity—return: } \beta_i^{\text{lcr}} \equiv \frac{\text{Cov}(\epsilon_{it}^c, \epsilon_{Mt}^r)}{\text{var}(\epsilon_{Mt}^r - \epsilon_{Mt}^c)} \quad (\text{A43})$$

$$\text{Net: } \beta_i^{\text{net}} \equiv \beta_i^{\text{ret}} + \beta_i^{\text{lcc}} - \beta_i^{\text{lrc}} - \beta_i^{\text{lcr}} \quad (\text{A44})$$

At the beginning of each month t , we sort stocks, separately, on β^{ret} , β^{lcc} , β^{lrc} , β^{lcr} , and β^{net} , estimated with the past 60 months (at least 24 months) from $t-60$ to $t-1$. Monthly decile returns are calculated for the current month t ($\beta^{\text{ret}}1$, $\beta^{\text{lcc}}1$, $\beta^{\text{lrc}}1$, $\beta^{\text{lcr}}1$, and $\beta^{\text{net}}1$), from month t to $t+5$ ($\beta^{\text{ret}}6$, $\beta^{\text{lcc}}6$, $\beta^{\text{lrc}}6$, $\beta^{\text{lcr}}6$, and $\beta^{\text{net}}6$), and from month t to $t+11$ ($\beta^{\text{ret}}12$, $\beta^{\text{lcc}}12$, $\beta^{\text{lrc}}12$, $\beta^{\text{lcr}}12$, and $\beta^{\text{net}}12$), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in $\beta^{\text{lcc}}6$ mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the $\beta^{\text{lcc}}6$ decile.

A.6.26 Srev, short-term reversal. At the beginning of each month t , we sort stocks into short-term reversal (Srev) deciles based on the return in month $t-1$. To be included in a decile in month t , a stock must have a valid price at the end of month $t-2$ and a valid return for month $t-1$. Monthly decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t+1$.

A.6.27 β^-1 , β^-6 , and β^-12 , downside beta. Following Ang, Chen, and Xing (2006), we define downside beta, β^- , as

$$\beta^- = \frac{\text{Cov}(r_i, r_m | r_m < \mu_m)}{\text{Var}(r_m | r_m < \mu_m)}, \quad (\text{A45})$$

in which r_i is stock i 's excess return r_m is the market excess return, and μ_m is the average market excess return. At the beginning of each month t , we sort stocks into deciles based on β^- , which is estimated with daily returns from prior 12 months from $t-12$ to $t-1$ (we only use daily observations with $r_m < \mu_m$). We require a minimum of 50 daily returns. Monthly decile returns are calculated for the current month t (β^-1), from month t to $t+5$ (β^-6), and from month t to $t+11$ (β^-12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in β^-6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the β^-6 decile.

A.6.28 Shl1, Shl6, and Shl12, the high-low bid-ask spread. The monthly Corwin and Schultz (2012) stock-level high-low bid-ask spread, Shl, are generated using codes from Shane Corwin's Web site. At the beginning of each month t , we sort stocks into deciles based on Shl for month $t-1$. Monthly decile returns are calculated for the current month t (Shl1), from month t to $t+5$ (Shl6), and from month t to $t+11$ (Shl12), and the deciles are rebalanced at the beginning of month $t+1$. Holding periods longer than one month like in Shl6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Shl6 decile.

A.6.29 Sba1, Sba6, and Sba12, The bid-ask spread. The monthly Hou and Loh (2016) stock-level bid-ask spread, Sba, are provided by Roger Loh for the sample period from 1984 to 2014 (excluding 1986 because of missing data). At the beginning of each month t , we sort stocks into deciles based on Sba for month $t - 1$. Monthly decile returns are calculated for the current month t (Sba1), from month t to $t + 5$ (Sba6), and from month t to $t + 11$ (Sba12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in Sba6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the Sba6 decile. The sample period for the Sba portfolios is from February 1984 to January 2015 (excluding February 1986 to January 1987).

A.6.30 β^{lev} 1, β^{lev} 6, and β^{lev} 12, the financial intermediary leverage beta. At the beginning of each quarter, we estimate a stock's financial intermediary leverage beta, β^{lev} , from regressing its quarterly returns in excess of the 3-month Treasury-bill rate on the quarterly nontraded leverage factor during the past 40 quarters (20 quarters minimum). We construct the leverage of financial intermediary using quarterly aggregate data on total financial assets and liabilities of security broker-dealers from table L.130 of the Federal Reserve Flow of Funds. To be consistent with the original data, we combine the repurchase agreement (repo) liabilities and the reverse repo assets into net repo liabilities. The financial intermediary leverage is measured as total financial assets/(total financial assets – total financial liabilities). The nontraded leverage factor is the seasonally adjusted log change in the level of leverage. The log changes are seasonally adjusted using quarterly seasonal dummies in expanding window regressions. Following Adrian, Etula, and Muir (2014), we start using the security broker-dealer data in the first quarter of 1968. The 3-month Treasury-bill rate data are from the Federal Reserve Bank database.

At the beginning of each month t , we sort stocks into deciles based on β^{lev} estimated at the beginning of the current quarter. Monthly decile returns are calculated for the current month t (β^{lev} 1), from month t to $t + 5$ (β^{lev} 6), and from month t to $t + 11$ (β^{lev} 12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in β^{lev} 6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the β^{lev} 6 decile. Because the financial intermediary leverage data start in 1968 and we need at least 20 quarters to estimate β^{lev} , the sample starts in January 1973.

A.6.31 β^{PS} 1, β^{PS} 6, and β^{PS} 12, the Pastor-Stambaugh beta. We estimate the liquidity risk beta, β^{PS} , as the sensitivity to innovations in aggregated liquidity:

$$r_{it} - r_{ft} = \beta_i^0 + \beta_i^{\text{PS}} L_t + \beta_i^{\text{M}} \text{MKT}_t + \beta_i^{\text{S}} \text{SMB}_t + \beta_i^{\text{H}} \text{HML}_t + \epsilon_{it}, \quad (\text{A46})$$

in which r_{it} is stock i 's return in month t , r_{ft} is the risk-free rate, L_t is the innovation in aggregated liquidity, and MKT_t , SMB_t , and HML_t are the Fama-French three factors. Data for innovations in aggregated liquidity are from Robert Stambaugh's Web site.

At the beginning of each month t , we sort stocks into deciles on their β^{PS} , which is estimated with monthly returns from month $t - 60$ to $t - 1$. We require a minimum of 24 monthly returns. Monthly decile returns are calculated for the current month t (β^{PS} 1), from month t to $t + 5$ (β^{PS} 6), and from month t to $t + 11$ (β^{PS} 12), and the deciles are rebalanced at the beginning of month $t + 1$. Holding periods longer than one month like in β^{PS} 6 mean that for a given decile in each month there exist six subdeciles, each initiated in a different month in the prior six months. We average the subdeciles returns as the monthly return of the β^{PS} 6 decile.

A.6.32 Pin, probability of information-based trading. At the beginning of each January in year t , we sort stocks into deciles based on the probability of information-based trading, Pin, available from Soeren Hvidkjaer's Web site for the period from 1983 to 2001. Monthly decile returns are calculated from January to December of year t , and the deciles are rebalanced at the beginning of year $t + 1$. The sample period for the Pin portfolios is from January 1984 to December 2002.

Appendix B. Delisting Adjustment

Following Beaver, McNichols, and Price (2007), we adjust monthly stock returns for delisting returns by compounding returns in the month before delisting with delisting returns from CRSP.

As discussed in Beaver, McNichols, and Price (2007), the monthly CRSP delisting returns (file `msedelist`) might not adjust for delisting properly. We follow their procedure to directly construct the delisting-adjusted monthly stock returns. For delisting that occurs before the last trading day in month t , we calculate the delisting-adjusted monthly return, DR_t , as

$$DR_t = (1 + \text{pmr}_{dt})(1 + \text{der}_{dt}) - 1, \quad (\text{B1})$$

in which pmr_{dt} is the partial month return from the beginning of the month to the delisting day d , and der_{dt} is the delisting event return from the daily CRSP delisting file (`dsedelist`).

We calculate the partial month return, pmr_{dt} , as follows:

- When the delisting date (item `DLSTDT`) is the same as the delisting payment date (item `DLPDT`), the monthly CRSP delisting return, mdr_t , includes only the partial month return:

$$\text{pmr}_{dt} = \text{mdr}_t. \quad (\text{B2})$$

- When the delisting date proceeds the delisting payment date, pmr_{dt} can be computed from the monthly CRSP delisting return and the delisting event return:

$$\text{pmr}_{dt} = \frac{1 + \text{mdr}_t}{1 + \text{der}_{dt}} - 1. \quad (\text{B3})$$

- If pmr_{dt} cannot be computed via the above methods, we construct it by accumulating daily returns from the beginning of month t to the delisting day d :

$$\text{pmr}_{dt} = \prod_{i=1}^d (1 + \text{ret}_{it}) - 1, \quad (\text{B4})$$

in which ret_{it} is the regular stock return on day i .

For delisting that occurs on the last trading day of month t , we include only the regular monthly return for month t , and account for the delisting return at the beginning of the following month: $DR_t = \text{ret}_t$ and $DR_{t+1} = \text{der}_{dt}$, in which ret_t is the regular full month return. Differing from Beaver, McNichols, and Price (2007), we do not account for these last-day delistings in the same month, because delisting generally occurs after the market closes. Also, delisting events are often surprises, and their payoffs cannot be determined immediately (Shumway 1997). As such, it might be problematic to incorporate delisting returns immediately on the last trading date in month t .

When delisting returns are missing, the delisting-adjusted monthly returns cannot be computed. Among nonfinancial firms traded on NYSE, Amex, and NASDAQ, there are 16,745 delistings from 1925 to 2016, with 86% of the delisting event returns available. One option is to exclude missing delisting returns. However, previous studies show that omitting these stocks can introduce biases in asset pricing tests (Shumway 1997; Shumway and Warther 1999). As such, we replace missing delisting returns using the average available delisting returns with the same stock exchange and delisting type (1-digit delisting code) during the past 60 months. We condition on stock exchange and delisting type because average delisting returns vary significantly across exchanges and delisting types. We also allow replacement values to vary over time because average delisting returns can vary greatly over time. Beaver, McNichols, and Price (2007) construct replacement values conditional on stock exchange and delisting type, but do not allow the replacement values to vary over time.

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